Blowin' in the wind: The evolvement of the wind merit order effect across Sweden's energy zones

Master Thesis

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Abstract

This paper analyses the impact of wind energy production on price levels and volatility across Sweden's four bidding zones. The findings include that wind power has a negative effect on price, being particularly strong during price spikes. At the same time, our thesis shows that a build-out of wind production capabilities is likely to increase volatility. Our research indicates that, while policy proposals should be mindful of additional volatility introduced by wind production, it should encourage further build-out to ensure against times of crisis.

Notes

This is the working version of a Master Thesis at the Stockholm School of Economics as of $March 17, 2026$. While we tried to bring this work-in-progress version closely to a final version, formatting issues or typos might have been missed. We ask to apologize these.

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We want to thank our supervisor, Prof. Rickard Sandberg, for his time and inputs. We also want to thank Prof. Elena Paltseva for her contributions towards this paper.

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Chapter 1

Introduction

On February 28, 2026, Israel and the United States of America launched air strikes on multiple locations within Iran (The Guardian 2026). While the aftermath of this conflict is not yet clear at time of writing, a significant spike in oil prices has followed these initial strikes (Marsh and Musafer 2026). This mirrors the price hike around the energy crisis in 2021/2022. As the dependence on fossil fuel shows its negative consequences, sustainable alternatives of energy production have moved back into the centre of the political discussion.

Sweden was the first country to establish an Environmental Protection Agency and hosted the world's first UN conference on the global environment (Swedish Institute 2026). The country positions itself as a leader of sustainability, aiming to make Sweden's energy fossil-free by 2045, supported by an ambitious build-out of windfarms and hydro power (Swedish Institute 2026). The government's objective is to lead the transformation away from fossil dependencies, highlighting the opportunities of green growth (Swedish Institute 2026). A well-functioning energy market is a prerequisite for these political ambitions and, therefore, a priority for Sweden's politics.

However, Sweden's electricity market is fractured. The gap between electricity prices in Sweden's different bidding zones is widening, starting in 2020 and increasingly since the European energy crisis in 2021 and 2022. In 2025, the Swedish Government tasked Svenska kraftnät to reevaluate the current configuration of Sweden's energy market and propose improvements, including a potential re-structuring of bidding zones (Svenska Kraftnät 2025). To ensure that meaningful policy suggestions can be developed, understanding the effects behind this price spread is essential.

This thesis aims to uncover the systematic change in Sweden's electricity market, focusing on the role of wind energy production. To achieve this, this study estimates the merit order effect of wind from 2015 to 2025, separately per bidding zone and per time window. Using a multivariate price model with appropriate controls, the estimation captures the response of prices to fluctuations in wind energy production. The analysis outlines the role of wind energy production in price formation over time, highlighting whether and how the sensitivity of prices towards wind energy has changed per bidding zone. Differentiating per zone is essential to understand the dynamics of the price spread and give targeted suggestions for market reform that account for local needs and energy mixes. This thesis places particular emphasis on the development of wind energy production coefficients on electricity prices during the energy crisis in 2021/2022 and the most recent period to indicate behaviour in times of market volatility and relevant future trends.

This thesis' research question is twofold:
How has the wind merit order effect evolved across Sweden's bidding zones, and what does that imply for efficient price formation under the current zonal structure?

The main contribution of this paper is the analysis of sensitivities over a longer time frame and their predictive power. Hence, the thesis provides the analytical foundation for the assessment of Svenska kraftnät. The evolvement of sensitivities of bidding zones over time has strong implications for Sweden's electricity market configuration as it outlines underlying mechanisms of price formation. This paper strengthens its analysis by additionally investigating the effect of wind energy on electricity price volatility. As consumers and firms must be able to plan, price levels and volatility are both essential for an efficient energy market and their understanding is a prerequisite for any policy that aims to reform the energy market.

While existing literature estimates sensitivities over shorter intervals or treats the relationship between wind and prices as stable, this paper provides a more nuanced picture of the electricity market. The analysis covers a decade defined by the rapid expansion of wind energy production capabilities, tests for structural breaks, and forms hypotheses about the future trends of sensitivities. The paper employs ARMAX-GARCH models on rolling windows to discover the trend of the underlying data. Understanding these trends, past and present, is essential to proposing target policy reforms.

In the paper, the terms electricity and energy are used interchangeably, referring consistently to the transportation of energy through electrical charges.

Chapter 2

Literature Review

The impact of wind energy production on electricity price level has been subject of multiple studies but knowledge about sensitivity changes over time remains fragmented. The following section outlines relevant literature, focusing on econometric studies. While ex-ante simulations are not the focus of the following analysis, Würzburg et al. (2013) provide an overview of previous work.

2.1 Econometric Studies

The impact of wind energy production of electricity prices has been a priority of research in energy markets and is outlined in the following section. While findings are consistent in showing that prices decrease with an increased production of wind energy, the size of the effect can vary across geographies and distributions of the energy price series (Würzburg et al. 2013; Macedo et al. 2021).

Sweden and Western Denmark A starting point of this thesis is the analysis of Sandberg (2024), who analyses the effect of wind and temperature on Swedish electricity prices from 2022 to 2023. The paper demonstrates a negative (price decreasing) effect by analysing four bidding zones and using various reference models, including fixed effects panel data model, piecewise linear model, and a dynamic model (Sandberg 2024). The thesis explicitly addresses the suggestion in Sandberg (2024) to investigate the gradual impact of wind on energy prices over time. Sandberg's insights are complimented by Macedo et al. (2021). The authors research the effect of wind power production on the mean and volatility of electricity prices in SE3, Sweden's bidding zone with the highest electricity flow, between 2016 and 2019, confirming wind power's merit order effect. Similar to the approach taken in this paper, the authors in Macedo et al.

(2021) use hourly data to assess each hour individually under a multivariate modelling framework. The authors highlight that electricity prices often show autocorrelation and heteroskedasticity, making it essential to account for volatility-clustering, mean-reversion, outliers, and seasonality. Macedo et al. (2021) highlights that, because of a current lack of large-scale storage capacity of electricity, the missing synchronization of production and consumption times can result in spikes. Jónsson et al. (2010) focuses on the effect of day-ahead wind power forecasts on day-ahead electricity price in Western Denmark between 2006 and 2007. The authors highlight the complexity of electricity spot prices due to its instantaneous nature, the almost inelastic demand, and the discontinuity and convexity of its supply function (Jónsson et al. 2010).By using a non-parametric regression model, they highlight the non-linear relationship between electricity price and wind power. Jónsson et al. (2010) use the penetration level of wind power, defined as the share of wind energy of total energy supply. The authors confirm previous studies and their finding that wind power has a relevant effect on day-ahead prices in the electricity market (Jónsson et al. 2010).

Germany Due to its size and geographic interconnection, Germany's energy market is a very interesting case study. Research on the effect of wind on electricity prices in Germany include Cludius et al. (2014), Jónsson et al. (2010), Ketterer (2014), and Paraschiv et al. (2014).

Ketterer (2014) uses a GARCH model on day-ahead prices from 2006 to 2012 to understand the effect of wind electricity generation on price level and volatility. She finds that wind power decreases electricity prices but increases volatility. This paper builds on the work by Ketterer (2014) in its methodology, applying it to the Swedish electricity market and indicating the evolvement of coefficients over time. Highlighting that the volatility of electricity prices has changed after a regulatory change in 2010, Ketterer (2014) emphasizes that market design can impact and potentially smoothen electricity price volatility. Ketterer (2014) uses forecasted values for her analysis of day-ahead prices, arguing that only the forecasted and not actual values were available to market participants at the time of price setting and, hence, only these should be included. Ketterer (2014) findings are supported by Cludius et al. (2014), investigating hourly energy prices between 2010 to 2012 in Germany. They model the day-ahead electricity spot price as dependent on wind electricity and solar electricity feed-in and load to indicate aggregate demand, showing that electricity generation by wind lowered market prices. Furthermore, Cludius et al. (2014) develop a forecasting tool for the merit order effect from 2013 to 2016, highlighting the importance of estimates on capacity extension of renewable energies over time to derive an adequate estimate. This thesis will follow a similar approach in predicting the future sensitivities of prices and, as a result, the merit order effect. The research by Cludius et al. (2014) is complemented by the work of Paraschiv et al. (2014). The authors conclude that the amount of electricity generated from renewables favours strong changes in price. In their analysis, they employ a dynamic fundamental model to indicate a continuous adaption of electricity prices to market fundamentals, including fossil fuel prices, expected demand, and prices for emission certificates into their study (Paraschiv et al. 2014). Similarly to the approach of this thesis, the authors outline the change of coefficients over time and highlight that the sensitivities of day-ahead electricity spot prices change over time (Paraschiv et al. 2014).

Global Other geographies that have been analysed on the effect of wind energy production on electricity prices include Italy, Australia, Greece, Portugal, and Spain Clò et al. (2015), Csereklyei et al. (2019), Macedo et al. (2020), Macedo et al. (2022), Maniatis and Milonas (2022), and Mwampashi et al. (2021). The following section outlines previous research and its implications.

Clò et al. (2015) analyse the merit order effect of solar and wind energy in Italy from 2005 to 2013. The authors prove the merit order effect and the decrease of energy prices along with an increase in wind energy. However, they highlight that the effect declines as solar and wind energy production increased (Clò et al. 2015). In Australia, Mwampashi et al. (2021) study the dynamics between electricity prices and wind electricity generation, concluding a negative effect of wind generation on prices by using a model of exponential generalized autoregressive conditional heteroscedasticity (eGARCH). The authors also indicate a positive (increasing) effect of wind production on price volatility, highlighting that the lower energy prices might come at the cost of increased price uncertainty and volatility. Their findings are in line with Csereklyei et al. (2019) that highlights Australia's increased electricity price volatility following additional wind electricity generation. These studies aid the methodology of this paper to enhance the understanding of how wind energy impacts price level and volatility. Maniatis and Milonas (2022) investigate the impact of electricity generation from wind and solar power on price and volatility in Greece between 2012 and 2018. They employ a GARCH-in-mean model on hourly data, confirming the merit-order effect and indicating that wind power generation increases volatility after controlling for regulatory mechanisms. They align their approach with Ketterer (2014) in using forecasted wind output and system load for their analysis as only the forecasted values are known to market participants when the price for the next day is set. This is supported by Macedo et al. (2020), analysing the effect of new integrating renewable energy on electricity prices in Portugal from 2011 to 2019. The authors use a SARMAX/eGARCH approach, concluding that, electricity generation from wind reduces the price level of electricity but increases its volatility (Macedo et al. 2020). In Spain, Macedo et al. (2022) analyse the effect on volatility and day-ahead mean electricity prices based on wind and solar electricity generation from 2015 to 2020. The authors conclude that a merit-order-effect exists during most hours of the day but varies in magnitude and increases price volatility at all hours analysed (Macedo et al. 2022).

Quantile regression approaches Other analyses include quantile regression approaches to differentiate the effect of renewable energies at different price levels of the supply curve Hagfors et al. (2016) and Maciejowska (2020). These methodologies were additionally employed in this paper to test initial conclusions from results and strengthen implications for policy.

Maciejowska (2020) employs a quantile regression analysis to determine the impact of renewable energy production on electricity price level and volatility, focusing on the German electricity market. With this analysis, the author aims to estimate the merit order effect on different quantiles of electricity prices to understand the impact on price dependent on where in the distribution the price is found. The author's results indicate that wind energy production has a negative impact on the electricity price level. The author demonstrates that the coefficient differs for different levels of demand, with wind having a strong price stabilising effect in times of high load, thereby reducing the probability of positive price spikes (Maciejowska 2020).

These results are confirmed by Hagfors et al. (2016). The authors employ a quantile regression approach to scrutinise the impact of renewables on EEX prices, quantifying the relationship between renewables and prices at a given hour and a given price quantile. They find that wind energy decreases the electricity price but highlight that, especially during nighttime, the marginal effects decline in absolute value with the quantiles and become less statistically significant (Hagfors et al. 2016). The lowest quantile, Q01, shows strong effects throughout all times of the day but is not statistically significant for most of them (Hagfors et al. 2016).

2.2 Scientific Extension

In light of policy discussions around Sweden's energy market structure, this paper shines light on recent dynamics in Sweden's energy market and interpolates future trends and their implications, Thereby, the analysis contributes to the Sweden's policy discussion in two ways, by providing context for the recent discussion around bidding zones and by outlining implications on a spread in bidding zone electricity prices on producers of wind energy. In its analysis, this paper builds upon previous analysis, most prominently Sandberg (2024), Ketterer (2014), Macedo et al. (2021), Macedo et al. (2020), and Maciejowska (2020). The paper combines a rolling window analysis with an ARMAX-GARCH model and a quantile regression model, thereby combining different perspectives on the energy market and providing detailed insights into the effects of wind energy on prices level and volatility over time and at different points of the price distribution.

Chapter 3

Electricity Markets and Nord Pool

The following section provides context on energy markets in general, including the merit order effect, and gives specific insights into the structure of Sweden's energy market. Furthermore, the different bidding zones and their differences are highlighted.

3.1 Introduction to Electricity Markets

In many regions, energy is traded on a central platform that connects buyers and suppliers. By submitting individual bids on prices that they are willing to buy and sell energy for, supply and demand curves are formed, and an equilibrium price is determined. To create these supply curves, a ranking of suppliers is necessary. This ordering is called a merit order, defined as the ranking of a market's energy suppliers based on their production costs, starting with the lowest-cost energy provider. Low-cost energy suppliers include renewable energy providers while, e.g., gas turbines are costly for energy production. (Zeier 2026). As demand in the market increases, more energy is needed and additional producers contribute to the total supply of energy despite their increasingly higher production prices, influencing energy costs (Zeier 2026). To encourage suppliers to enter the market in times of increasing demand, the market clearing price must be at least as high as the production cost of the providers that should enter the market. Therefore, the market equilibrium price is equal to the marginal production cost of the most expensive energy supplier that is needed to serve a respective energy demand. By relying on the merit order, producers enter the markets sequentially according to their increasing production costs, thereby using market resources efficiently (Zeier 2026). Since renewable energy sources like solar or wind energy are at the lower end of the price curve, their expansion and prevalence in the market changes the supply curve of energy, shifting the merit order curve and leading to a reduction in price (Zeier 2026). This shift makes higher-cost energy production technologies less competitive. Figure 3.1 displays a merit order curve for Sweden exemplarily, highlighting the effect that a loss of wind production power supply (e.g., in times without wind) has on prices if demand remains unchanged. This shift in price dependent on the supply of certain energies is called the merit order effect.

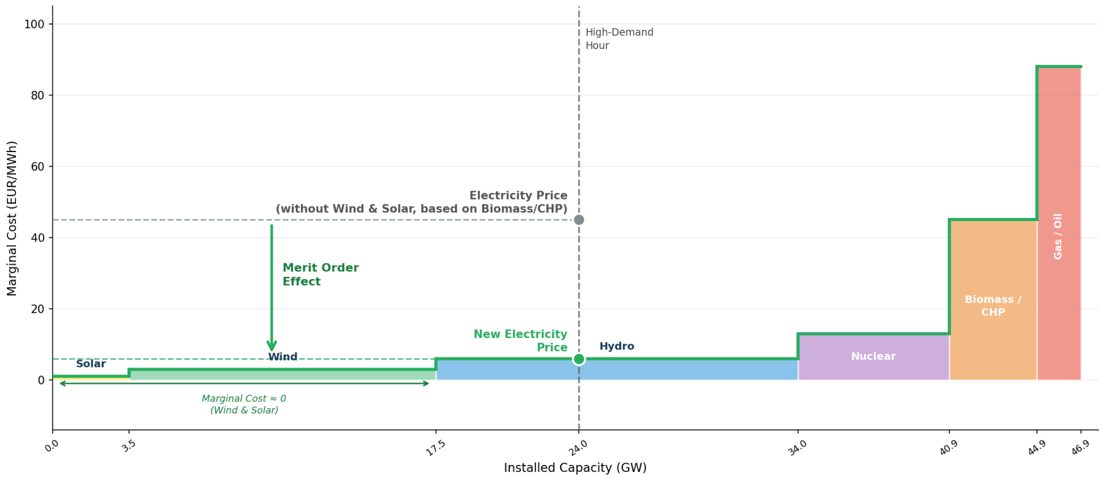
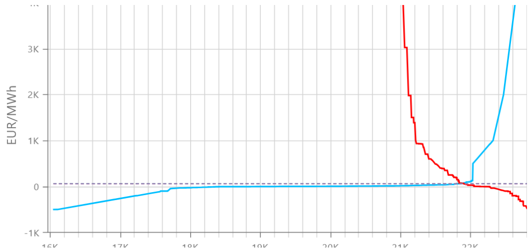
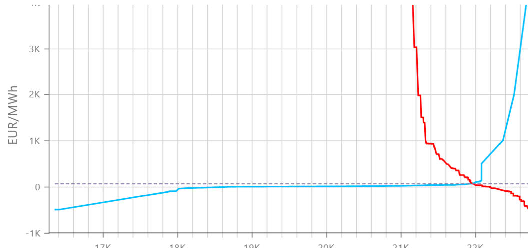


Figure 3.1: Exemplary Merit Order Curve for Sweden

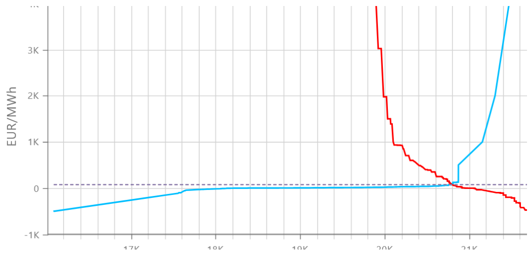
From the demand side, changes in temperature, sunlight, or European market coupling (e.g., Germany's decision to shut down nuclear power plants) impact the need for electricity and change the position of the demand curve (Sandberg 2024). The supply side is influenced by network constraints in energy transmission, European market coupling, or changes in wind power production (Sandberg 2024). Hence, the final market price is dependent on forecasted consumption, production, and transmission capacity for every hour on the day, determined one day in advance on the day-ahead market. The intersection of supply and demand curves determines market price and volume (Zeier 2026). The supply offers below (or equal to) and the demand offers above (or equal to) the market clearing price are accepted. The market clearing price is the amount actually paid per unit of energy (Zeier 2026). Figure 3.2 shows Sweden's supply and demand curves at different times.



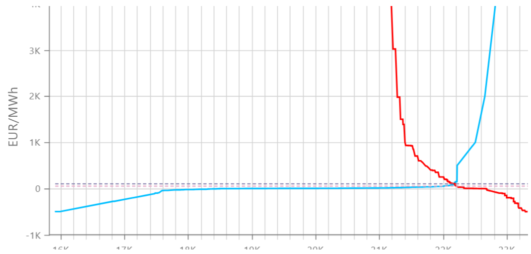
(a) Delivery period: 00:00-00:15



(b) Delivery period: 06:00-06:15



(c) Delivery period: 12:00-12:15



(d) Delivery period: 18:00-18:15

Figure 3.2: Merit Order Curves for Sweden (31.12.2025)

3.2 Nord Pool

Nord Pool is a power exchange platform that follows the previously outlined energy market principles, enabling market participants to submit their bids and offers (Zeier 2026). Sweden is part of the Nord Pool electricity market, one of the world's most integrated electricity markets. Nord Pool covers Sweden, Norway, Denmark, and the Baltic countries Lithuania, Estonia, and Latvia (Macedo et al. 2021; Nord Pool 2026). Sweden is divided into four different bidding zones, SE1, SE2, SE3, and SE4, since 2011 while there are five zones in Norway, one in Finland, and two in Denmark (Unger et al. 2018). Figure **??** outlines the entire area covered by the Nord Pool market and highlights the borders of the individual zones.

captionSweden's bidding zones as traded on Nord Pool (31.12.2025)

The Nord Pool market consists of a day-ahead market and an intra-day market for electricity
parenciteunger$_{l}ong - term_{2}017$.
$autoreftab : cross_{b}order_{f}lows_{2}025outlinestheinterconnectionbetweenthedifferentzonesinSwedenar$
$autoreffig : NordPool_{z}ones, Sweden'selectricitymarketisconnectedtoDenmark(DK1andDK2), No$

Table 3.1: Net Exchange by Swedish Bidding Zone and Trading Partner, 2025 (GWh)

Zone	SE1	SE2	SE3	SE4	Finland	Norway	Denmark	Germany	Lithuania	Poland
SE1	—	+10, 574	—	—	+5, 910	−772	—	—	—	—
SE2	−10, 574	—	+50, 550	—	—	−253	—	—	—	—
SE3	—	−50, 550	—	+29, 026	+4, 034	+4, 516	+3, 408	—	—	—
SE4	—	—	−29, 026	—	—	—	+7, 204	+2, 419	+3, 893	+3, 267
Sweden (total)	—	—	—	—	+9, 944	+3, 491	+10, 612	+2, 419	+3, 893	+3, 267

Note: Values are annual net exchanges in GWh (positive = net export from the row zone to the column zone). Norway includes NO1, NO3, and NO4; Denmark includes DK1 and DK2. The Sweden total row excludes intra-Swedish flows and sums external partner columns.

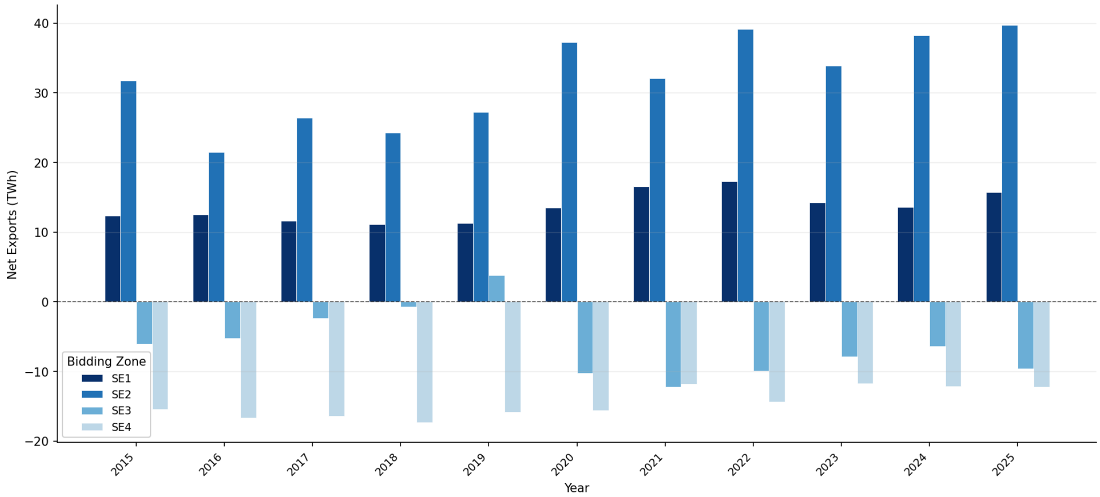
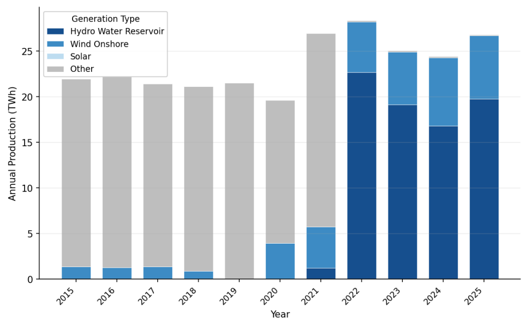


Figure 3.3: Energy net-export by bidding zone, 2015-2025

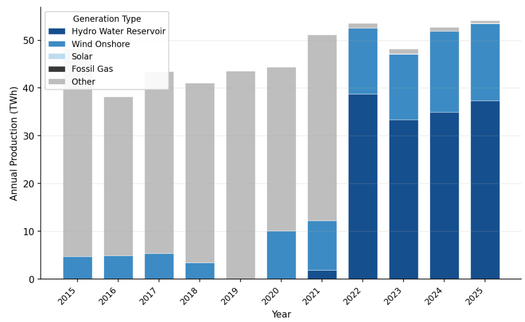
This paper focuses on energy traded on the day-ahead market which constitutes 99.8% of all energy traded in Sweden in 2025 (Nord Pool 2026). On the day-ahead market, electricity is traded at 12:00 central European time (midday) for all 24 hours of the following day separately, using the marginal price principle (Sorknæs et al. 2019). Since October 2025, Nord Pool has switched to 15-minute intervals instead of hourly intervals for electricity trading, allowing the market more flexibility and a better match of supply and demand. The final market price is set by compiling all orders at which participants are willing to sell, starting at -500 EUR to +4000 EUR. A similar process compiles the buy orders. To eliminate price differences between zones, cross-border energy flows via interconnections are used. An objective of this research is to outline why these cross-border transactions are not sufficient to eliminate or reduce electricity price arbitrage opportunities between zones (Unger et al. 2018). Once all bids and offers have been submitted on the Nord Pool market, a market clearing price is calculated, not taking into account the physical constraints of the transmission system (Unger et al. 2018). For each individual bid zone, a supply and demand curve is calculated that constitutes the equilibrium price, taking into account the congestion in the energy grid (Unger et al. 2017; Unger et al. 2018).

SE1 and SE2, the bidding zones in northern Sweden, tend to have a deficit in electricity consumption, while SE3 and SE4 in southern Sweden have a deficit in electricity production (Macedo et al. 2021). This makes the southern bidding zones net importers and the northern bidding zones net exporters of electricity Figure 3.3. In general, Sweden is a net-exporter of energy in 2025.

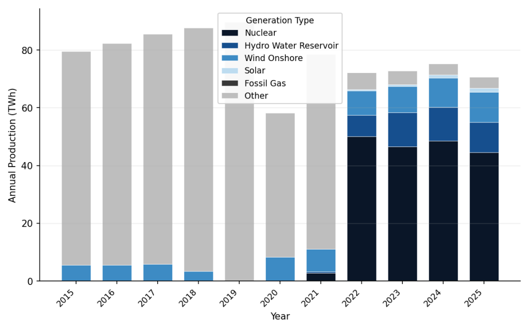
Sweden's bidding zones also differ in their means of electricity generation. SE1 and SE2 are strongly dependent on hydropower, while SE3 incorporates a significant share of nuclear energy in its energy composition **??** (Macedo et al. 2021). The low marginal costs of nuclear power generation result in price differences in electricity production between SE3 and other bidding zones (Macedo et al. 2021). Figure 3.4 outlines the difference in the mix of energy production between the zones, which implies different marginal cost curves and, as a result, different equilibrium prices per bid zone.



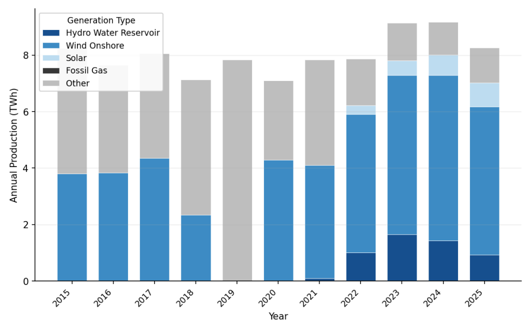
(a) SE1



(b) SE2



(c) SE3



(d) SE4

Figure 3.4: Swedish energy production mix by zone

Until 2020, the zones' electricity prices moved very closely together. However, since then and particularly after the start of the energy crisis in 2021/2022, the per-zone equilibrium prices started to diverge, resulting in higher electricity prices for some areas than others. Figure 3.5 depicts the change in energy price over time, including monthly averaged values. Figure 3.6 highlights how volatility of prices has changed over time. This paper aims to understand the mechanisms behind this change, focusing on the role of wind energy production on price level and price volatility.

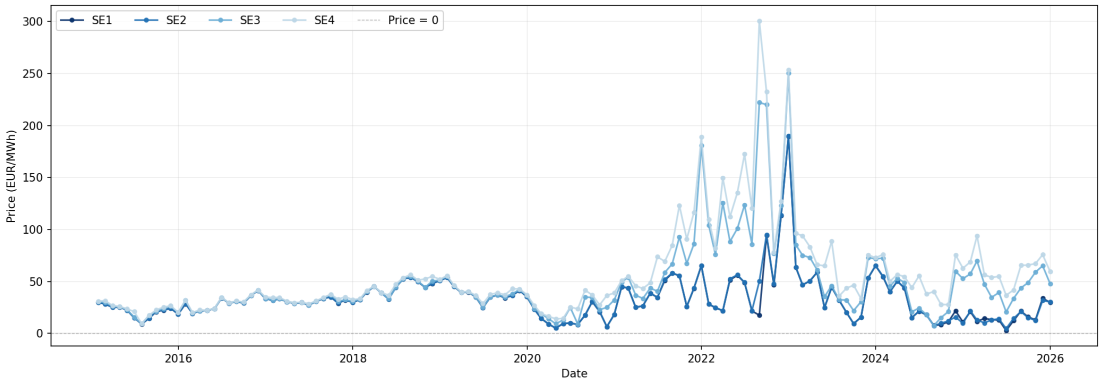


Figure 3.5: Price Level (raw), monthly average

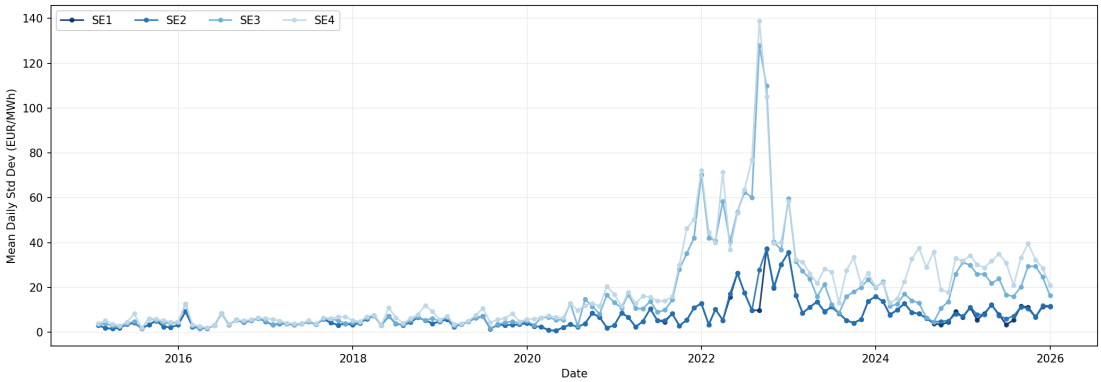


Figure 3.6: Price Volatility (raw), monthly average

Chapter 4

Data and Methodology

The following section outlines the data and methodology used throughout this paper.

4.1 Data

The analysis is based on hourly data across Sweden's four bidding zones from 1 January 2015 to 31 December 2025. This accounts to approximately 96,000 hourly observations per zone. Market data are sourced from Nord Pool and fossil fuel prices from Bloomberg Terminal. The baseline estimation window used for model selection is 1 January 2024 to 31 December 2025. The full 2015–2025 period is used for the rolling window analysis described in chapter 5.

4.1.1 Variable Selection

Day-Ahead Spot Price The dependent variable is the hourly day-ahead spot price for each bidding zone, measured in EUR/MWh and sourced from Nord Pool. Day-ahead prices are chosen given that majority of electricity exchanged in the Nordics is traded on the day-ahead Elspot market instead of the intraday market. Würzburg et al. (2013) notes that explanatory variables should match the characteristics of the day-ahead price and ideally consist of day-ahead forecasts. It is the foreseen rather than realised values of production and load that are available to market participants at the time of price setting (Jónsson et al. 2010). This motivates the use of forecasted values for wind production and consumption.

Wind Power Production Forecast The main variable of interest is the hourly wind power production forecast for each bidding zone, measured in MWh and sourced from Nord Pool. Forecasted rather than actual wind generation is used because actual generation is determined after the price has been set and is therefore an outcome variable rather than an input to price formation (Jónsson et al. 2010).The forecast is exogenous to the spot price by construction. Wind power enters the merit order at near-zero marginal cost and displaces more expensive technologies which puts downward pressure on the market clearing price. β_{wind} is therefore expected to be negative.

Consumption Forecast We include hourly consumption forecasts for each bidding zone as a proxy for electricity demand, measured in MWh and sourced from Nord Pool. Clò et al. (2015) and Jónsson et al. (2010) show that wind power has a stronger price impact when demand is high which makes it important to control for demand when estimating the merit order effect. The consumption forecast is exogenous to the spot price (Ketterer 2014). Higher demand pushes the market equilibrium further up the merit order curve, so β_{cons} is expected to be positive.

Net Exchange Flows We include hourly bilateral net exchange flows between each bidding zone and its neighbours, measured in MWh and sourced from Nord Pool. By including bilateral net exchange flows, we control for cross-border electricity trade which helps mitigate demand and supply imbalances across zones (Würzburg et al. 2013). Each zone has a distinct set of neighbours: SE1 trades with Finland and Norway; SE2 with SE1, Norway, and Finland; SE3 with SE2, SE4, Finland, Norway, and Denmark; and SE4 with SE3, Denmark, Germany, Lithuania, and Poland. Positive values indicate net export from the zone to the neighbour. The expected sign of $\beta_{\text{exch},n}$ varies by zone and neighbour, reflecting the direction of typical electricity flow.

Hydro Reservoir Levels We include weekly hydro reservoir levels for each bidding zone, measured in GWh and sourced from Nord Pool. By including hydro reservoir levels, we control for the availability of hydro power generation. Hydro power is the dominant generation source in SE1 and SE2 and plays a significant role in price formation across the Nordic system (Huisman et al. 2015). Higher reservoir levels imply greater available generation capacity, reducing the need for more expensive technologies and puts downward pressure on prices. Actual hydro generation is not used to avoid endogeneity as producers dispatch hydro in response to price signals. β_{hydro} is therefore expected to be negative.

Oil and Gas Prices Finally, we include the prices of US light crude oil and natural gas, sourced from Bloomberg Terminal. These are included to control for the cost of fossil fuel based electricity generation. Both series enter with a 24-hour lag to reflect the commodity prices available to market participants when submitting bids for the day-ahead auction (O'Mahoney and Denny n.d.). Higher fossil fuel prices increase the cost of electricity production which puts upward pressure on the spot price (Woo et al. 2011). US light crude oil is used rather than Brent crude to mitigate endogeneity concerns, as European electricity prices are unlikely to influence a globally traded commodity (Ketterer 2014). Both β_{oil} and β_{gas} are therefore expected to be positive.

4.1.2 Variable Specifications and Treatment

Day-Ahead Spot Price

The day-ahead spot price is log-transformed after a level shift from adding a constant and fully deseasonalised using hour-of-day×day-of-week and hour-of-day×month-of-year interaction dummies, a Swedish public holiday indicator, and year dummies. This subsection justifies each of these decisions in turn.

Log transformation and level shift The raw price distribution is right-skewed (skewness = 1.88), driven by infrequent but large upward spikes during periods of scarcity. A small number of hours record negative prices, reflecting excess generation that cannot be curtailed. Prior to log transformation, each observation is shifted upward by $|p_{\min}| + 0.01$ so that the minimum value equals 0.01.[1] Log transformation then stabilises variance across the price distribution, reduces the disproportionate influence of extreme spikes, and yields elasticity interpretations for all regression coefficients.

Deseasonalisation Spot prices exhibit statistically significant seasonal patterns along all three calendar dimensions: hour-of-day ($\eta^2 = 0.106$), day-of-week ($\eta^2 = 0.069$), and month-of-year ($\eta^2 = 0.052$), each significant at the 0.1% level. If these deterministic patterns are not removed from the dependent variable, they contaminate the regression residuals and lead to misleading inference. The challenge is choosing a specification that is rich enough to remove the relevant structure without over-fitting.

Diagnosing residual structure with the ACF The autocorrelation function (ACF) is the primary diagnostic tool for evaluating the adequacy of a deseasonalisation

[1]The shift is applied once to the full sample. It preserves the ordinal structure of the series and has negligible effect on variation given the small magnitude of negative prices.

specification. For a time series y_t, the sample ACF at lag k is defined as

$$\hat{\rho}(k) = \frac{\sum_{t=k+1}^{T}(y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{T}(y_t - \bar{y})^2}, \tag{4.1}$$

measuring the linear correlation between y_t and its own value k periods earlier. A well-deseasonalised series should contain no systematic calendar-driven autocorrelation. Hence, persistent spikes at lags that are multiples of 24 or 168 indicate remaining daily or weekly structure that the deseasonalisation has failed to remove.

Figure 4.1 shows the ACF of the basic deseasonalised log price series after removing separate hour-of-day, day-of-week, holiday, month, and year dummies, but before ARMA modelling. The picture is immediately problematic. The ACF is 0.90 at lag 1 and decays slowly, reflecting strong persistence in electricity prices. More critically, there is a sharp spike of 0.52 at lag 24 (green marker), the daily cycle, followed by a comb pattern that resurfaces at lags 48, 72, and at multiples of 24 up to and beyond lag 168 (orange marker). These spikes cannot be attributed to genuine stochastic price dynamics. They are the signature of deterministic seasonal patterns that the basic dummy specification has not fully absorbed.

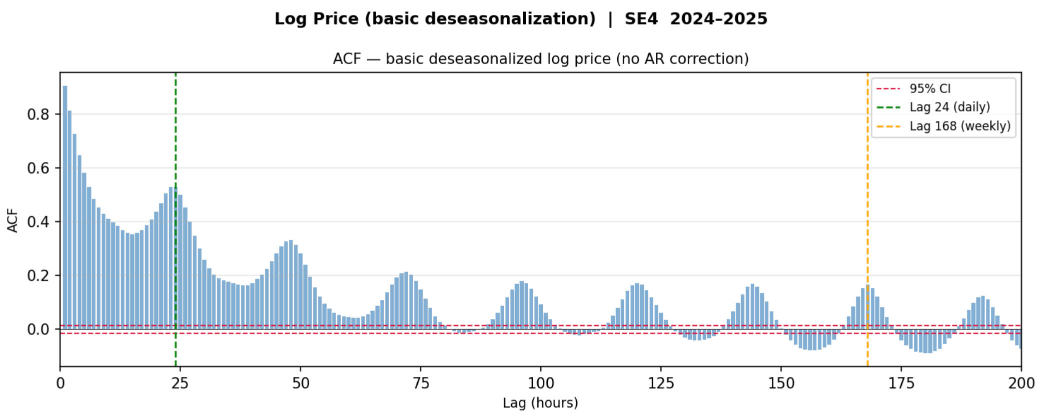


Figure 4.1: ACF of the log deseasonalised spot price (SE4, 2024–2025)

Note: The autocorrelation at lag 1 is 0.90 and decays slowly, reflecting strong stochastic price persistence common to all four bidding zones. The spike at lag 24 (green) and the repeating comb structure at multiples of 24 indicate remaining deterministic calendar patterns addressed by the deseasonalisation specification described in subsection 4.1.2.

Separating stochastic persistence from seasonal structure Reading the ACF of the raw deseasonalised series is complicated by the dominant stochastic autocorrelation: the slow decay from lag 1 spreads across all subsequent lags and obscures which spikes are truly seasonal. To isolate the seasonal residual, we filter out the dominant autoregressive component by fitting an AR(1) process to the deseasonalised series,

$$\tilde{p}_t = \phi_1 \tilde{p}_{t-1} + \varepsilon_t, \tag{4.2}$$

where $\tilde{p}_t$ denotes the deseasonalised log price and ε_t is the innovation. If the deseasonalisation were complete and the AR(1) correctly specified, the residuals $\hat{\varepsilon}_t = \tilde{p}_t - \hat{\phi}_1 \tilde{p}_{t-1}$ would be approximately white noise with a flat ACF. Systematic spikes in the ACF of $\hat{\varepsilon}_t$ therefore reveal remaining seasonal structure that the deseasonalisation has not removed. These are precisely the patterns we need to target. Controlling for AR(1) in this diagnostic step does not mean AR(1) is the final model specification; it is used only as a lens to expose residual calendar patterns more clearly.

Choosing the deseasonalisation specification Figure 4.2 shows the ACF of AR(1) residuals under three progressively richer deseasonalisation specifications. Under the basic specification, the comb pattern remains prominent: spikes at multiples of 24 persist throughout, and a cluster at lag 168 confirms the unresolved weekly cycle. The R^2 of the basic specification is 25.1% (43 parameters). Adding hour×day-of-week interaction dummies, thereby allowing the intraday price profile to differ across weekdays and weekends, substantially reduces the amplitude of the 24-hour spikes. However, it leaves the weekly cluster at lag 168 largely intact, raising R^2 by only 2.1 percentage points to 27.2% (181 parameters). The weekly residual reveals that the peak-hour profile varies not only across days of the week but also across seasons: the morning ramp in a winter weekday is structurally different from that of a summer weekday, in a way that hour×DOW dummies alone cannot capture. Adding hour×month interaction dummies resolves this: under the full specification, the ACF of AR(1) residuals collapses to noise across all lags including lag 168, and R^2 rises by a further 9.4 percentage points to 36.6% (457 parameters). Given 17,533 hourly observations per two-year window, the parameter-to-observation ratio remains well within acceptable bounds. Therefore, the full interaction specification is retained.

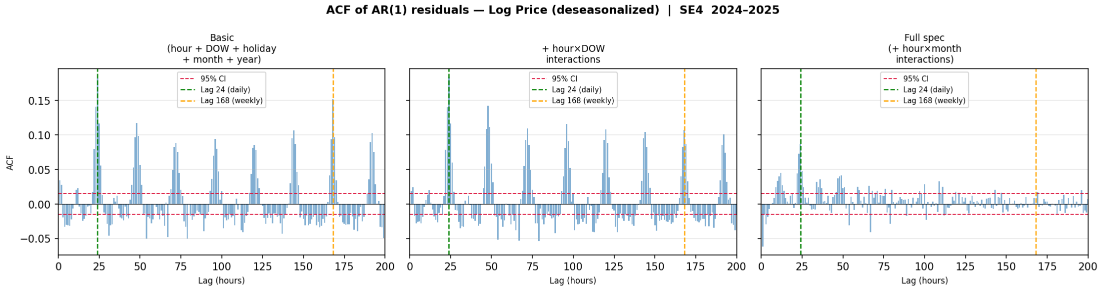


Figure 4.2: ACF of AR(1) residuals of log spot price under alternative deseasonalisation specifications (SE4, 2024–2025)

Note: Vertical markers indicate lag 24 (daily cycle, green) and lag 168 (weekly cycle, orange). Dashed red lines represent 95% confidence bands.

Wind Power Production Forecast

Wind power forecast is included in the regression in log levels without deseasonalisation. This is justified theoretically and empirically.

Empirically, the seasonal component of wind forecast is modest. A variance decomposition of log wind forecast for SE4 over 2024–2025 shows that hour-of-day dummies explain 0.8% of variance, day-of-week dummies explain 0.4%, and month-of-year dummies explain 6.9%. Jointly, all seasonal dummies account for 8.1% of total variance in log wind forecast, leaving 91.9% attributable to weather-driven variation.

The boxplots in Figure 4.3 confirm this picture. Across all 24 hours of the day, the interquartile ranges overlap almost completely and the mean line is flat, indicating no systematic intraday wind cycle. The day-of-week pattern is similarly absent. The only detectable seasonal pattern is monthly: mean wind is approximately twice as high in January as in July, reflecting the Baltic storm season. This within-year variation is real, but since the dependent variable, the log spot price, is already deseasonalised using full hour×day-of-week and hour×month interaction dummies, the seasonal component of wind forecast is orthogonalized from the price residual at the estimation stage. The coefficient on log wind therefore captures how wind deviations from any level affect prices after seasonal price patterns have been removed, which is the quantity of interest. The approach in Ketterer (2014) is in line with our methodology.

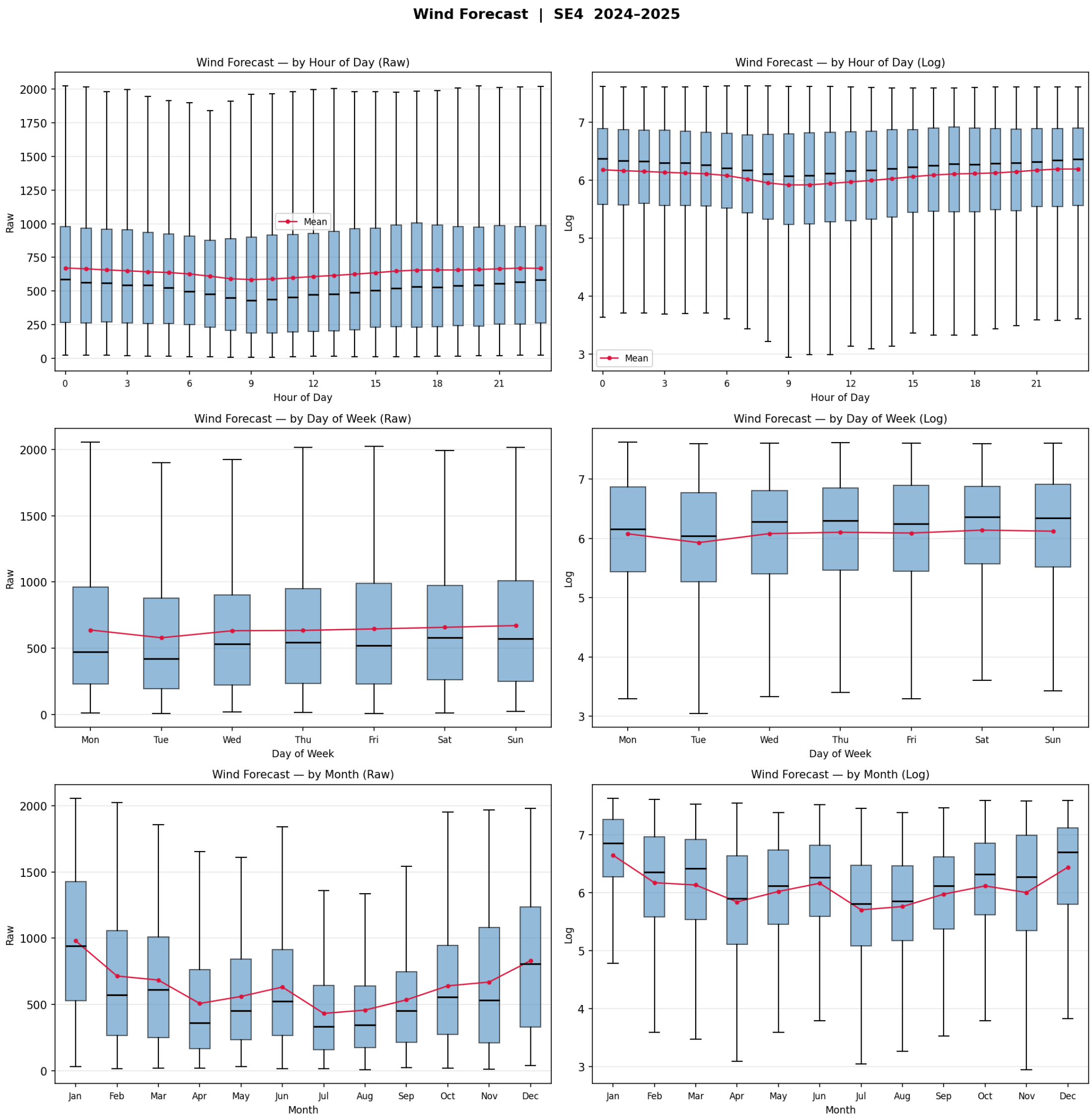


Figure 4.3: Wind Forecast (SE4, 2024–2025)

Consumption Forecast

Log electricity consumption forecast is included in the regression after full deseasonalisation using hour-of-day×day-of-week and hour-of-day×month-of-year interaction dummies. Again, this is justified theoretically and empirically.

Empirically, consumption is overwhelmingly seasonal. A variance decomposition of log consumption for SE4 over 2024–2025 shows that hour-of-day dummies alone explain 13.6% of variance, day-of-week dummies explain 5.9%, and month-of-year dummies explain 68.7%. Together, all seasonal dummies with the main-effect account for 88.2% of the total variance in log consumption. Replacing the separate dummy sets with full hour×day-of-week and hour×month interaction dummies raises explained variance to

91.0%, indicating that the intraday profile differs systematically across day types and seasons in a way that additive main effects cannot fully capture.

The boxplots in Figure 4.4 confirm this picture. The hour-of-day panel shows a clear intraday cycle: mean consumption troughs around 04:00–05:00 and rises to a broad plateau between 09:00 and 20:00, with the mean nearly 25% higher at midday than in the early-morning hours. The day-of-week panel reveals a step-down on weekends, with Saturday and Sunday means approximately 10–15% below the weekday level, reflecting the absence of industrial and commercial load. The monthly panel is the most striking: mean consumption in January is roughly 65% higher than in July, tracing the U-shaped heating cycle characteristic of Scandinavian electricity markets. This within-year variation is not noise but a deterministic, calendar-driven pattern that would dominate the regression if left in the series.

Hence, the theoretical case for deseasonalisation is straightforward. Including raw or insufficiently cleaned consumption as a regressor would mean that its seasonal variation, which co-moves with seasonal price variation, absorbs part of the price variation that should be attributed to economic fundamentals. By applying the same interaction-dummy specification to consumption as to the dependent variable, we ensure that the ARMAX coefficient on deseasonalised consumption captures demand-driven price pressure after all shared calendar patterns have been removed from both series.

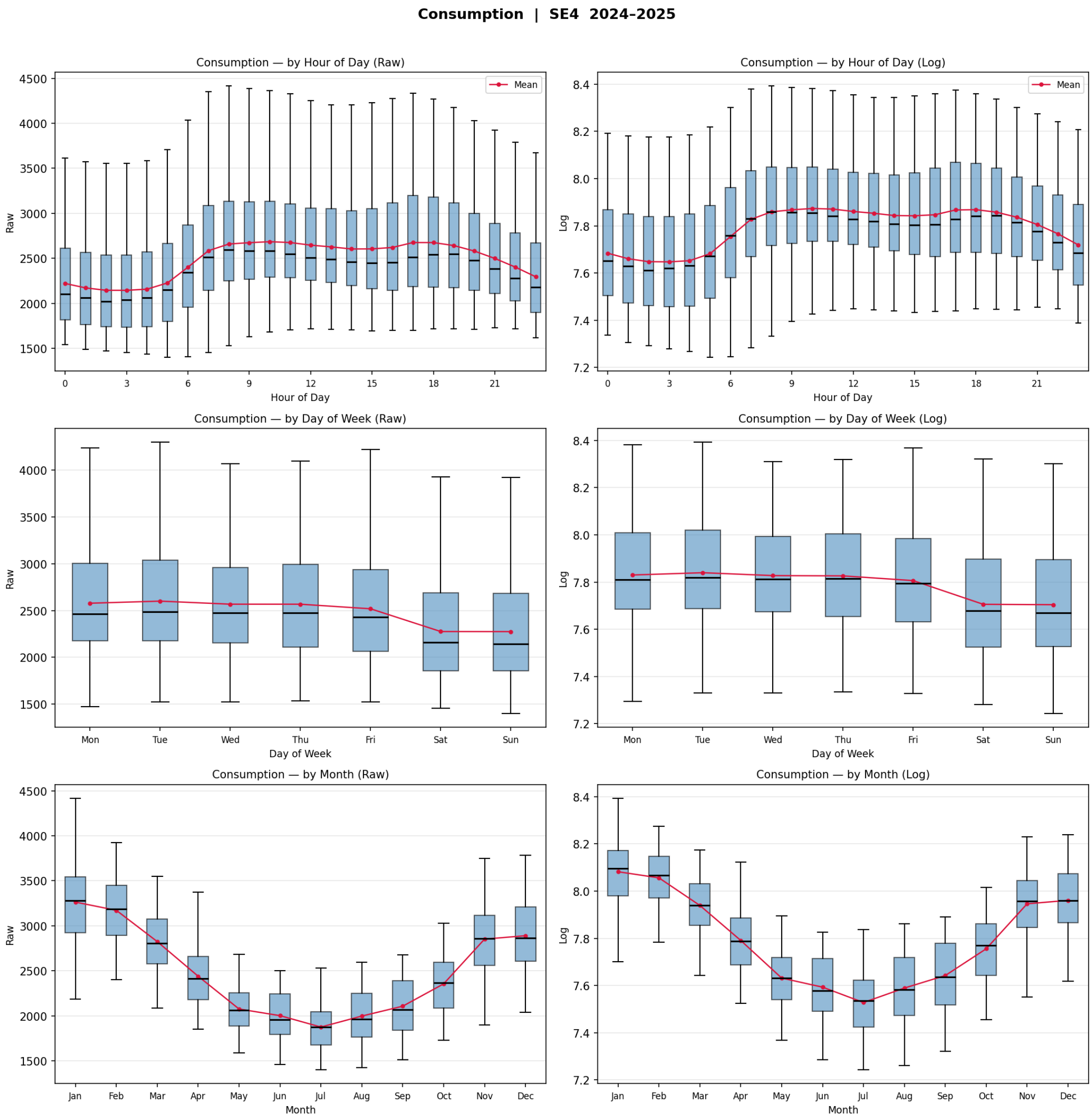


Figure 4.4: Consumption (SE4, 2024–2025)

Net Exchange Flows

Net exchange, defined as the hourly net flow into SE4 from all interconnected zones, measured in MW, is included in the regression in levels without logarithmic transformation or deseasonalisation. Both choices are justified on structural grounds.

Log transformation is precluded by the sign of the variable. Net exchange is a signed quantity: positive values indicate net imports and negative values indicate net exports. In the SE4 sample over 2024–2025, 0.6% of hourly observations are negative, reflecting hours in which SE4 is a net exporter. Because $\log(x)$ is undefined for $x \leq 0$, and because a sign-preserving transformation such as the inverse hyperbolic sine would alter the economic interpretation of the coefficient, the variable is retained in levels.

The decision not to deseasonalise is supported empirically. A variance decomposition shows that hour-of-day dummies explain 5.1% of variance, day-of-week dummies 4.6%, and month-of-year dummies 25.4%. The full interaction specification, including hour×day-of-week and hour×month dummies, accounts for 39.1% of total variance. While statistically significant, this leaves 60.9% of net exchange variation unexplained by calendar effects. The majority of the variation in net exchange therefore reflects non-deterministic factors: price differentials across bidding zones, available hydro generation in Norway and northern Sweden, and short-term grid constraints. Deseasonalising would remove not only the calendar component, but also distort the residual's economic content.

The boxplots in Figure 4.5 illustrate this. The hour-of-day panel shows a mild morning peak around 07:00–09:00 as cross-border imports ramp up with the demand morning peak, but the mean line moves only modestly and the interquartile ranges are wide and overlapping throughout the day. The day-of-week panel similarly shows a small weekday-to-weekend step-down, consistent with lower industrial demand, reducing the need for imports. The most pronounced pattern is monthly: mean net exchange is highest in winter (January–February, around 1,800–2,000 MW) and lowest in summer (June–July, around 1,000–1,100 MW), reflecting the seasonal swing in heating demand. However, within each month, the dispersion is large, confirming that the seasonal mean is superimposed on substantial high-frequency variation driven by fundamentals.

Since the dependent variable is fully deseasonalised, any remaining deterministic seasonal pattern in net exchange that co-moves with the price seasonal would in principle be absorbed by the ARMAX error rather than the net exchange coefficient. In practice, the seasonal mean of net exchange, driven by the heating cycle, mirrors the seasonal mean of consumption, which is itself fully deseasonalised. The residual demand-driven import requirement after removing consumption seasonality is therefore what the net exchange coefficient primarily captures, and retaining net exchange in levels preserves this interpretation. This approach is in line with the work by Ketterer (2014).

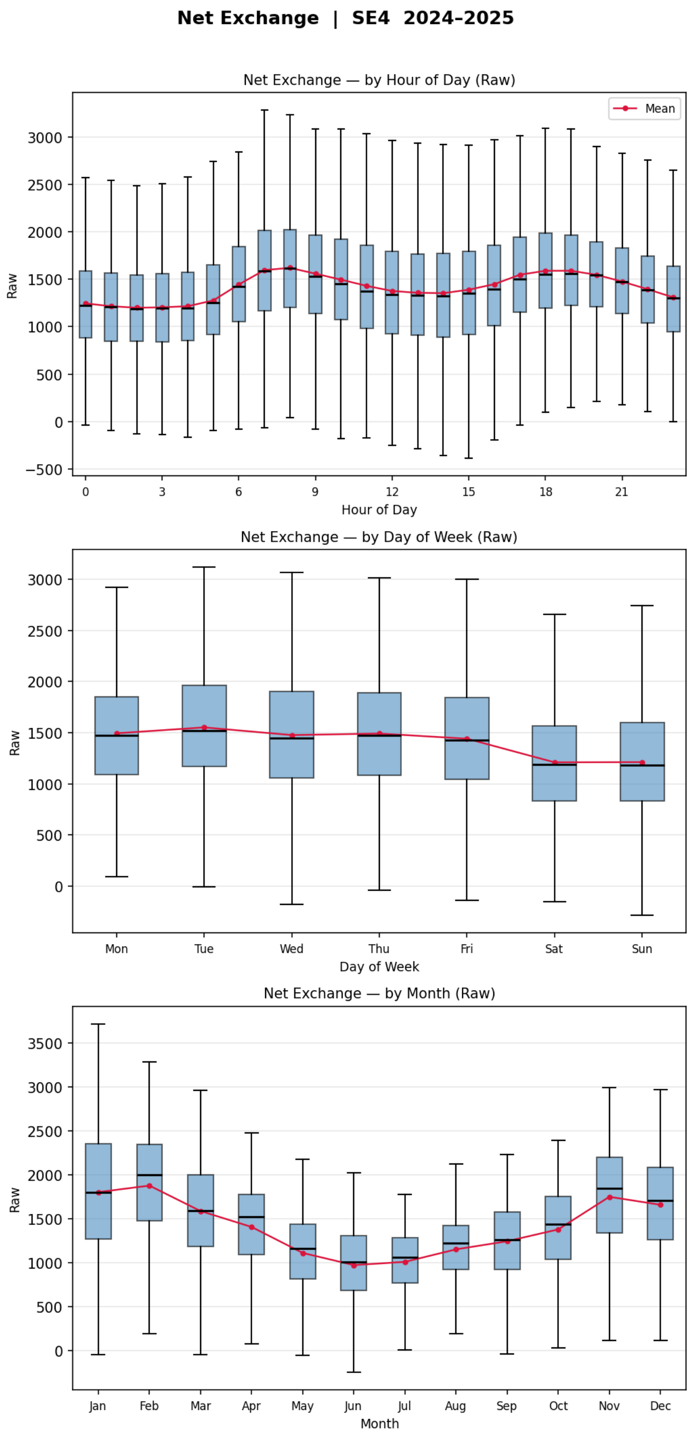


Figure 4.5: Net Exchange (SE4, 2024–2025)

Hydro Reservoir Levels

Log hydro reservoir levels are included in the regression after partial deseasonalisation using month-of-year dummies only. The treatment differs from consumption in two respects: hour-of-day and day-of-week dummies are excluded, and no interaction terms are used. Both restrictions are justified by the underlying data frequency.

The hydro reservoir series is sourced at weekly frequency with approximately 52 observations per year and forward-filled to match the hourly panel.[2] As a direct consequence, every hour within a given week carries an identical reservoir value. There is therefore no intraday or intra-week variation in the series by construction: hour-of-day and day-of-week dummies explain 0.0% of variance (ANOVA F $\approx$ 0.01, $p = 1.00$ for both), and adding interaction terms changes the R^2 by less than 0.01 pp. Including these dummies would be numerically harmless but since they provide no economic value, they are excluded.

The meaningful seasonal dimension is monthly. Month-of-year dummies explain 50.9% of variance in log hydro reserves, and adding any intraday terms leaves this unchanged. The boxplots in Figure 4.6 illustrate the pattern clearly: reservoir levels peak in January–February following autumn rainfall and early snowmelt, decline sharply through spring as reservoirs are drawn down, reach a trough in July–August, and partially recover through autumn. This cycle reflects the hydro production logic of the Nordic system: reservoirs are filled by precipitation and snowmelt and depleted by turbine dispatch to meet demand. The within-month dispersion is substantial, indicating that year-to-year variation in precipitation and dispatch decisions generates considerable residual variation beyond the average seasonal path.

We apply partial deseasonalisation with month dummies only to remove the predictable average seasonal level from log hydro reserves. The residual series `hydro_log_ds` retains both the within-month variation and the cross-year deviations from the seasonal mean, which are the economically relevant signals: a reservoir level above its seasonal norm implies greater generation capacity and downward pressure on prices, and vice versa.

[2]Weekly observations do not align perfectly with calendar year boundaries. The final observation of each year is extended forward to fill the remaining hours until the first observation of the following year begins, and symmetrically at the start of the sample. This introduces a small number of repeated boundary values but does not affect the substance of the analysis.

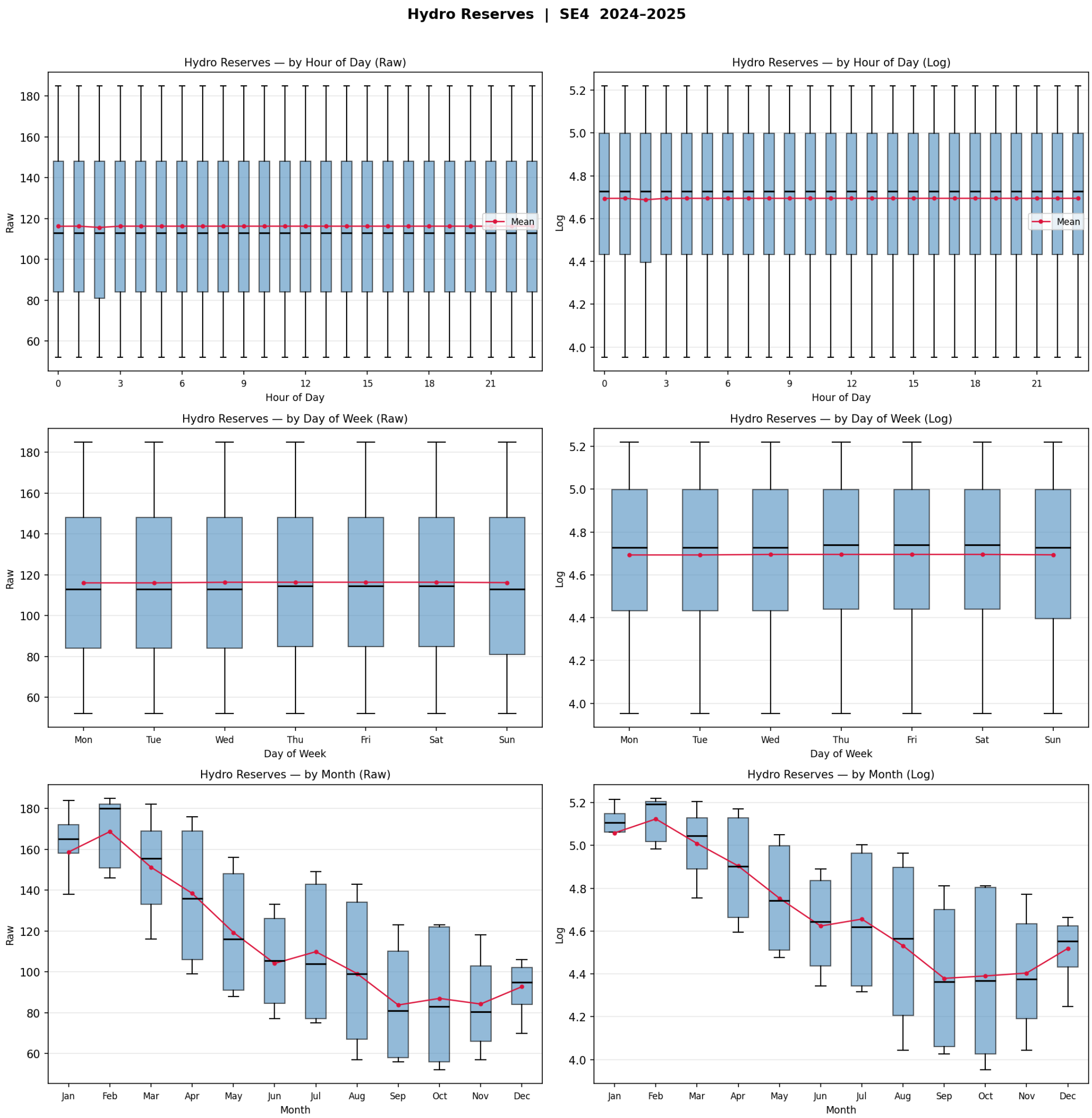


Figure 4.6: Hydro Reserves (SE4, 2024–2025)

Oil Price

The crude oil price is included in the regression as a log-transformed variable without deseasonalisation. The series enters with a 24-hour lag, reflecting the fact that commodity market prices are known to electricity market participants before the day-ahead spot auction clears.[3]

The decision not to deseasonalise follows the same logic applied to wind power forecast and net exchange. A variance decomposition shows that month-of-year dummies explain 27.0% of variance in log oil price, while hour-of-day and day-of-week dummies explain

[3]The 24-hour lag is applied during data construction: each hourly observation in the panel is assigned the oil price from the corresponding hour of the previous calendar day.

0.0% by construction. The daily oil price is broadcast identically to all 24 hours of a trading day, and weekend observations carry forward the Friday settlement price. The remaining 73% of variance reflects non-seasonal drivers: geopolitical events, OPEC supply decisions, and global macroeconomic conditions.

The boxplots in Figure 4.7 confirm this structure. The hour-of-day and day-of-week panels show perfectly flat mean lines with identical distributions across all groups, a mechanical consequence of the daily data frequency. The monthly panel reveals a modest seasonal pattern with mean oil price being highest in the first quarter and around April, declining through summer, and reaching its lowest level in the fourth quarter. However, within each month the dispersion is large relative to the cross-month variation, consistent with the 73% non-seasonal share.

Crucially, since the dependent variable, the log spot price, is fully deseasonalised prior to estimation, the Frisch–Waugh–Lovell theorem guarantees that any seasonal component in log oil price is orthogonalized from the price residual at the estimation stage, regardless of whether it is removed in preprocessing. Hence, the ARMAX coefficient on log oil captures the full oil price signal, and the 27% seasonal share does not bias the estimate. Applying month dummies to oil but not to net exchange, whose seasonal share is larger at 39%, would be methodologically inconsistent. Treating both variables symmetrically, and relying on the estimation procedure to handle their seasonal components implicitly, produces a coherent and uniform treatment across all controls that are not subject to full deseasonalisation.

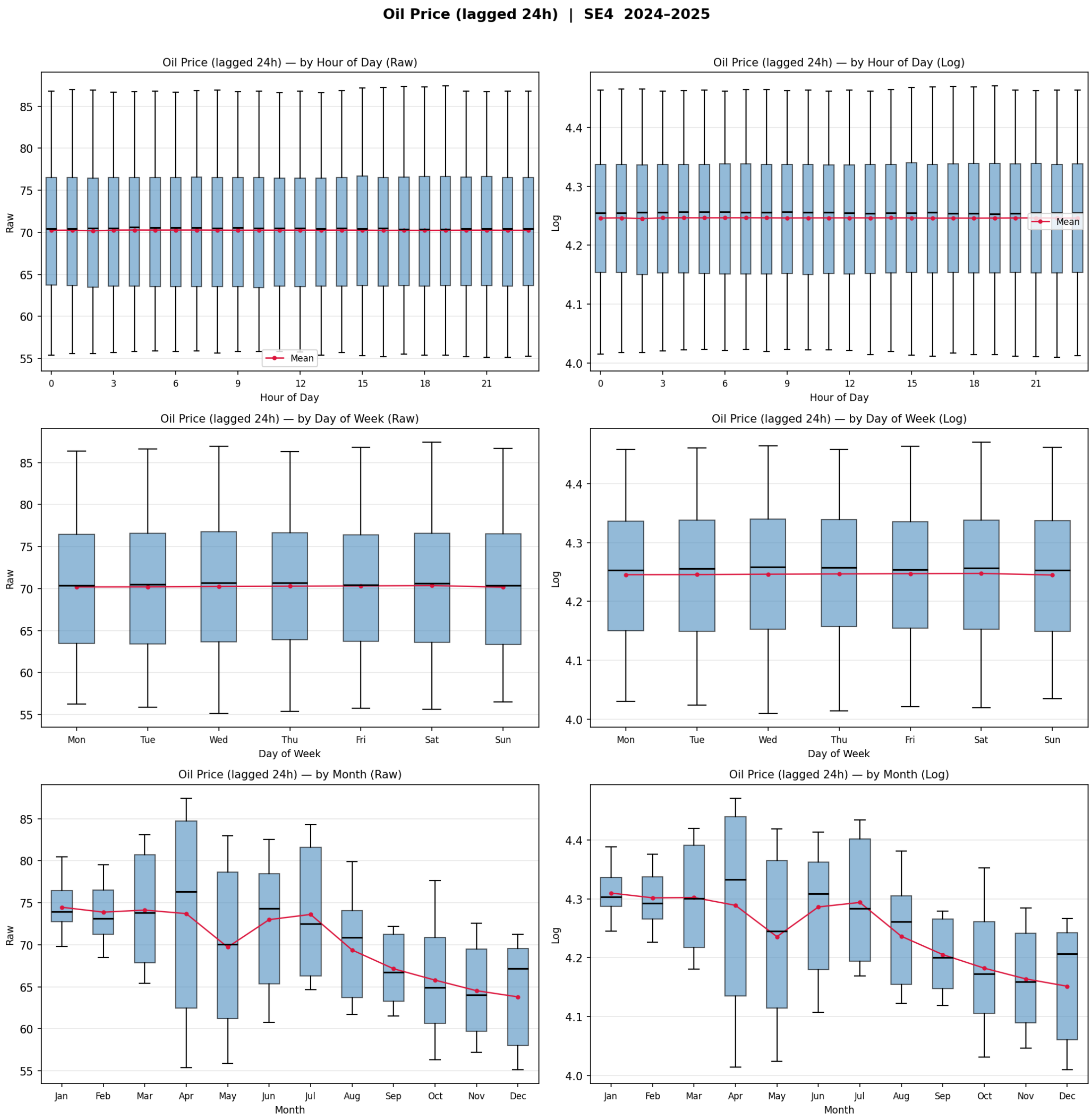


Figure 4.7: Oil Price (SE4, 2024–2025, lagged 24h)

Gas Price

This section is still work in progress.

4.2 Descriptive Statistics and Visualisations

As discussed, we use day-ahead hourly values for our analysis. This enables us to get a precise understanding of the dynamics in the energy market while controlling for seasonality. Figure 4.8 and Figure 4.9 show the raw data series, including a daily average weighted by energy volume.

Table 4.1: Descriptive statistics, 2015–2025

	SE1					SE2					SE3					SE4				
	Mean	SD	Min	Max	N	Mean	SD	Min	Max	N	Mean	SD	Min	Max	N	Mean	SD	Min	Max	N
Spot price (EUR/MWh)	32.79	33.20	−74.11	590.00	96,432	33.01	33.79	−60.04	590.00	96,432	46.85	56.79	−60.04	799.97	96,432	54.92	64.61	−60.04	799.97	96,432
Wind forecast (MW)	439	492	1	2,832	96,095	1,124	1,048	6	7,017	96,095	876	659	18	3,559	95,951	520	403	7	2,155	95,951
Consumption forecast (MW)	1,179	199	650	1,798	96,336	1,859	372	903	3,613	96,336	9,758	2,132	5,130	18,215	96,336	2,667	639	1,301	5,091	96,336
Net exchange (MW)	−1,546	892	−4,526	776	96,432	−3,677	1,328	−8,752	556	96,432	580	1,529	−4,661	6,853	96,432	1,732	634	−496	3,972	96,432
Hydro reserves (GWh)	8,481	3,176	1,381	13,759	96,432	9,713	3,744	1,504	15,037	96,432	1,793	407	727	2,600	96,432	120	41	36	216	96,432
Oil price (USD/bbl)	62.19	17.30	12.88	124.77	2,854	*common across zones*					*common across zones*					*common across zones*				
Gas price (EUR/MWh)	35.86	38.16	3.62	311.00	2,671	*common across zones*					*common across zones*					*common across zones*				

Notes: Zone-specific variables are at hourly frequency; N reports the number of hourly observations. Oil and gas prices are at daily frequency and common across zones; their N reports the number of trading days. Statistics for oil and gas are reported once under SE1. Net exchange is the aggregate net flow into the zone from all neighbours; positive values indicate net import. Zone-specific variables are sourced from Nord Pool; oil and gas prices from Bloomberg Terminal. Sample period: 1 January 2015 to 31 December 2025.

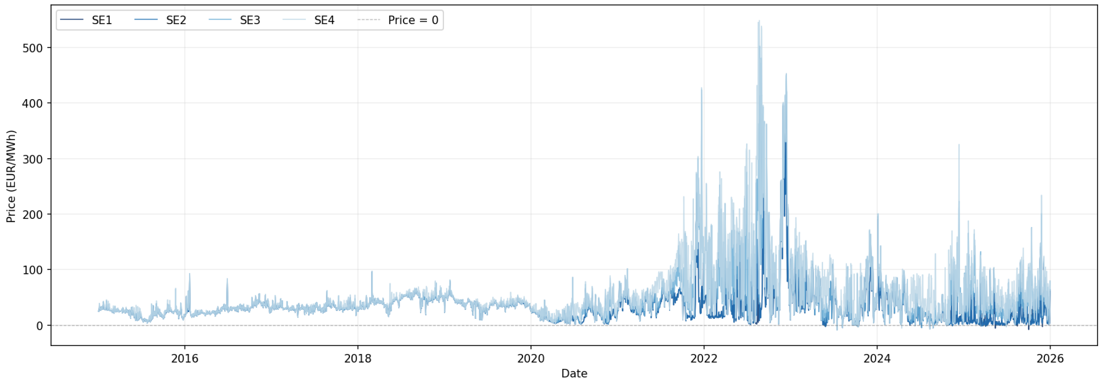


Figure 4.8: Price Series Level, volume weighted daily average, raw data

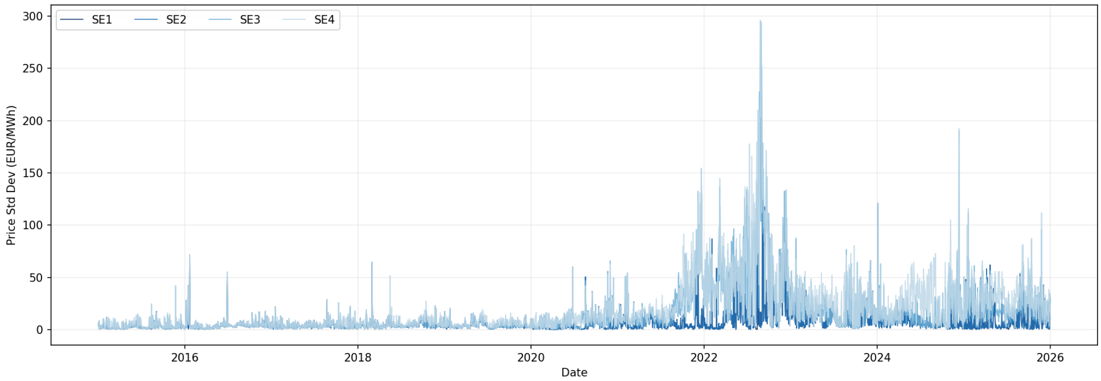
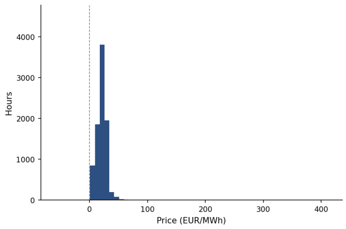
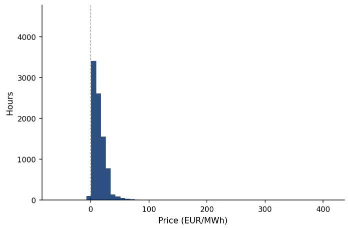


Figure 4.9: Price Series Volatility, daily standard deviation, raw data,

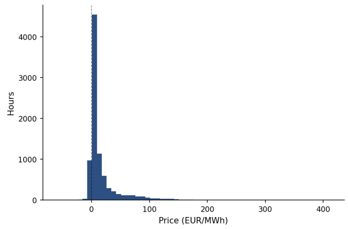
It is noticeable that the distribution of energy prices changes quite significantly between 2015 and 2025. Particularly, the distribution becomes increasingly right-skewed, indicating more extreme positive price spikes in recent years. That is in line with the observations from the time series.



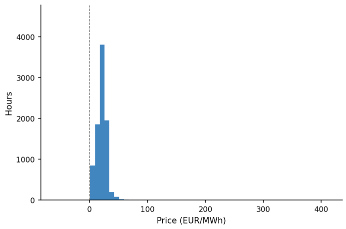
(a) SE1, 2015



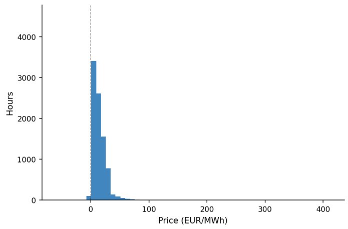
(b) SE1, 2020



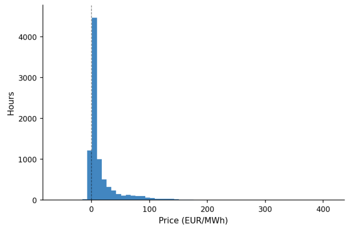
(c) SE1, 2025



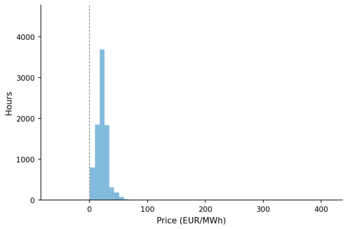
(d) SE2, 2015



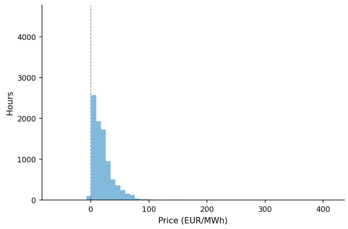
(e) SE2, 2020



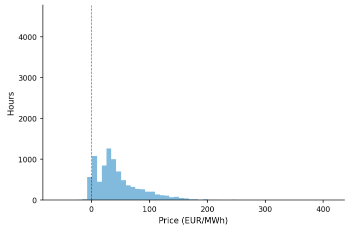
(f) SE2, 2025



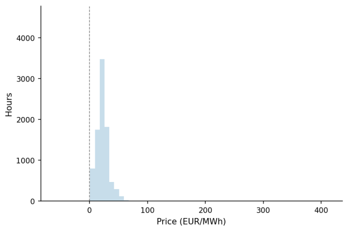
(g) SE3, 2015



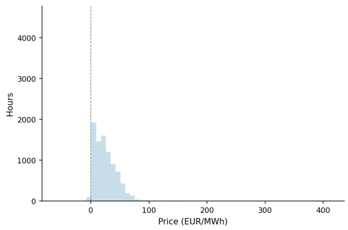
(h) SE3, 2020



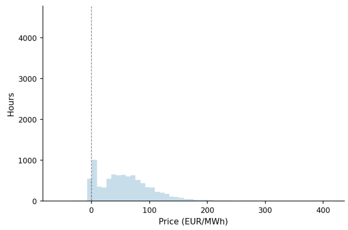
(i) SE3, 2025



(j) SE4, 2015



(k) SE4, 2020



(l) SE4, 2025

Figure 4.10: Price Histograms by Bidding Zone and Year

4.3 Data Treatment

In line with literature, we apply a logarithmic transformation on our data series to capture non-linear impacts while handling spikes and stabilising variance (Paraschiv et al. 2014). To apply a logarithmic transformation, we ensure that all values are positive by adding a large enough constant to every value in the data series (Rintamäki et al. 2017). In line with Ketterer (2014), we remove outliers from our time series that exceed three times the standard deviation. Since we observe a strong change in the behaviour of energy prices after the start of the energy crisis, we decided to introduce a structural break into the dataset after logarithmic transformation and apply the outlier removal to each side of the structural break separately. This way, we avoid that volatile values that are normal under the new reality of an energy crisis are flagged as outliers and removed from our analysis, potentially invalidating results.

Figure 4.12 shows the outlier removal process exemplary for SE1. Graphs for the other bidding zones were included in the appendix.

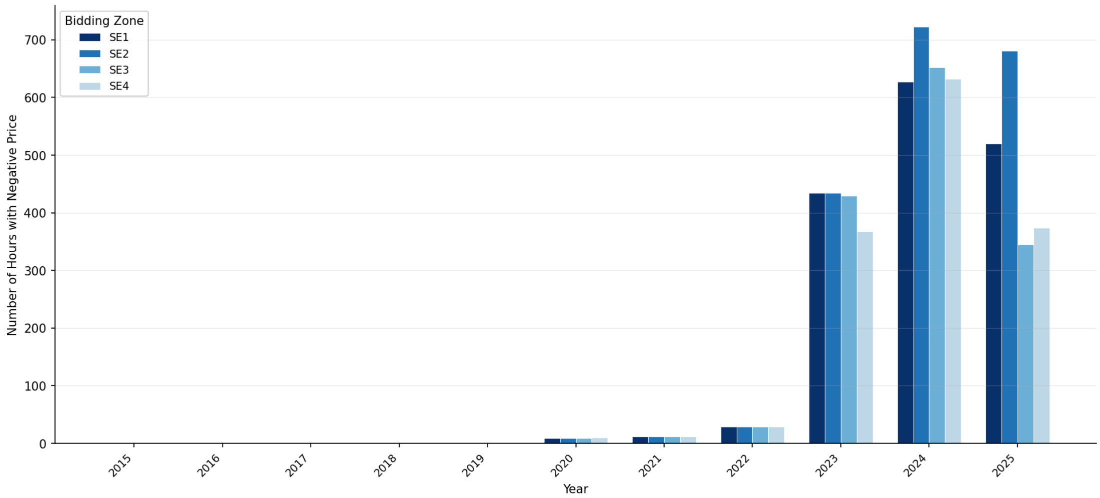


Figure 4.11: Amount of negative prices per zone, 2015-2025

Table 4.2: Outlier removal by bidding zone and regime (structural break: 2021-10-01)

	Pre-break (2015–2021)			Post-break (2021–2025)			Total		
Zone	N	Outliers	%	N	Outliers	%	N	Outliers	%
SE1	59,195	621	1.049	37,280	619	1.660	96,475	1,240	1.285
SE2	59,195	641	1.083	37,280	564	1.513	96,475	1,205	1.249
SE3	59,195	472	0.797	37,280	148	0.397	96,475	620	0.643
SE4	59,195	572	0.966	37,280	44	0.118	96,475	616	0.639
All zones	236,780	2,306	0.974	149,120	1,375	0.922	385,900	3,681	0.954

Notes: Outliers are capped at the per-weekday $\pm 3\sigma$ bound within each regime. Percentages display the share of observations exceeding the bound.

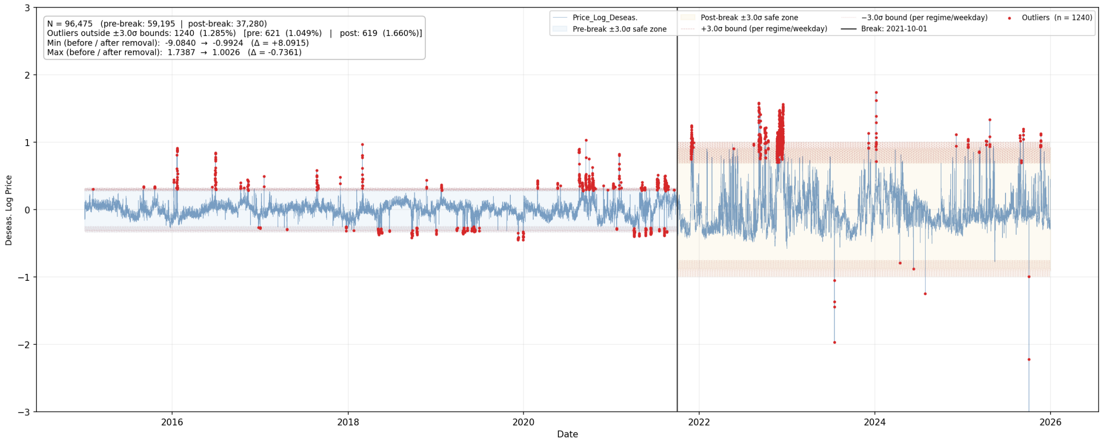


Figure 4.12: SE1: Outlier removal

4.4 Methodology

4.4.1 Motivation for the Modelling Framework

The properties of hourly electricity prices impose two requirements on the empirical model. First, the price series exhibits strong autocorrelation. Figure 4.1 shows the ACF of the log deseasonalised spot price for SE4 over 2024–2025, which is representative of all four zones. The autocorrelation at lag 1 is 0.90 and decays slowly across subsequent lags, indicating that the current price is strongly related to its recent past. Table 4.3 confirms this formally.

Table 4.3: Ljung-Box test for serial correlation in the log deseasonalised spot price

Lag	SE1	SE2	SE3	SE4
1	16,262.95***	16,216.19***	15,742.09***	15,124.84***
2	31,077.45***	30,857.81***	29,589.99***	27,914.72***
3	44,592.14***	44,164.20***	41,941.26***	38,926.43***
4	57,007.67***	56,409.63***	53,172.80***	48,614.78***
5	68,507.13***	67,785.50***	63,492.13***	57,272.94***
10	117,292.37***	115,529.34***	106,462.17***	91,553.87***
15	155,422.72***	151,782.13***	139,199.06***	116,564.14***
20	183,649.41***	177,771.94***	164,041.53***	135,253.51***

Notes: Ljung-Box Q-statistics for the log deseasonalised spot price series, estimated over the sample 1 January 2024 to 31 December 2025. The null hypothesis is no serial correlation. The series is tested prior to any model estimation. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The null hypothesis of no serial correlation is rejected at the 1% level at all lag lengths and across all four zones. This persistence motivates an autoregressive mean equation. If not included, the residuals carry systematic structure that biases inference on the coefficients of interest. Second, the variance of the residuals is not constant over time. To test this, Engle's ARCH test is applied to the residuals of an AR(1) process fitted to the log deseasonalised price series. Filtering the mean dynamics first ensures that the test captures volatility clustering rather than autocorrelation in levels.Table 4.4 reports the results.

Table 4.4: Engle's ARCH test for conditional heteroscedasticity in AR(1) residuals

Lag	**SE1**	**SE2**	**SE3**	**SE4**
1	870.17***	541.48***	164.33***	164.57***
2	879.12***	910.59***	670.61***	350.05***
3	934.86***	1,060.27***	815.63***	413.44***
4	971.42***	1,062.13***	815.56***	422.25***
5	974.83***	1,063.88***	818.00***	456.67***
10	1,035.72***	1,105.57***	882.15***	576.06***
15	1,068.78***	1,116.15***	932.52***	655.95***
20	1,084.12***	1,144.43***	969.64***	738.24***

Notes: Engle's LM test statistics for ARCH effects in the residuals of an AR(1) process fitted to the log deseasonalised spot price series, estimated over the sample 1 January 2024 to 31 December 2025. The null hypothesis is no ARCH effects. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The null hypothesis of no ARCH effects is rejected at the 1% level at all lag lengths and across all four zones, confirming that volatility clusters. This means that periods of large price movements are followed by further large movements. Similarly, calm periods tend to persist. A model that assumes a constant error variance is therefore misspecified on this data. The pattern where the variance of the error terms depends on its own past is known as conditional heteroscedasticity and must be accounted for. Together, the autocorrelation in the price series itself and clustering in squared residuals motivate an ARMAX-GARCH-X framework. The mean equation captures price persistence through the autoregressive and moving average terms, and the effect of market fundamentals on the log day-ahead spot price through the exogenous regressors. The variance equation captures the time-varying behaviour of price uncertainty and allows us to test whether wind power production affects its volatility. Estimating both equations jointly ensures that inference on the wind coefficient in the variance equation is not distorted by misspecification of the mean.

4.4.2 The Mean Equation

To account for the serial correlation documented in Table 4.3, the mean equation is specified as an ARMAX model. For each bidding zone $z \in \{\mathrm{SE1}, \mathrm{SE2}, \mathrm{SE3}, \mathrm{SE4}\}$, the mean equation is:

$$\begin{aligned} \ln P_{z,t} = \alpha_z &+ \sum_{i=1}^{p} \phi_i \ln P_{z,t-i} + \beta_{\text{wind}} \ln W_{z,t} + \beta_{\text{hydro}} \ln H_t + \beta_{\text{cons}} \ln C_{z,t} \\ &+ \sum_{n \in \mathcal{N}(z)} \beta_{\text{exch},n} X_{z,n,t} + \beta_{\text{oil}} \ln O_t + \beta_{\text{gas}} \ln G_t + \sum_{j=1}^{q} \theta_j \varepsilon_{z,t-j} + \varepsilon_{z,t} \end{aligned} \tag{4.3}$$

where $P_{z,t}$ is the day-ahead spot price in zone z at time t, $W_{z,t}$ is the forecasted wind power production, H_t is the hydro reservoir level, $C_{z,t}$ is the forecasted consumption, and $X_{z,n,t}$ is the net energy exchange between zone z and neighbouring zone or country n, measured in MWh, where $\mathcal{N}(z)$ denotes the set of neighbours of zone z. Positive values indicate net export from zone z to neighbour n. As the bilateral exchange variables take negative values in hours of net import, they are retained in levels rather than log-transformed. O_t and G_t are the prices of oil and gas respectively. The autoregressive terms ϕ_i capture price persistence and the moving average terms θ_j capture residual serial correlation not absorbedby the AR structure. The ARMA orders p and q are selected via a grid search over information criteria, as described in chapter 5.

The coefficient of interest is β_{wind}, which measures the elasticity of the log day-ahead spot price with respect to log wind power production. A 1% increase in wind power production is associated with a $|\beta_{\text{wind}}|$% change in the day-ahead spot price, holding all other covariates constant. The model is estimated separately for each zone, so that β_{wind} and the bilateral exchange coefficients $\beta_{\text{exch},n}$ are allowed to differ freely across SE1, SE2, SE3, and SE4. The model is estimated by quasi-maximum likelihood (QML) with Bollerslev-Wooldridge robust standard errors, which remain valid even when the error term is not normally distributed.

4.4.3 The Variance Equation

The ARCH test results in Table 4.4 establish that the error term $\varepsilon_{z,t}$ from Equation 4.3 is conditionally heteroscedastic. The conditional variance is therefore modelled explicitly.

For each zone z, the variance equation is specified as a GARCH-X model:

$$h_{z,t} = \left(\omega_z + \alpha\varepsilon^2_{z,t-1} + \delta h_{z,t-1}\right) \exp\left(\sum_k \gamma_k x_{z,k,t}\right) \tag{4.4}$$

where $h_{z,t}$ is the conditional variance of $\varepsilon_{z,t}$, ω_z is a zone-specific constant, $\varepsilon^2_{z,t-1}$ is the squared error from the previous period, and $h_{z,t-1}$ is the lagged conditional variance. The set $\{x_{z,k,t}\}$ contains the same exogenous controls as the mean equation: log wind production, log hydro reservoir levels, log consumption, bilateral net exchange flows, log oil price, and log gas price. The parameter α captures to what degree a large price shock in the previous period increases current variance and δ captures the persistence of the conditional variance over time. For the variance process to be stationary, $\alpha + \delta < 1$ is required. If this condition holds, shocks to the conditional variance have only a temporary effect and the variance reverts to its long-run mean. The coefficient of interest in the variance equation is γ_{wind}, which measures the effect of wind production on price volatility. A positive γ_{wind} implies that higher wind production increases price uncertainty. The multiplicative specification guarantees that the conditional variance $h_{z,t}$ remains positive by construction regardless of the sign of any γ_k. In the standard additive GARCH-X formulation, a negative coefficient on any control variable can push the conditional variance below zero for certain covariate values which violates the non-negativity requirement of a variance. The model is estimated jointly with the mean equation by QML, with Bollerslev-Wooldridge robust standard errors applied throughout.

4.4.4 Zone-Level Estimation Strategy

The model is estimated separately for each of the four bidding zones. Formally, $z \in \{SE1, SE2, SE3, SE4\}$. This zone-by-zone approach allows β_{wind} and all other coefficients to differ freely across zones, reflecting the structural heterogeneity across Sweden's bidding zones documented in chapter 4. An aggregate specification would impose common parameters across zones, obscuring the cross-zone differences that are central to the research question.

The ARMA order is selected once on the baseline sample of 1 January 2024 to 31 December 2025 and then held fixed across all zones and all rolling windows. The purpose of the order selection is to absorb the autocorrelation structure in the residuals, ensuring that the exogenous coefficients are not biased by residual serial correlation. A common specification across zones also facilitates direct comparison of $\hat{\beta}_{\text{wind}}$ across SE1, SE2, SE3, and SE4. We acknowledge the limitations of this approach. The selected specification is based on the most recent two years of data and may not be optimal for earlier windows. Especially for windows from the pre-crisis period when price dynamics differed substantially. Re-optimising the order for each window would mitigate this concern. However, this is computationally infeasible at the rolling window frequency used here: with one-year windows and monthly shifts, the analysis covers approximately 120 windows per zone and yields around 480 separate specifications across all four zones.

4.4.5 Quantile Regression

This section is still work in progress.

Chapter 5

Results

5.1 Model Selection and Baseline Estimation

The ARMA order is selected by estimating Equation 4.3 over the sample 1 January 2024 to 31 December 2025 and conducting a grid search over $p, q \in \{0, 1, 2, 3\}$, with $d = 0$ imposed throughout, given that the deseasonalised log price series is stationary by construction. Full grid search results for each bidding zone are reported in Table A.4.1–Table A.4.4 in section A.4.

Across all four zones, the information criteria converge on an AR(1) mean structure. ARMA(1,1) achieves the lowest or near-lowest BIC among stable, well-identified specifications across all zones. Higher-order specifications exhibit near-explosive AR polynomials and are discarded. A common specification is applied across zones to ensure that cross-zone differences in estimated coefficients are not confounded by differences in the mean equation structure.

Table 5.1 reports the full coefficient estimates from the ARMAX(1,1)-GARCH-X(1,1) specification across all four zones. the wind coefficient $\hat{\beta}_{\text{wind}}$ is negative and significant at the 1% level in all zones, confirming the merit-order effect of wind energy on the log day-ahead spot price throughout Sweden. The effect varies substantially across zones. SE2 exhibits the strongest sensitivity ($\hat{\beta}_{\text{wind}} = -0.072$), implying that a 1% increase in wind power production is associated with a 0.072% decrease in the day-ahead spot price. SE1 exhibits the weakest effect ($\hat{\beta}_{\text{wind}} = -0.020$), consistent with its hydro-dominated generation mix and structural position as a net exporter. SE3 and SE4 fall between these extremes at -0.075 and -0.080, respectively.

Consumption has a positive effect on the log day-ahead spot price across all zones, significant at the 1% level throughout. The effect is largest in SE3 and SE4 ($\hat{\beta}_{\text{cons}} = 1.477$

and 1.557), reflecting their structural supply deficits, as price impact is amplified in zones that depend on imports to meet consumption. SE1 and SE2 show smaller elasticities of 0.128 and 0.475 respectively. Hydro reservoir levels carry a negative coefficient in SE2 and SE4 (significant at the 5% level), but are insignificant in SE1 and SE3, where nuclear baseload and net exporter position reduce the marginal role of hydro in price formation.

Bilateral net exchange coefficients are negative and significant across most interconnections, consistent with standard supply logic: net exports to a neighbouring zone are associated with lower local prices. Full bilateral exchange results are reported in Table A.2.1–Table A.2.4 in section A.2. Gas prices are significantly negative in SE1 through SE3, while the oil price coefficient is significant only in SE1 and SE2, suggesting limited pass-through in the southern zones at the daily frequency. The AR coefficient confirms stationarity across all zones ($\hat{\phi}_1 < 1$ throughout), with SE3 and SE4 exhibiting stronger autoregressive dynamics than SE1 and SE2, consistent with the more volatile price behaviour observed in the southern zones.

Table 5.1: ARMAX(1,1)-GARCH-X(1,1) baseline estimates, 1 January 2025 to 31 December 2025

	SE1	**SE2**	**SE3**	**SE4**
Mean equation				
Constant	-1.324^{**}	0.881	2.306^{**}	1.410
	(0.531)	(0.696)	(1.042)	(1.216)
Log wind ($\hat{\beta}_{\text{wind}}$)	-0.0148^{***}	-0.1488^{***}	-0.1361^{***}	-0.1002^{***}
	(0.0034)	(0.0152)	(0.0139)	(0.0124)
Log hydro	−0.024	0.226	−0.383	-0.548^{**}
	(0.081)	(0.177)	(0.308)	(0.273)
Log consumption	0.128^{*}	0.475^{***}	1.477^{***}	1.557^{***}
	(0.068)	(0.125)	(0.164)	(0.142)
Log oil price	0.273^{**}	−0.057	−0.365	−0.134
	(0.125)	(0.215)	(0.335)	(0.398)
Log gas price	0.042	0.040	0.042	−0.094
	(0.048)	(0.112)	(0.148)	(0.162)
AR(1)	0.960^{***}	0.945^{***}	0.931^{***}	0.902^{***}
	(0.009)	(0.008)	(0.006)	(0.008)
MA(1)	0.230^{***}	0.193^{***}	0.126^{***}	0.144^{***}
	(0.025)	(0.018)	(0.017)	(0.016)
Variance equation				
ω	2.540	−2.135	-10.927^{***}	-14.950^{***}
	(4.170)	(4.019)	(2.564)	(2.794)
α ARCH(1)	0.990^{***}	0.665^{***}	0.119^{**}	0.201^{***}
	(0.125)	(0.085)	(0.060)	(0.059)
δ GARCH(1)	0.161^{***}	0.222^{***}	0.637^{***}	0.498^{***}
	(0.042)	(0.040)	(0.172)	(0.191)
γ_{wind} Log wind	-0.470^{***}	-0.827^{***}	−0.089	−0.004
	(0.059)	(0.063)	(0.062)	(0.041)
γ_{hydro} Log hydro	0.719	-1.858^{*}	−0.135	2.699^{***}
	(1.147)	(1.128)	(0.890)	(0.668)
γ_{cons} Log consumption	5.691^{***}	5.090^{***}	-1.398^{*}	-2.765^{***}
	(1.595)	(1.169)	(0.763)	(0.686)
γ_{oil} Log oil price	−1.465	0.918	1.169	2.556^{***}
	(1.397)	(1.380)	(0.854)	(0.922)
γ_{gas} Log gas price	−0.641	−0.448	0.606	0.061
	(0.606)	(0.608)	(0.380)	(0.431)
Persistence ($\alpha + \delta$)	$1.151^{\dagger}$	0.887	0.755	0.699
N	8,736	8,736	8,736	8,736
Log-likelihood	15,517	9,435	5,976	4,530
AIC	−30,989	−18,821	−11,901	−9,013
BIC	−30,834	−18,651	−11,717	−8,843

Notes: ARMAX(1,1)-GARCH-X(1,1) estimates with all exogenous controls entering both the mean and variance equations. The variance equation uses a multiplicative GARCH-X specification. Bilateral net exchange coefficients are reported separately in Table A.2.1–Table A.2.4 in section A.2. Robust standard errors in parentheses. †Persistence exceeds 1 for SE1, indicating a non-stationary variance process for this window. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

5.2 Rolling Window Results

Figure 5.1 reports $\hat{\beta}_{wind}$ across all four zones over the full sample period. Two features stand out. First, the wind coefficient is negative in every zone and every window, confirming that the merit-order effect of wind on the log day-ahead spot price is a persistent feature of Sweden's electricity market throughout the period 2015–2025. Second, the cross-zone dispersion in $\hat{\beta}_{wind}$ widens substantially over the sample: the four zones track closely through 2015–2019, diverge sharply at the 2021–2022 window midpoints, and recover at different rates thereafter. The post-crisis levels of $\hat{\beta}_{wind}$ are uniformly more negative than the pre-crisis levels in all four zones, indicating that the sensitivity of prices to wind production has increased structurally over the decade. The zones diverge most clearly in the magnitude of the crisis-period spike and in the degree of post-crisis recovery, which we describe in turn.

In the pre-crisis windows, SE1 and SE2 show the weakest sensitivity with $\hat{\beta}_{wind}$ close to zero, while SE3 and SE4 are already more negative at comparable windows. Through 2015–2018 the zones track closely and the cross-zone spread remains narrow, with all four $\hat{\beta}_{wind}$ estimates falling between -0.014 and -0.104. From 2019 onward, all four zones deepen, with SE3 and SE4 leading the move. The crisis period centred on the 2021–2022 windows produces a sharp deepening across all zones; SE3 and SE4 reach their most negative estimates over the full sample at -0.447 and -0.488 respectively in the 2022 window, while SE1 deepens comparatively little, reaching -0.160 in the same window. Post-crisis, all zones recover partially, but the recovery is uneven. SE1 returns close to its pre-crisis range, while SE2, SE3, and SE4 recover more slowly and, in the most recent windows, show renewed deepening rather than continued normalisation. The net result is that the gap between SE1 and the remaining three zones is wider in the 2025 window than at any point in the pre-crisis period, suggesting that the structural divergence in price sensitivity across Sweden's bidding zones has compounded rather than resolved over the decade.

Results reporting of variance results to be added.

Full numerical estimates of $\hat{\beta}_{\text{wind}}$ and $\hat{\gamma}_{\text{wind}}$ across all windows and zones are reported in Table A.3.1 and Table A.3.2 in section A.3.

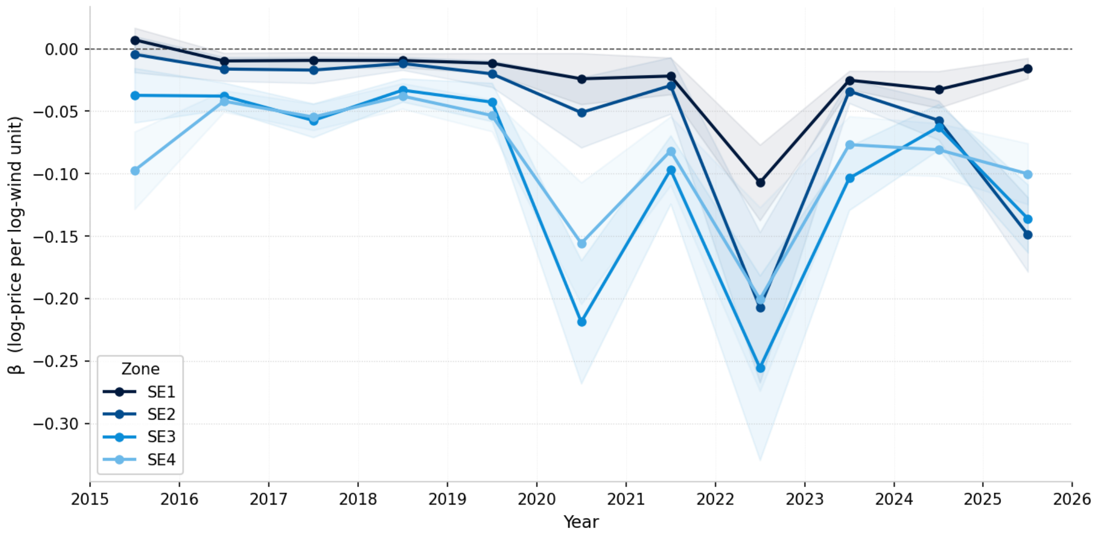


Figure 5.1: Rolling-window wind coefficient, all zones, 1 year window, 1 year shift

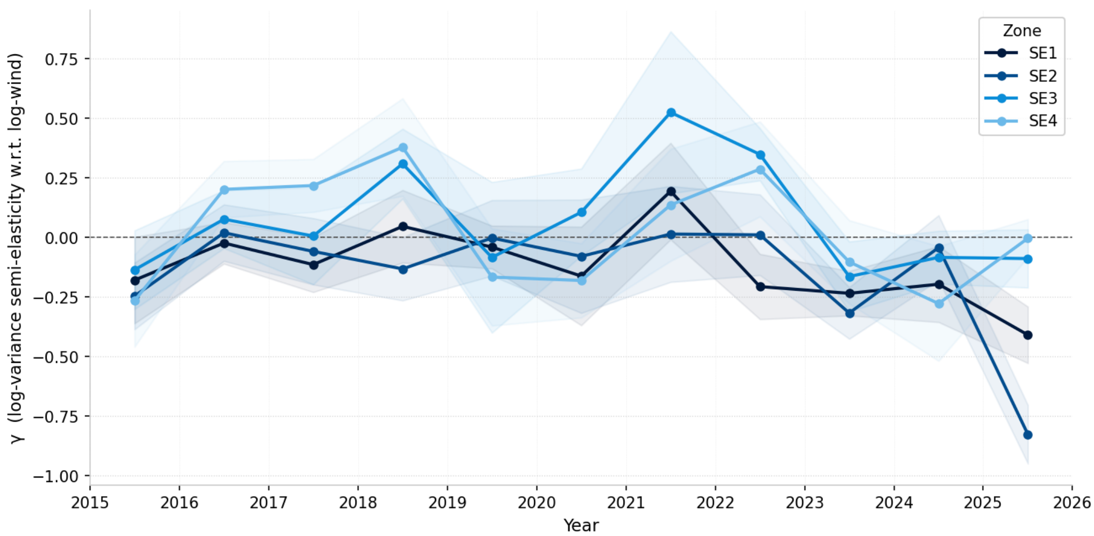


Figure 5.2: Rolling-window volatility wind coefficient, all zones, 1 year window, 1 year shift

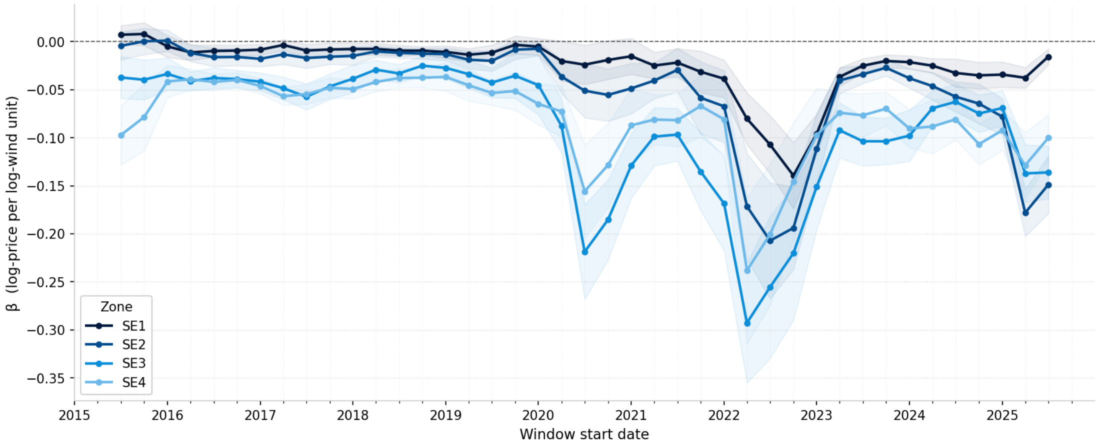


Figure 5.3: Rolling-window wind coefficient, all zones, 1 year window, quarterly shift

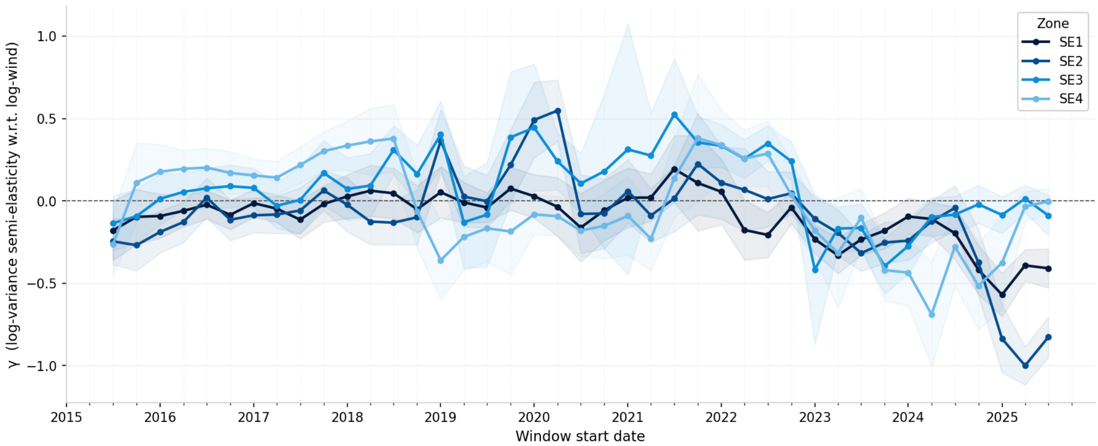


Figure 5.4: Rolling-window volatility wind coefficient, all zones, 1 year window, quarterly shift

5.3 Structural break

Results reporting of structural break results to be added.

The following section outlines the different rolling windows of wind coefficients for price level and volatility. Only the graphs that were selected by the algorithm were included while the whole data series is found in section A.5.

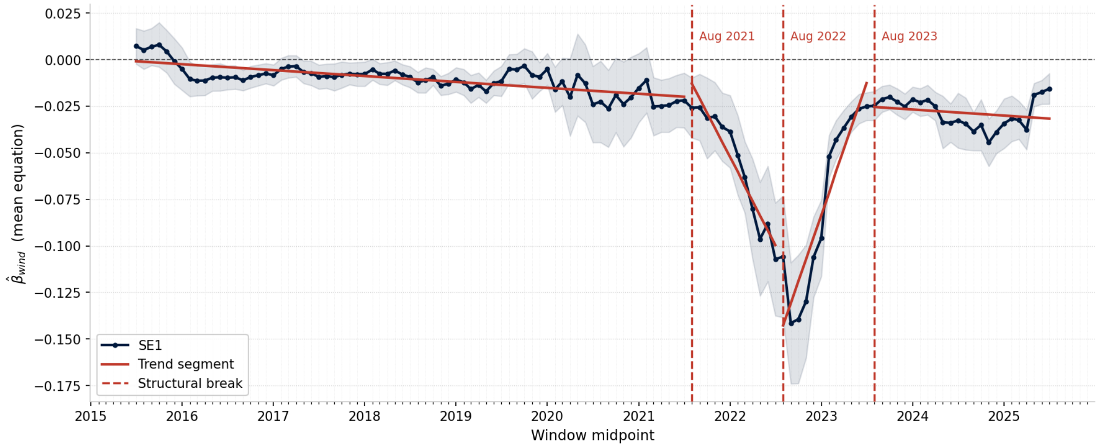


Figure 5.5: SE1: Structural break, 3 break, rolling-window wind coefficient, 1 year window, 1 month shift, this was the chosen one

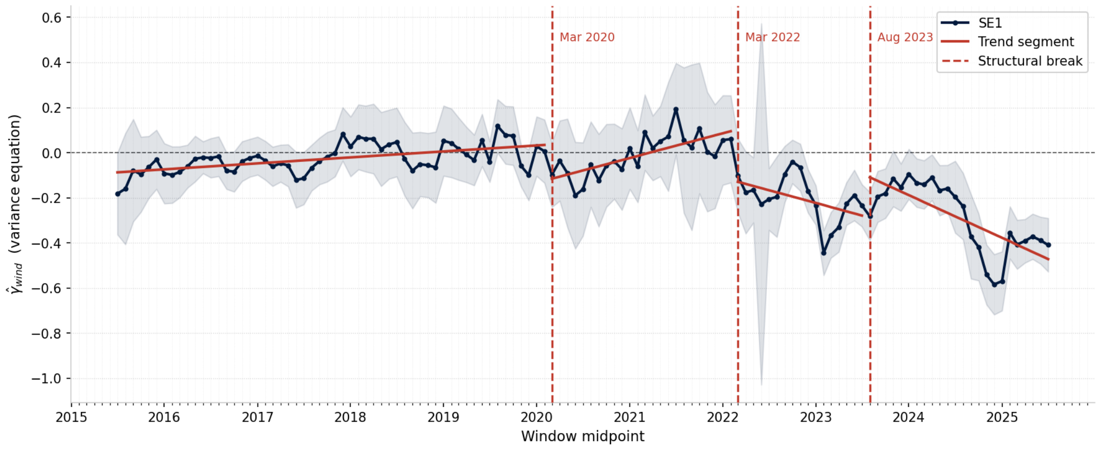


Figure 5.6: SE1: Structural break, 3 break, rolling-window wind var coefficient, 1 year window, 1 month shift, this was the chosen one

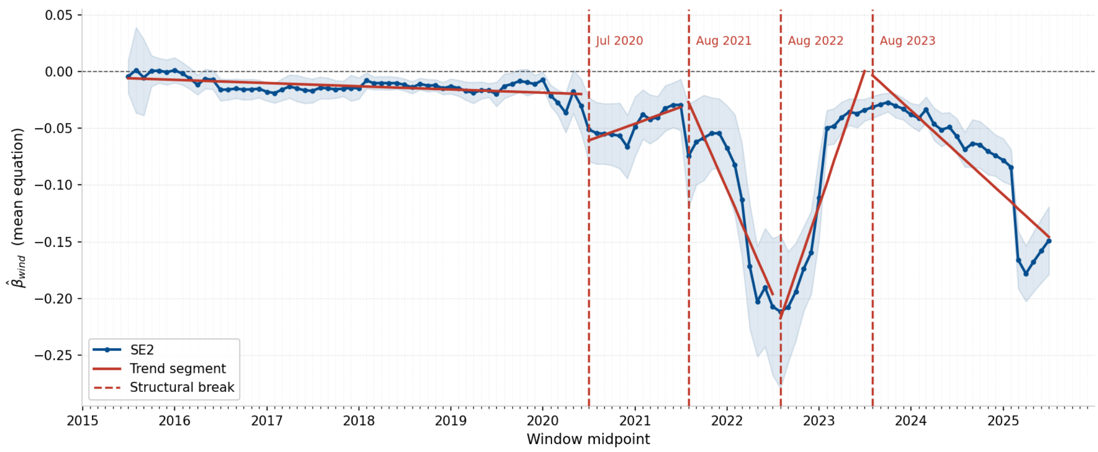


Figure 5.7: SE2: Structural break, 4 break, rolling-window wind coefficient, 1 year window, 1 month shift, this was the chosen one

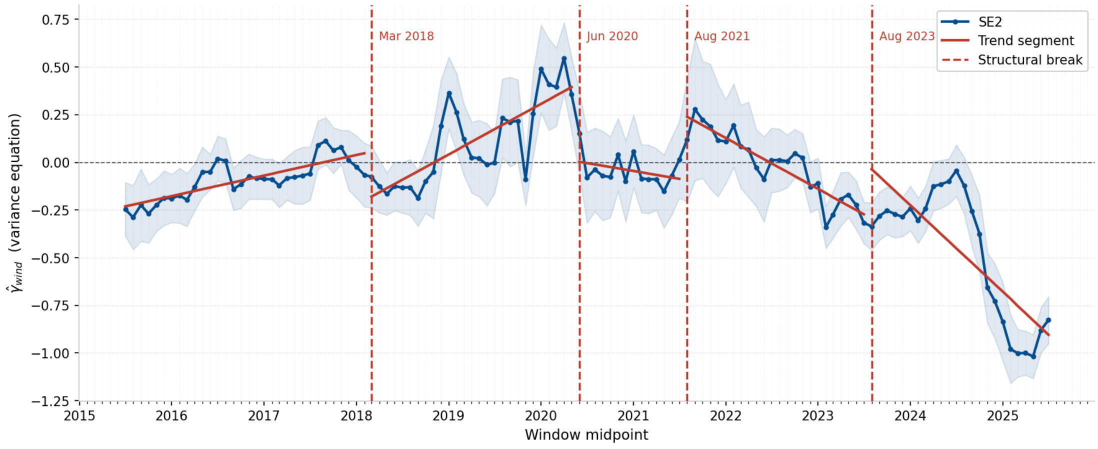


Figure 5.8: SE2: Structural break, 4 break, rolling-window wind var coefficient, 1 year window, 1 month shift, this was the chosen one

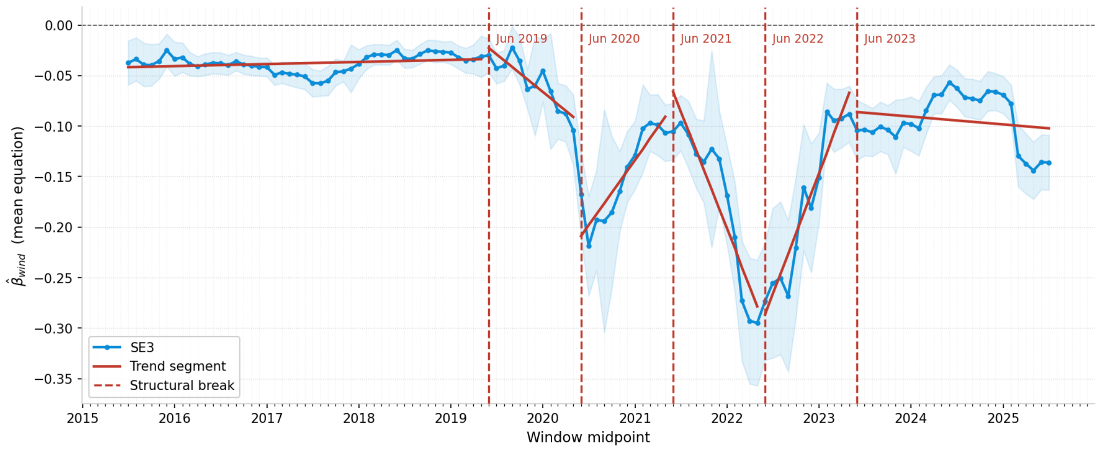


Figure 5.9: SE3: Structural break, 5 break, rolling-window wind coefficient, 1 year window, 1 month shift, this was the chosen one

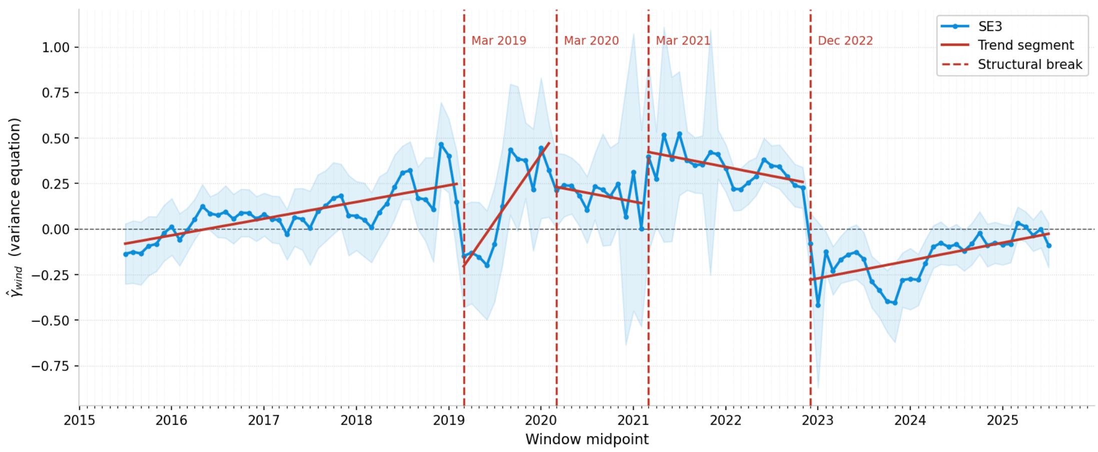


Figure 5.10: SE3: Structural break, 4 break, rolling-window wind var coefficient, 1 year window, 1 month shift, this was the chosen one

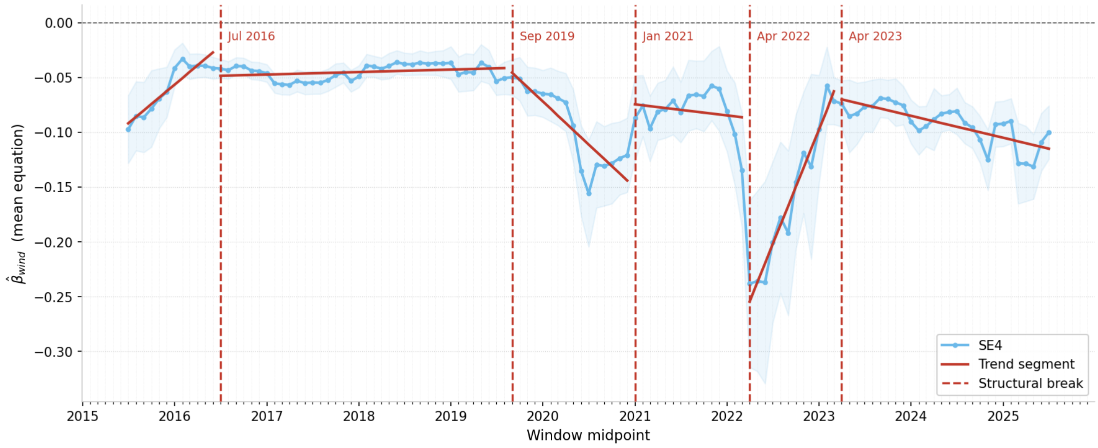


Figure 5.11: SE4: Structural break, 5 break, rolling-window wind coefficient, 1 year window, 1 month shift, this was the chosen one

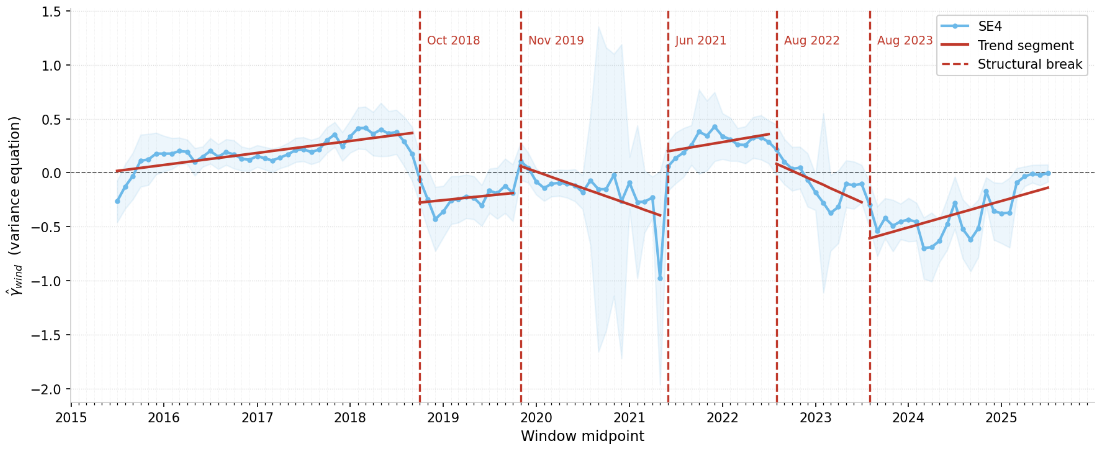


Figure 5.12: SE4: Structural break, 5 break, rolling-window wind var coefficient, 1 year window, 1 month shift, this was the chosen one

5.4 Quantile regression

Results reporting of quantile regression results to be added.

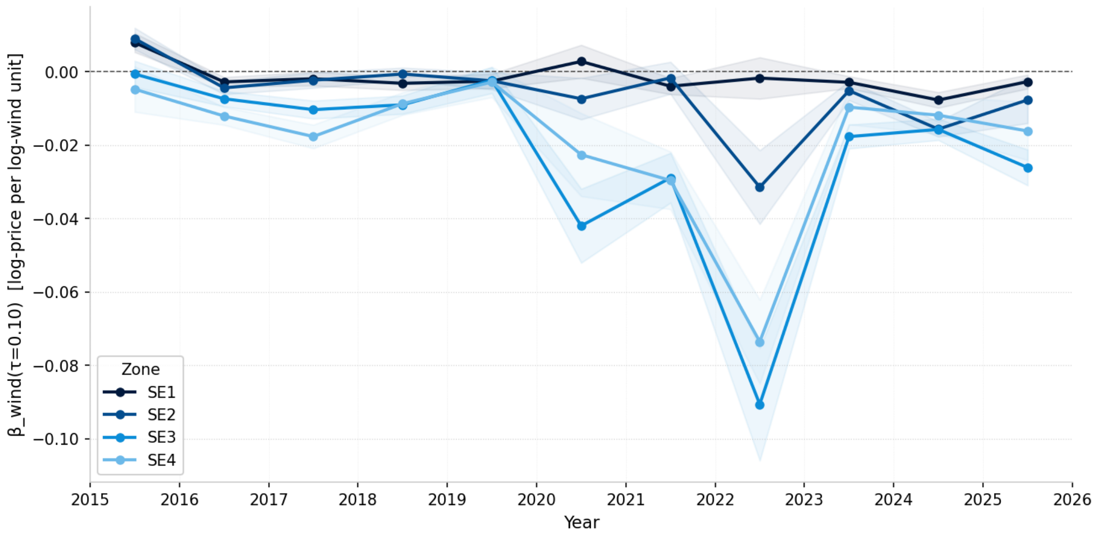


Figure 5.13: Rolling-window wind coefficient $\hat{\beta}_{\text{wind}}$ at the 10th conditional quantile ($\tau = 0.10$)

Note: Each line is one price zone (SE1–SE4); shaded bands are ±1.96 standard errors. Annual non-overlapping windows, 2015–2025.

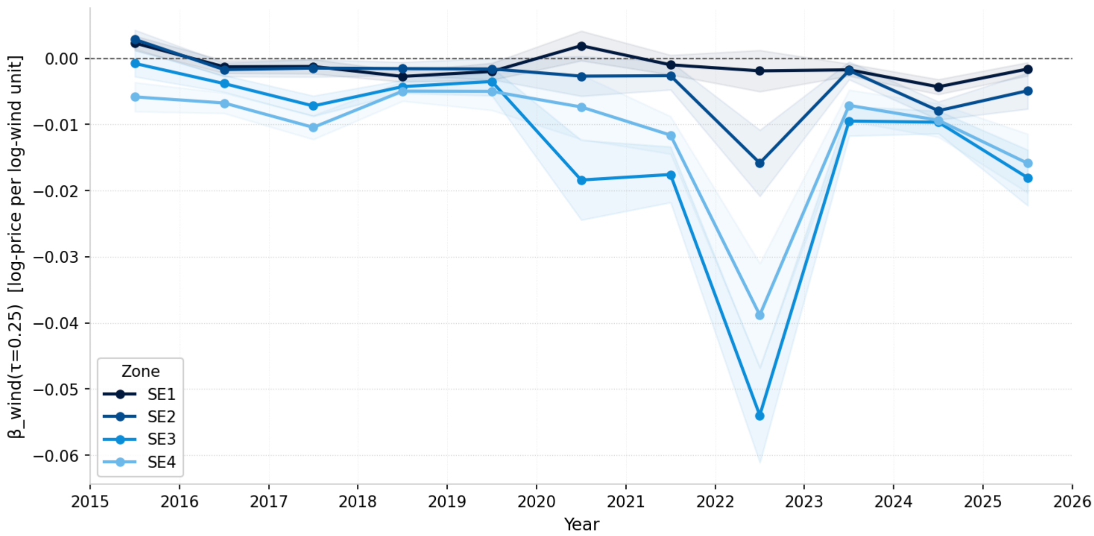


Figure 5.14: Rolling-window wind coefficient $\hat{\beta}_{\text{wind}}$ at the 25th conditional quantile ($\tau = 0.25$)

Note: Each line is one price zone (SE1–SE4); shaded bands are ±1.96 standard errors. Annual non-overlapping windows, 2015–2025.

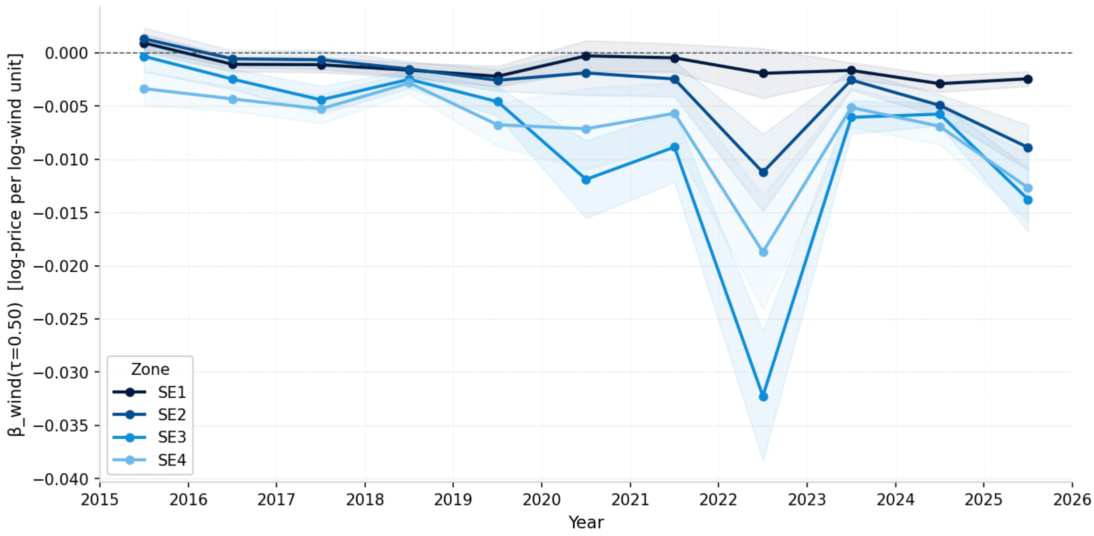


Figure 5.15: Rolling-window wind coefficient $\hat{\beta}_{\text{wind}}$ at the median conditional quantile ($\tau = 0.50$)

Note: Each line is one price zone (SE1–SE4); shaded bands are ±1.96 standard errors. Annual non-overlapping windows, 2015–2025.

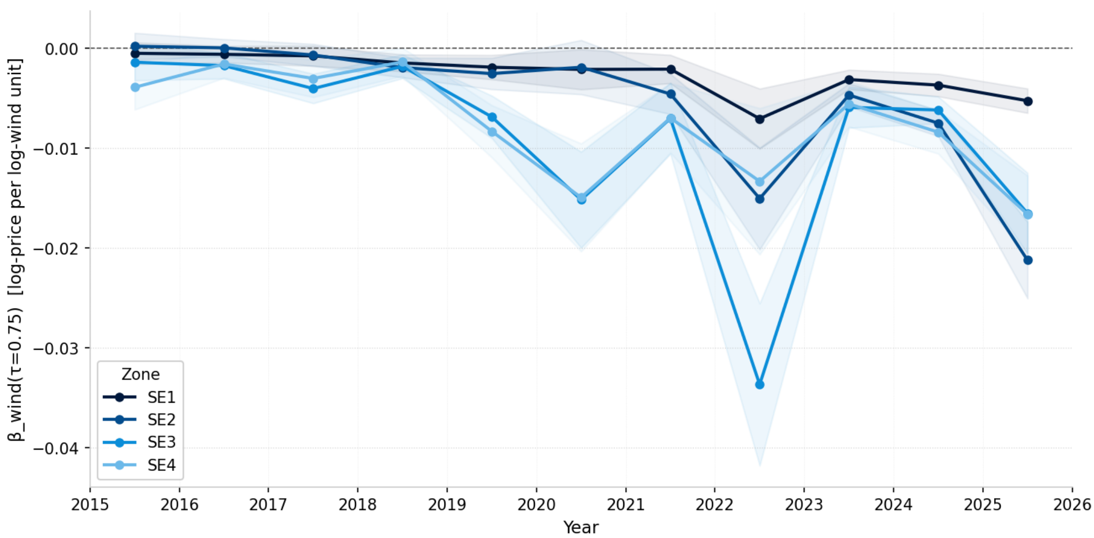


Figure 5.16: Rolling-window wind coefficient $\hat{\beta}_{\text{wind}}$ at the 75th conditional quantile ($\tau = 0.75$)

Note: Each line is one price zone (SE1–SE4); shaded bands are ±1.96 standard errors. Annual non-overlapping windows, 2015–2025.

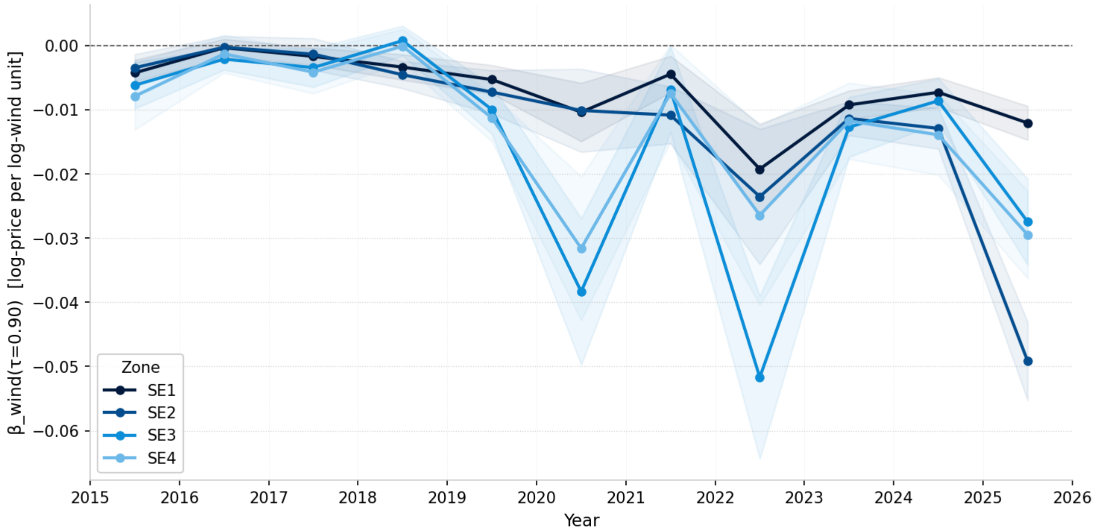


Figure 5.17: Rolling-window wind coefficient $\hat{\beta}_{\text{wind}}$ at the 90th conditional quantile ($\tau = 0.90$)

Note: Each line is one price zone (SE1–SE4); shaded bands are ±1.96 standard errors. Annual non-overlapping windows, 2015–2025.

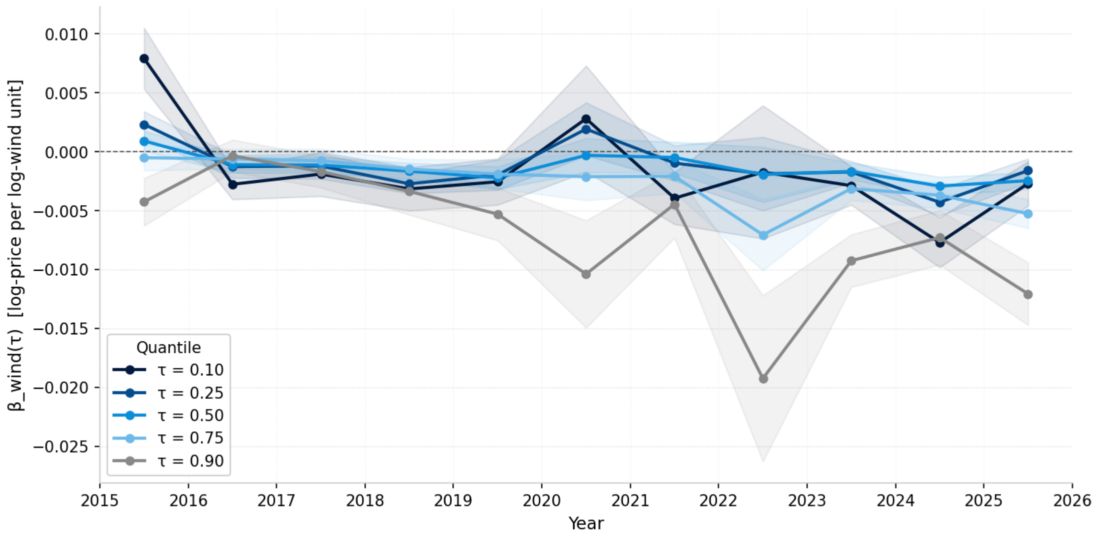


Figure 5.18: SE1: Rolling-window wind coefficient $\hat{\beta}_{\text{wind}}(\tau)$ across five conditional quantiles

Note: $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. Darker shades correspond to lower quantiles; shaded bands are ±1.96 standard errors. Annual non-overlapping windows, 2015–2025.

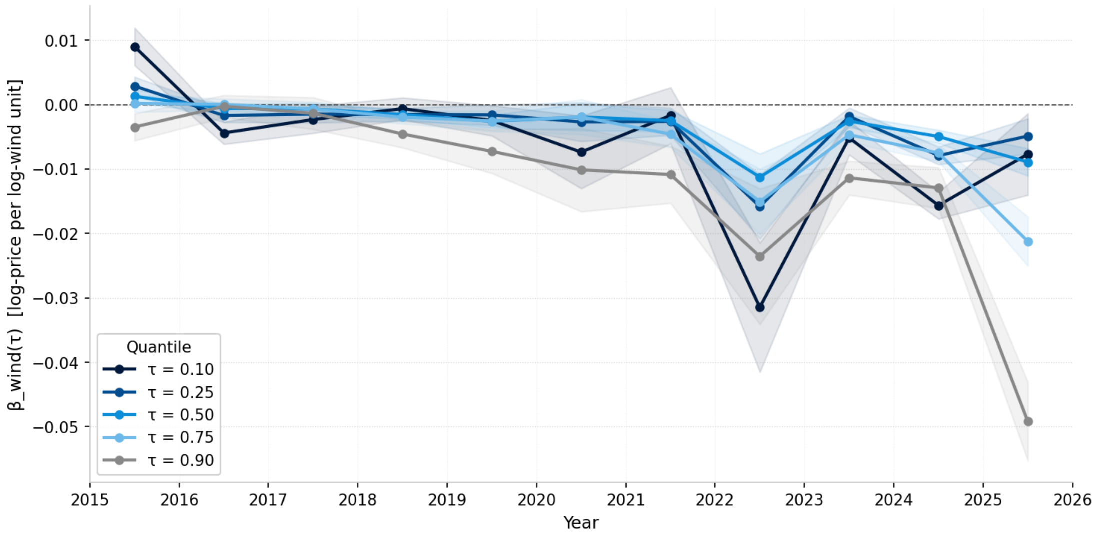


Figure 5.19: SE2: Rolling-window wind coefficient $\hat{\beta}_{\text{wind}}(\tau)$ across five conditional quantiles

Note: $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. Darker shades correspond to lower quantiles; shaded bands are ± 1.96 standard errors. Annual non-overlapping windows, 2015–2025.

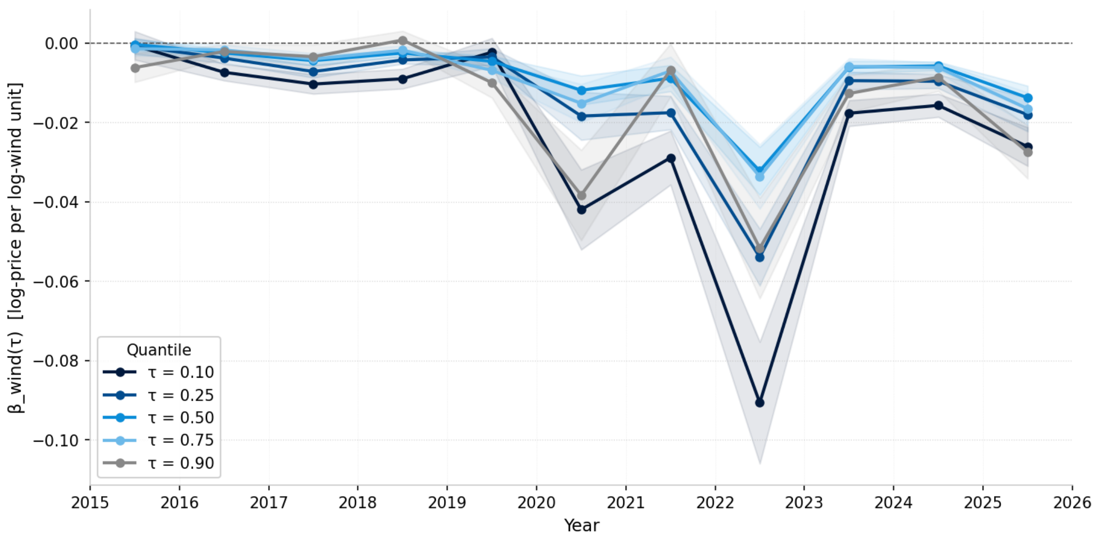


Figure 5.20: SE3: Rolling-window wind coefficient $\hat{\beta}_{\text{wind}}(\tau)$ across five conditional quantiles

Note: $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. Darker shades correspond to lower quantiles; shaded bands are ± 1.96 standard errors. Annual non-overlapping windows, 2015–2025.

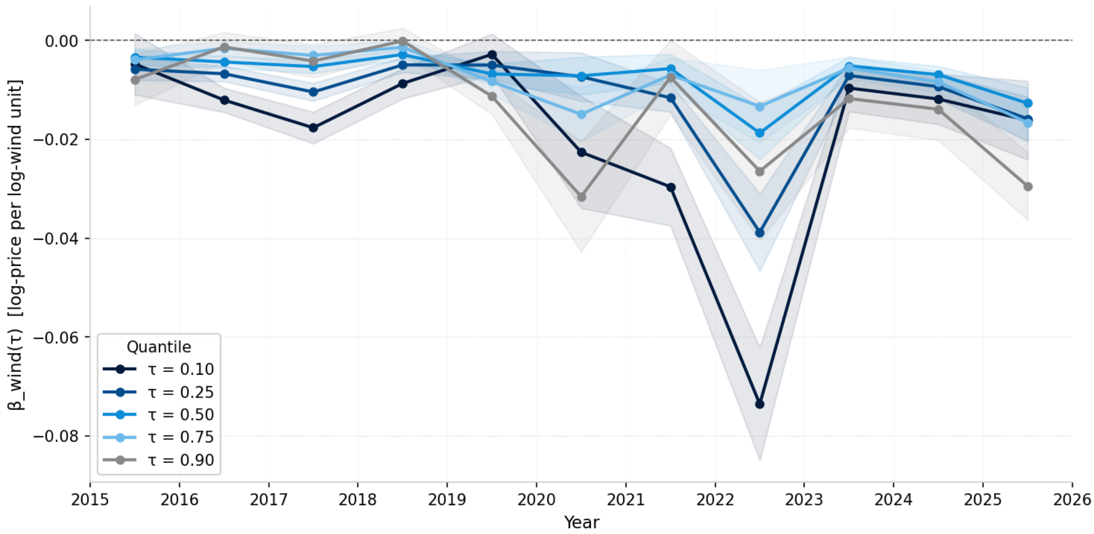
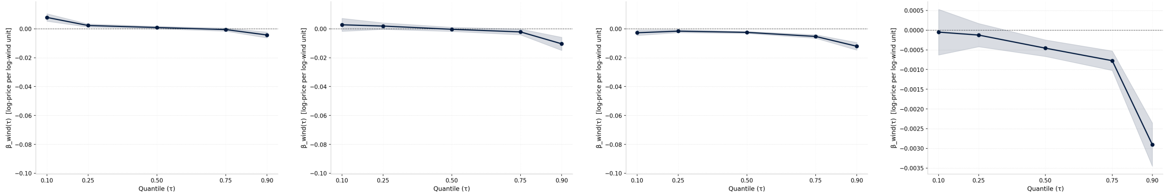


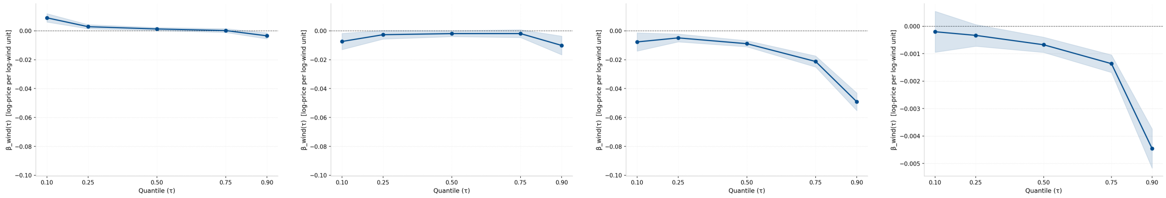
Figure 5.21: SE4: Rolling-window wind coefficient $\hat{\beta}_{\text{wind}}(\tau)$ across five conditional quantiles

Note: $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. Darker shades correspond to lower quantiles; shaded bands are ± 1.96 standard errors. Annual non-overlapping windows, 2015–2025.

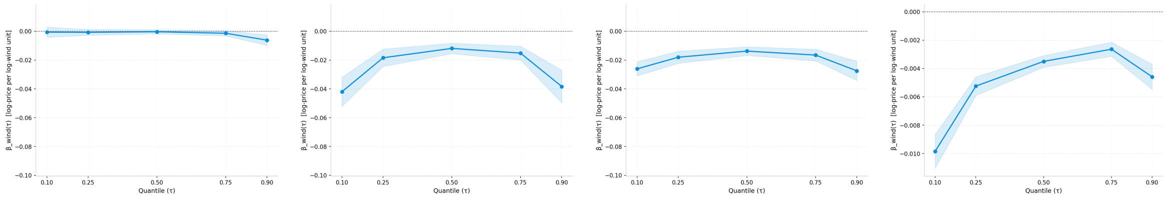
Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$ are reported for $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, estimated separately for each zone and each calendar year from 2015 to 2025 as well as over the full sample period. Full coefficient tables are reported in section A.6 (Table A.6.1–Table A.6.4), and the corresponding plots are shown in Figure 5.22, with all remaining years in section A.6 (Figure A.6.1–Figure A.6.4).



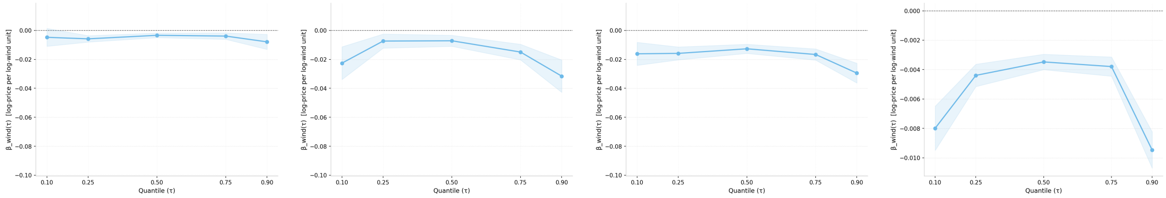
(a) SE1, 2015 **(b)** SE1, 2020 **(c)** SE1, 2025 **(d)** SE1, 2015–2025



(e) SE2, 2015 **(f)** SE2, 2020 **(g)** SE2, 2025 **(h)** SE2, 2015–2025



(i) SE3, 2015 **(j)** SE3, 2020 **(k)** SE3, 2025 **(l)** SE3, 2015–2025



(m) SE4, 2015 **(n)** SE4, 2020 **(o)** SE4, 2025 **(p)** SE4, 2015–2025

Figure 5.22: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$ across price quantiles for selected years and the full sample period

Note: Quantiles $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. Each panel plots the wind coefficient estimate against the quantile, with 95% confidence bands. A negative coefficient indicates that wind production reduces the spot price at that quantile. Results are shown for snapshot years 2015, 2020, and 2025 alongside the full 2015–2025 estimate. Remaining years are reported in section A.6.

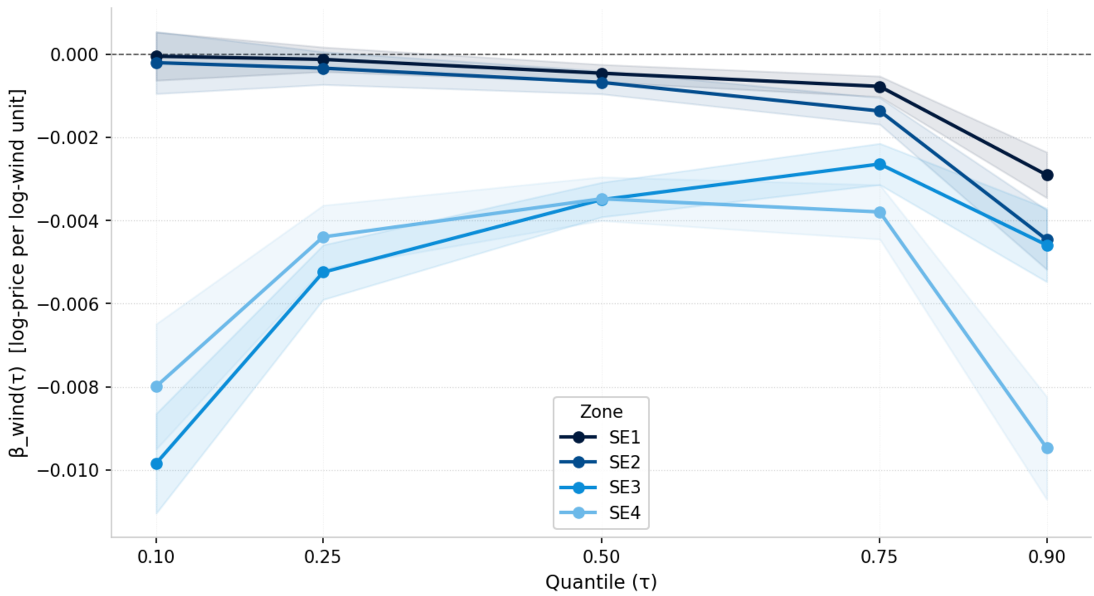


Figure 5.23: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$ across all four bidding zones, full sample 2015–2025

Note: Each line shows the wind coefficient at quantiles $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$ for one zone, with 95% confidence bands. The merit-order effect is strongest at the tails of the price distribution and increases in magnitude from north to south.

5.5 Robustness checks

This section is still work in progress.

5.6 Limitations of the empirical model

This section is still work in progress.

5.7 Forecasting

This section is still work in progress.

Chapter 6

Discussion

6.1 Economic Interpretation

General Interpretation It can be observed that β_{wind} is persistently negative, across all zones and different times. Therefore, it can be concluded that wind unambiguously leads to a decrease in prices between 2015 and 2025. This is in line with previous findings as outlined in chapter 2. However, the impact of wind differs significantly between zones. The following analysis outlines the results of the time period 2024–2025 but is similarly applicable for other time periods as well as indicated by the rolling window coefficients. The results highlight that SE1 is not as strongly affected by fluctuations in wind power production while SE4 is heavily influenced by it (-0.035 compared to -0.123). These results are significant on the 1% level (**??**). The direction of these coefficient makes sense, indicating that a positive change in wind energy that is supplied to the market as a price-decreasing effect by shifting the supply curve throught the merit order effect and adjusting the equilibrium price downwards, ceteris paribus. Looking at the data, we observe that except for hydro reservoir levels and oil and gas prices, all the variables we selected to include are significant at the 1% level. The results for the other variables are intuitive with a positive and significant coefficient for logarithmic consumption. As outlined in chapter 3, increasing demand through a positive change in the consumption forecast shifts the demand curve, thereby, ceteris paribus, increasing the equilibrium price. Similarly, the coefficients of oil and gas indicate a negative effect on price, showing that an increased supply of either of these commodities shifts the supply curve, thereby lowering the equilibrium price.

Rolling Window Interpretation When considering the wind coefficients per zone over a longer time window, different trends become clear. While the coefficients of bidding zones moved closely together before 2019 (Figure 5.3), they started to respond differently to wind energy production fluctuations afterwards. The zones already displayed different sensitivities with regard to wind energy production before 2019 but these differences amplified later. This divergence in sensitivities towards wind production between zones fits the divergence in price levels observed between zones (Figure 3.5). Particularly SE3 and SE4 are noteworthy due to their large spike in negative coefficient of β_{wind} around 2020 and 2022. These results have been confirmed by different specifications of rolling windows, including yearly, quarterly, and monthly shifts. The interpretation of this is that for SE4 and SE4, the sensitivity of prices towards wind energy production increased significantly within these years. While the result is somewhat intuitive for 2022, considering the global energy crisis at the time, it surprising in 2020. The negative spike in β_{wind} during 2022 can be understood in the context of the price hike in fossil fuel, leading to a different form of the supply curve and impacts on the merit order effect, resulting in larger absolute price reductions through more wind power production. For the time of the Covid-19 pandemic in 2020, a change in demand could provide an initial explanation for this change in sensitivity. Particularly noticeable is the asymmetric recovery after the crisis period of 2021/2022. As shown in Figure 5.3, SE1 normalizes across time to before-crisis-levels while all other bidding zones do not. A reason might be the lack of sufficient transmission capacities between energy zones, coupled with growing wind power penetration in the South, resulting in a structural change. In this context, it is noteworthy that the increasingly negative β_{wind} of SE4 is accompanied by a rising share in wind production capacities (Figure 3.4). That could hint towards cannibalization in the market, as wind production reduces the price that economically supports its build-out and infrastructure investments. Overall, we see a consistently negative coefficient of wind energy on electricity prices although the magnitude varied substantially over time. Further details can be found in Table A.3.1. From the table it becomes clear that the negative coefficient of wind production is almost consistently signficant at 1%. However, the magnitude differs within bidding zones across years and between bidding zones. The sensitivity ranges from -0.0093 to -0.1072 for SE1 in 2018 and 2022, respectively. The differences for other bidding zones are similarly striking with all of them having their highest (most negative) coefficient in 2022 (-0.2072 for SE2, -0.2556 for SE3, and -0.2006 for SE4) (Table A.3.1.

When investigating the impact of wind energy on volatility, the trend is not as clear as indicated by previous research (chapter 2). Particularly noteworthy is the huge increase in the volatility coefficient in SE3 in 2021 during the energy crisis and the recently sharp decline of the volatility coefficient in 2025 for SE2 (**??**. These results have been confirmed by yearly, quarterly, and monthly shifts. As seen in Figure 5.4, the magnitude of the volatility coefficient varies substantially throughout time and is not consistently significant. However, it is observable that in the most recent years following 2023, the volatility coefficients for the different zones are almost consistently negative and significant. Particularly SE1 and SE2 show a strong negative volatility coefficient, indicating that in these zones wind energy production reduces the volatility of energy prices. This contrasts some of the literature outlined in chapter 2 but is in line with the findings of Maciejowska (2020). Table A.3.2 shows the coefficients of volatility over time. However, here it becomes clear that they are not consistently significant across years and oscillate between positive and negative coefficients within zones. This underscores the take-away from **??** and Figure 5.4 in that trends are not as clear for the impact of wind production on volatility as it is clear for price level. However, particularly the strongly negative and significant at 1% coefficient for SE1 from 2022 to 2025 is noteworthy, indicating that an increase in wind power reduces price volatility substantially (-0.2065 in 2022, -0.2344 in 2023, -0.1964 in 2024, and $-.4092$ in 2025). After 2023, all coefficients of volatility across zones turn negative but only a slight majority of them is statistically significant.

Therefore, this paper is in line with previous literature in confirming the merit order effect of wind energy on prices, helping to lower electricity prices overall. However, the effect on volatility is more complex and not simply volatility-inducing. Contrary to that, initial results would indicate that, under certain circumstances, wind production can even have decreasing effect on volatility, thereby enhancing price stability and energy supply security.

Structural Breaks Interpretation We test for structural breaks to separate different periods in our analysis and derive accurate estimations for future trends. When testing for structural breaks in the rolling windows of the wind price level coefficient and volatility coefficient, it becomes clear that structural breaks are chosen in a way that the large deviations during the 2021/2022 energy crisis are treated as deviations from general trends (Figures A.5.3, A.5.7, A.5.12, A.5.16, A.5.21, A.5.25, A.5.30, and A.5.35). Following the latest break, it is observable that almost all trends for price level and price volatility point downwards with different magnitude aside from the volatility and price level coefficient of SE4 that are both increasing. These results indicate that, aside from SE4, wind energy has a increasingly negative effect on prices while at the same time lowering price volatility. These trends highlights the findings from the rolling window analysis and imply that the observed negative price and volatility effects of wind energy might become stronger going forward.

Quantile regression Interpretation When considering the coefficients on the different quantiles of the price distribution, it becomes clear that higher prices that are part of the highest quantiles are subject to stronger (more negative) coefficients of wind energy production. These results are meaningful as they imply that wind production is particularly strong in lowering prices when they are high, particularly in times of positive price spikes such as during a crisis due to energy supply shocks. This is in line with the merit order effect initially discussed in that wind energy can crowd out significantly more expensive sources of energy but provides an interesting perspective on the nuance of the effect of wind energy on different areas of the price distribution. When considering the effects of wind energy on prices over time, it becomes clear that particularly in recent years the upper end of the price distribution has developed a significantly stronger sensitivity towards wind production than it was the case previously (Figures 5.18, 5.19, 5.20, 5.21). These results are consistent for SE1, SE2, and SE3, while SE4 shows a different pattern (Figure 5.21. Noticeably, it is the lowest quantile that has the strongest sensitivity towards wind energy during the crisis period in 2021/2022. This is surprising as it means that wind production shifts the merit order curve particularly strong in the beginning, indicating that the next-best energy supplier replaced by additional wind power production exhibits relatively high prices compared to the alternatives in other zones that show lower sensitivities. When considering Figure 5.22, a general pattern becomes clear in that the effect of wind production is strongest on prices in the upper and lower tails of the price distribution. This inverse u-shaped coefficients across the price distribution is intuitive in that results are almost consistently negative except for 2015, where results are not entirely significant. In the following years, it is noteworthy that, compared to the other parts of the distribution, the highest quantile shows a increasingly negative coefficient Figure 5.22. When observing the quantile regression across the entire time period from 2015 to 2025, a similar trend is present with a particularly strong inverse u-shape for SE3 (Figure 5.23). This indicates that, considering the entire time period, SE3 and SE4 have strong negative sensitivities in price towards an increase in wind production on the upper and lower ends of the distribution while SE1 and SE2 only show that for the upper tail of their price distribution (Figure 5.23). This indicates the different energy supplies available per zone and the differences in merit order curves that result in different sensitivities towards wind dependent on where on the merit order curve a change in wind production supply occurs. The appendix outlines the coefficients of the price distribution on a yearly basis (section A.6). Particularly worth highlighting is the enormous change in the coefficients of the distribution for SE3 and SE and, to a smaller extend, SE2 (Figure A.6.2-Figure A.6.4). The strong negative coefficient during the energy crisis in the upper and lower tail of the distribution shows that for these two regions the effect of additional wind was particularly pronounced. In light of their different composition of the energy production mix (Figure 3.4), these results are surprising but in line with previous findings in this analysis.

These findings, in combination with the results of the ARMAX-GARCH model, imply that wind energy can actually lower prices and volatility in times of crisis, reflected through the highest prices of the overall price distribution.

6.2 Policy Implications

The Swedish Government tasked Svenska kraftnät to re-assess the structure of Sweden's bidding zones, considering a reduction of number o zones. The results of this paper highlight that prices are diverging as are sensitivities towards wind energy between zones. This provides an argument against a consolidation of bidding zones. As long the market fundamentals between zones differ that substantially with regard to the impact that wind (and other energy sources) have on price setting in the day-ahead market, a consolidation could lead to more inefficiencies and an enlarged vulnerability. Such a change in bidding zone area should be accompanied by substantial investments in transmission infrastructure to ensure that excess supply and demand can easily be traded across zones and bring prices and sensitivities together. Only by providing the physical infrastructure that enables a seamless exchange of energy can the price sensitivities be brought closer together, indicating a convergence in merit order curves. If transmission constraints were non-binding and wind surpluses from SE1 and SE2 could freely flow towards Sweden's South, the merit order effect from wind energy would equalize across zones.

The widening gap in price levels between SE1 and sE2 on one side and SE3 and SE4 on the other side (Figure 3.5) has distributional implications for consumers and firms: The south faces more volatile and higher prices while they show similar sensitivities towards wind with SE2 (Table 5.1. The different coefficients in price level and volatility between zones, particularly SE4, indicate that no near convergence of prices can be expected. The last estimates of the sensitivity of price level towards wind production for 2025 show -0.0156 for SE1, -0.1489 for SE2, -0.1361 for SE3, and -0.1002 for SE4. These differences are quite substantial, particularly between SE1 and SE4 on one side and SE2 and SE3 on the other side, indicating that a convergence in price is unlikely considering the significant differences in sensitivities.

This paper highlights a potential risk for these investments. A stronger negative coefficient β_{wind} works against the investment case for new wind capabilities in SE3 and SE4, making it potentially unattractive for investors to enlarge wind production capabilities. Policy makers should bear these dynamics in mind when designing policy, ensuring that the very incentives that lead to the build-out of wind energy in the first place are not cannabalized by the increasingly negative coefficient of wind energy.

From a perspective of energy security, this paper indicates that wind production capabilities might become an increasing necessity during times of volatility. With a negative coefficient on price level and volatility in three out of four bidding zones in the years since the last structural break, wind energy lowers energy prices and seems to insure against exogenously induced market volatility by lowering price level and decreasing price volatility when prices are in the highest quantiles, likely in times of crisis. These implications are non-trivial, as they contradict the commonly held belief that wind power is inherently insecure and threatening to grid stability and constant energy supply. This paper shows hat, while global economies are still dependent on fossil fuels and their exogenously induced price shocks, wind energy provides an alternative that is price and volatility reducing in the current energy market environment.

Overall, we see the structural break imposed by the 2022 energy crisis. The price sensitivities post-crisis have not returned to its pre-crisis level in SE1 and SE2. This is highly relevant for policy proposals that need to account for this sustained changed relationship between wind and energy prices when suggesting bidding zone reforms or wind infrastructure investments.

6.3 Limitations

We use fixed ARMA (2,1) specifications across all rolling windows. As noted before, these are optimal for the years 2024 and 2025 but might misspecified windows of previous years that contain different price dynamics.

While we tested for a structural break in the data, we ultimately manually imposed it at 2021-10-01 by analysing the data. After running the estimations with different (and without) structural breaks in the initial data cleaning process, we see that it does not change the results, neither directionally nor in any significant form.

While the paper's findings towards price levels are very interesting and relevant for policy research, price volatilities are just as relevant to understand how dynamic prices are evolving across time. Future research could investigate further how the crisis period of 2021-2022 affected the volatility of energy prices as a response to change sin wind production capabilities, providing more nuanced recommendations for energy policy proposals.

Finally, while the model controls for European market coupling, it does not directly model variables like the German nuclear exit directly but only indirectly through the energy exchange with the respective Swedish bidding zones.

6.4 Directions for Further Research

For further research and to aid policy makers, it would be relevant to understand how the change in wind production dynamics effect wind producers themselves, including the implications that a potential decline in price levels due to a permanent shift of the supply curve have on their incentives for further wind power build-out. In combination with the current gap in prices and sensitivities between Sweden's energy zones, this future research would extend on our work and answer questions relevant to the future of Sweden's electricity market policies and its build-out of renewable energy sources. It would build on this paper's findings by explicitly modelling marginal cost and profit curves of different wind energy producers and indicating the implications a cannibalization has on the repayment length of investments and their overall profitability.

Furthermore, it would be relevant to investigate a cross-country panel of countries in the Nordics, including Norway, Denmark, and Finland, to understand whether the same dynamics observed within the Swedish electricity market also hold across Nord Pool's other electricity trading areas.

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Chapter A

Appendix

A.1 Outlier removal visualization (SE2–SE4)

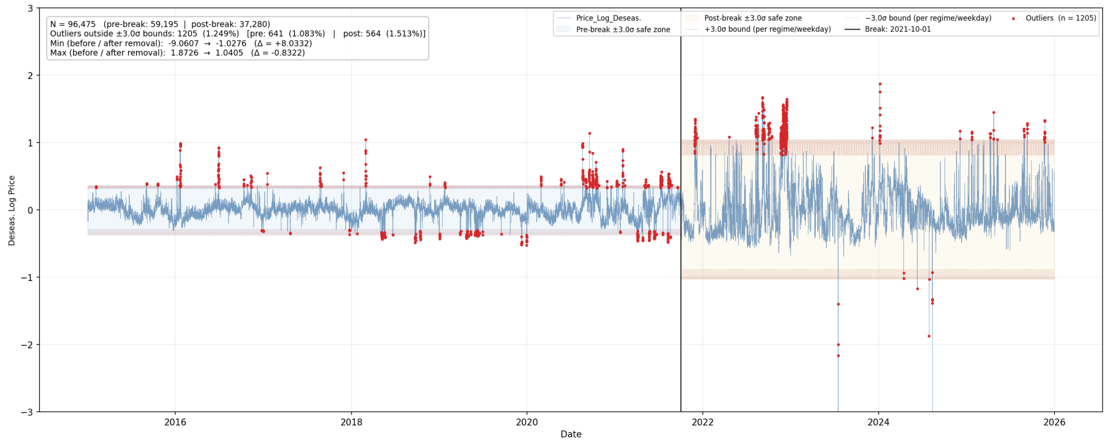


Figure A.1.1: SE2: Outlier removal

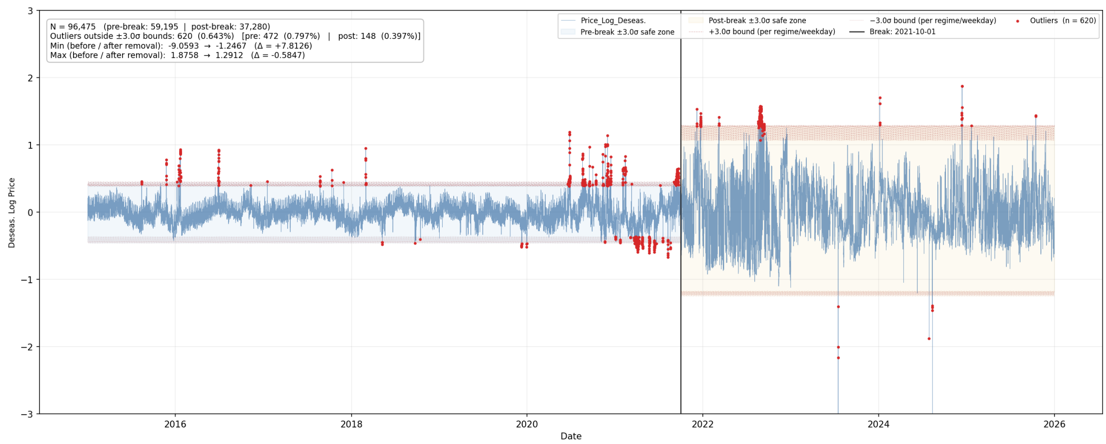


Figure A.1.2: SE3: Outlier removal

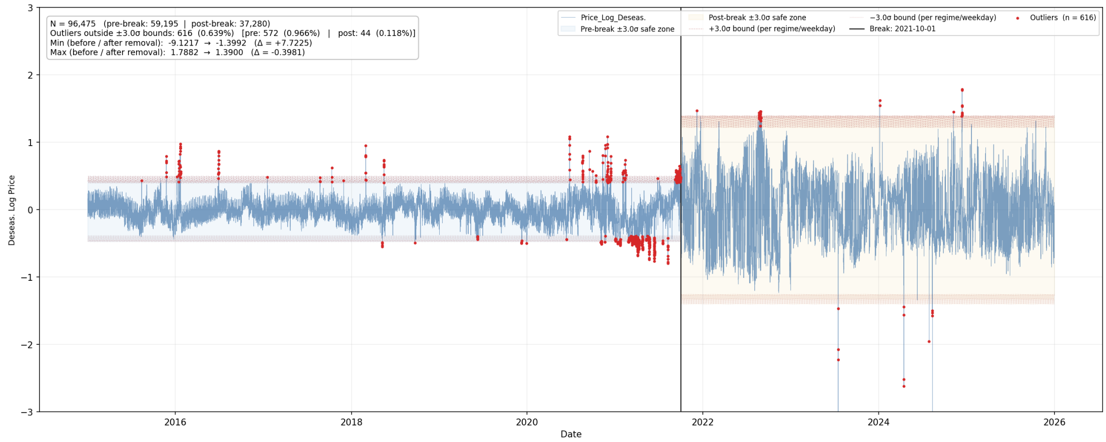


Figure A.1.3: SE4: Outlier removal

A.2 Bilateral Net Exchange Coefficients

The following tables report the bilateral net exchange coefficients from the mean and variance equations of the ARMAX(1,1)-GARCH-X(1,1) baseline models. Results are reported separately for each bidding zone. The corresponding main model estimates are reported in **??** and Table 5.1.

Table A.2.1: Bilateral net exchange coefficients, SE1, 2025

	Coefficient	**Std. Error**
Mean equation		
Net export: FI	$-1.97 \times 10^{-5***}$	(3.34×10^{-6})
Net export: NO4	$-3.89 \times 10^{-5***}$	(8.83×10^{-6})
Net export: SE2	$-2.06 \times 10^{-5***}$	(3.12×10^{-6})
Variance equation		
γ Net export: FI	-0.0008^{***}	(9.67×10^{-5})
γ Net export: NO4	-0.0006	(0.0004)
γ Net export: SE2	-0.0007^{***}	(0.0001)

Notes: Bilateral net exchange coefficients for SE1 from the ARMAX(1,1)-GARCH-X(1,1) model estimated over 2025, reported in Table 5.1. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.2.2: Bilateral net exchange coefficients, SE2, 2025

	Coefficient	**Std. Error**
Mean equation		
Net export: NO3	$-5.83 \times 10^{-5***}$	(5.53×10^{-6})
Net export: NO4	$-5.83 \times 10^{-5***}$	(5.82×10^{-6})
Net export: SE1	$-5.41 \times 10^{-5***}$	(5.22×10^{-6})
Net export: SE3	$-5.00 \times 10^{-5***}$	(5.46×10^{-6})
Variance equation		
γ Net export: NO3	0.0003^{**}	(0.0001)
γ Net export: NO4	-0.0001	(0.0003)
γ Net export: SE1	-0.0002	(0.0001)
γ Net export: SE3	-5.00×10^{-5}	(0.0001)

Notes: Bilateral net exchange coefficients for SE2 from the ARMAX(1,1)-GARCH-X(1,1) model estimated over 2025, reported in Table 5.1. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.2.3: Bilateral net exchange coefficients, SE3, 2025

	Coefficient	**Std. Error**
Mean equation		
Net export: DK1	$-5.19 \times 10^{-5***}$	(9.93×10^{-6})
Net export: FI	6.92×10^{-6}	(8.11×10^{-6})
Net export: NO1	$-2.72 \times 10^{-5***}$	(6.41×10^{-6})
Net export: SE2	$-2.59 \times 10^{-5***}$	(5.70×10^{-6})
Net export: SE4	$-4.43 \times 10^{-5***}$	(4.50×10^{-6})
Variance equation		
γ Net export: DK1	-0.0003^{**}	(0.0001)
γ Net export: FI	-0.0003^{***}	(9.21×10^{-5})
γ Net export: NO1	4.75×10^{-5}	(0.0001)
γ Net export: SE2	-0.0002^{**}	(7.90×10^{-5})
γ Net export: SE4	0.0002^{**}	(7.67×10^{-5})

Notes: Bilateral net exchange coefficients for SE3 from the ARMAX(1,1)-GARCH-X(1,1) model estimated over 2025, reported in Table 5.1. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.2.4: Bilateral net exchange coefficients, SE4, 2025

	Coefficient	**Std. Error**
Mean equation		
Net export: DK2	$-4.27 \times 10^{-5***}$	(9.49×10^{-6})
Net export: GER	$-6.85 \times 10^{-5***}$	(1.37×10^{-5})
Net export: PL	$-3.66 \times 10^{-5**}$	(1.46×10^{-5})
Net export: SE3	$2.35 \times 10^{-5***}$	(5.45×10^{-6})
Variance equation		
γ Net export: DK2	-9.92×10^{-5}	(0.0001)
γ Net export: GER	-0.0005^{***}	(0.0002)
γ Net export: PL	-0.0001	(0.0002)
γ Net export: SE3	-0.0004^{***}	(5.07×10^{-5})

Notes: Bilateral net exchange coefficients for SE4 from the ARMAX(1,1)-GARCH-X(1,1) model estimated over 2025, reported in Table 5.1. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

A.3 Rolling Window Coefficient Tables

Table A.3.1: Rolling-window estimates of $\hat{\beta}_{\text{wind}}$, mean equation

Year	**SE1**	**SE2**	**SE3**	**SE4**
2015	0.0072	−0.0044	−0.0373***	−0.0974***
2016	−0.0097***	−0.0162***	−0.0379***	−0.0420***
2017	−0.0093***	−0.0171***	−0.0574***	−0.0546***
2018	−0.0093***	−0.0118***	−0.0331***	−0.0379***
2019	−0.0116***	−0.0201***	−0.0427***	−0.0534***
2020	−0.0240**	−0.0511***	−0.2188***	−0.1558***
2021	−0.0218***	−0.0295**	−0.0968***	−0.0818***
2022	−0.1072***	−0.2072***	−0.2556***	−0.2006***
2023	−0.0251***	−0.0340***	−0.1037***	−0.0767***
2024	−0.0327***	−0.0573***	−0.0627***	−0.0809***
2025	−0.0156***	−0.1489***	−0.1361***	−0.1002***
Overall	−0.0070***	−0.0229	−0.0496***	−0.0573***

Notes: Each entry reports $\hat{\beta}_{\text{wind}}$ from the mean equation of the ARMAX(1,1)-GARCH-X(1,1) model estimated over the calendar year indicated, using a one-year rolling window. Overall reports the estimate over the full 2015–2025 sample. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.3.2: Rolling-window estimates of $\hat{\gamma}_{\text{wind}}$, variance equation

Year	**SE1**	**SE2**	**SE3**	**SE4**
2015	-0.1793^{*}	-0.2458^{***}	-0.1358	-0.2653^{***}
2016	-0.0236	0.0198	0.0765	0.2019^{***}
2017	-0.1140^{*}	-0.0586	0.0059	0.2178^{***}
2018	0.0464	-0.1322^{*}	0.3091^{***}	0.3785^{***}
2019	-0.0405	-0.0024	-0.0838	-0.1664
2020	-0.1621	-0.0794	0.1056	-0.1806^{**}
2021	0.1939^{*}	0.0142	0.5247^{***}	0.1364
2022	-0.2065^{***}	0.0108	0.3486^{***}	0.2867^{***}
2023	-0.2344^{***}	-0.3172^{***}	-0.1641^{**}	-0.1029
2024	-0.1964^{**}	-0.0427	-0.0836	-0.2776^{**}
2025	-0.4092^{***}	-0.8267^{***}	-0.0887	-0.0035
Overall	0.1173	0.0866	0.2651^{**}	-0.1359

Notes: Each entry reports $\hat{\gamma}_{\text{wind}}$ from the variance equation of the ARMAX(1,1)-GARCH-X(1,1) model estimated over the calendar year indicated, using a one-year rolling window. The multiplicative GARCH-X specification means $\hat{\gamma}_{\text{wind}}$ measures the effect of log wind production on log conditional variance. Overall reports the estimate over the full 2015–2025 sample. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

A.4 Grid search over ARMA specifications

Table A.4.1: SE1: ARMA(p, q) search results, QML robust standard errors, sample 01 Jan 2024 to 31 Dec 2025

	ARMA(p,q) specification														
	(0,1)	(0,2)	(0,3)	(1,0)	(1,1)	(1,2)	(1,3)	(2,0)	(2,1)	(2,2)	(2,3)	(3,0)	(3,1)†	(3,2)‡	(3,3)‡
a_1				0.958***	0.939***	0.933***	0.935***	1.178***	1.081***	0.829***	1.749***	1.186***	2.059***		
				(0.004)	(0.005)	(0.006)	(0.006)	(0.020)	(0.046)	(0.148)	(0.021)	(0.021)	(0.021)		
a_2								−0.231***	−0.138***	0.098	−0.753***	−0.268***	−1.324***		
								(0.020)	(0.044)	(0.139)	(0.020)	(0.029)	(0.038)		
a_3												0.031**	0.262***		
												(0.014)	(0.019)		
m_1	0.841***	1.180***	1.265***		0.237***	0.252***	0.248***		0.103**	0.356**	−0.581***		−0.880***		
	(0.003)	(0.015)	(0.020)		(0.019)	(0.023)	(0.023)		(0.046)	(0.147)	(0.028)		(0.013)		
m_2		0.631***	1.009***			0.043***	0.037**			0.068*	−0.187***				
		(0.007)	(0.015)			(0.016)	(0.018)			(0.036)	(0.020)				
m_3			0.475***				−0.014				−0.088***				
			(0.010)				(0.018)				(0.014)				
AIC	−36892	−46597	−51711	−58917	−59860	−59885	−59886	−59870	−59881	−59884	−60103	−59885	−60139	—	—
BIC	−36799	−46496	−51603	−58824	−59759	−59776	−59769	−59769	−59772	−59767	−59978	−59776	−60023	—	—
Log-lik	18,458	23,312	25,870	29,470	29,943	29,957	29,958	29,948	29,954	29,957	30,067	29,956	30,085	—	—
N	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	—	—

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Mean equation includes exogenous controls (not shown). `vce(robust)` corresponds to QML sandwich standard errors.

† ARMA(3,1) exhibits AR–MA root cancellation with near-explosive AR polynomial roots; fit improvement is likely spurious.

‡ ARMA(3,2) and ARMA(3,3) failed to converge (Stata `arima` returned a non-zero error code); no estimates available.

Table A.4.2: SE2: ARMA(p, q) search results, QML robust standard errors, sample 01 Jan 2024 to 31 Dec 2025

	ARMA(p,q) specification														
	(0,1)	(0,2)	(0,3)	(1,0)	(1,1)	(1,2)	(1,3)	(2,0)	(2,1)	(2,2)	(2,3)	(3,0)	(3,1)†	(3,2)	(3,3)†
a_1				0.950***	0.927***	0.920***	0.925***	1.162***	1.047***	0.761***	1.742***	1.171***	2.049***	0.042	2.394***
				(0.004)	(0.006)	(0.007)	(0.007)	(0.020)	(0.056)	(0.168)	(0.023)	(0.021)	(0.022)	(0.309)	(0.059)
a_2								−0.226***	−0.116**	0.148	−0.746***	−0.268***	−1.311***	0.929***	−1.963***
								(0.020)	(0.053)	(0.156)	(0.022)	(0.029)	(0.039)	(0.017)	(0.104)
a_3												0.037**	0.257***	−0.110	0.565***
												(0.015)	(0.020)	(0.293)	(0.047)
m_1	0.831***	1.157***	1.242***		0.236***	0.250***	0.243***		0.123**	0.409**	−0.588***		−0.887***	1.128***	−1.240***
	(0.003)	(0.017)	(0.021)		(0.019)	(0.023)	(0.024)		(0.053)	(0.167)	(0.030)		(0.014)	(0.312)	(0.065)
m_2		0.621***	0.976***			0.038**	0.026			0.076*	−0.193***			0.131	0.278***
		(0.009)	(0.015)			(0.019)	(0.023)			(0.040)	(0.020)			(0.311)	(0.052)
m_3			0.456***				−0.021				−0.084***				0.071***
			(0.010)				(0.019)				(0.017)				(0.024)
AIC	−31188	−40598	−45388	−51742	−52645	−52663	−52666	−52643	−52659	−52662	−52886	−52665	−52916	−52667	−52949
BIC	−31087	−40489	−45271	−51641	−52536	−52546	−52542	−52535	−52542	−52538	−52754	−52548	−52792	−52535	−52809
Log-lik	15,607	20,313	22,709	25,884	26,337	26,346	26,349	26,336	26,344	26,347	26,460	26,347	26,474	26,350	26,492
N	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Mean equation includes exogenous controls (not shown). `vce(robust)` corresponds to QML sandwich standard errors.

† ARMA(3,1) and ARMA(3,3) exhibit AR–MA root cancellation with near-explosive AR polynomial roots; fit improvement is likely spurious.

Table A.4.3: SE3: ARMA(p, q) search results, QML robust standard errors, sample 01 Jan 2024 to 31 Dec 2025

	ARMA(p,q) specification														
	(0,1)	(0,2)	(0,3)	(1,0)	(1,1)	(1,2)	(1,3)	(2,0)	(2,1)	(2,2)	(2,3)	(3,0)	(3,1)	(3,2)[†]	(3,3)[†]
a_1				0.940***	0.913***	0.899***	0.893***	1.185***	1.204***	1.130***	0.470**	1.183***	0.981***	2.346***	2.847***
				(0.004)	(0.005)	(0.005)	(0.006)	(0.012)	(0.036)	(0.083)	(0.190)	(0.013)	(0.207)	(0.048)	(0.000)
a_2								−0.261***	−0.279***	−0.211***	0.381**	−0.254***	−0.014	−1.827***	−2.767
								(0.011)	(0.034)	(0.075)	(0.171)	(0.018)	(0.245)	(0.091)	(0.000)
a_3												−0.006	−0.059	0.479***	0.915***
												(0.011)	(0.055)	(0.043)	(0.000)
m_1	0.816***	1.110***	1.195***		0.254***	0.282***	0.290***		−0.021	0.053	0.714***		0.202	−1.180***	−1.705***
	(0.003)	(0.012)	(0.012)		(0.011)	(0.013)	(0.013)		(0.038)	(0.082)	(0.189)		(0.207)	(0.056)	(0.011)
m_2		0.596***	0.941***			0.080***	0.092***			0.023	0.211***			0.229***	0.570***
		(0.007)	(0.009)			(0.011)	(0.012)			(0.024)	(0.054)			(0.054)	(0.022)
m_3			0.449***				0.026**				0.059***				0.216***
			(0.007)				(0.010)				(0.016)				(0.011)
AIC	−20157	−28749	−33262	−37821	−38918	−39008	−39015	−39015	−39014	−39013	−39015	−39014	−39012	−39277	−39737
BIC	−20048	−28632	−33137	−37712	−38802	−38884	−38883	−38899	−38890	−38881	−38875	−38890	−38880	−39137	−39597
Log-lik	10,093	14,389	16,647	18,924	19,474	19,520	19,525	19,523	19,523	19,523	19,525	19,523	19,523	19,657	19,887
N	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Mean equation includes exogenous controls (not shown). `vce(robust)` corresponds to QML sandwich standard errors.

[†] ARMA(3,2) and ARMA(3,3) exhibit AR–MA root cancellation with near-explosive AR polynomial roots; fit improvement is likely spurious.

Table A.4.4: SE4: ARMA(p, q) search results, QML robust standard errors, sample 01 Jan 2024 to 31 Dec 2025

	ARMA(p,q) specification														
	(0,1)	(0,2)	(0,3)	(1,0)	(1,1)	(1,2)	(1,3)	(2,0)	(2,1)	(2,2)	(2,3)	(3,0)	(3,1)	(3,2)[†]	(3,3)[†]
a_1				0.905***	0.873***	0.858***	0.848***	1.076***	1.196***	1.190***	0.941***	1.072***	1.183***	2.418***	2.613***
				(0.004)	(0.005)	(0.006)	(0.007)	(0.014)	(0.054)	(0.083)	(0.249)	(0.014)	(0.154)	(0.083)	(0.005)
a_2								−0.188***	−0.297***	−0.291***	−0.080	−0.163***	−0.282*	−1.965***	−2.525***
								(0.013)	(0.048)	(0.073)	(0.214)	(0.020)	(0.167)	(0.154)	(0.009)
a_3												−0.023**	−0.003	0.545***	0.881***
												(0.012)	(0.032)	(0.071)	(0.005)
m_1	0.772***	1.024***	1.075***		0.191***	0.212***	0.224***		−0.125**	−0.118	0.131		−0.111	−1.369***	−1.570***
	(0.004)	(0.013)	(0.015)		(0.014)	(0.015)	(0.016)		(0.060)	(0.084)	(0.250)		(0.153)	(0.097)	(0.015)
m_2		0.540***	0.814***			0.061***	0.076***			0.002	0.056			0.403***	0.729***
		(0.007)	(0.010)			(0.011)	(0.013)			(0.020)	(0.053)			(0.095)	(0.026)
m_3			0.395***				0.030***				0.024				0.144***
			(0.008)				(0.011)				(0.018)				(0.015)
AIC	−11189	−18290	−21709	−25107	−25679	−25728	−25737	−25729	−25737	−25735	−25735	−25737	−25735	−25958	−26241
BIC	−11120	−18213	−21623	−25037	−25601	−25642	−25644	−25652	−25652	−25642	−25634	−25651	−25642	−25857	−26132
Log-lik	5,604	9,155	10,865	12,562	12,849	12,875	12,881	12,875	12,880	12,880	12,881	12,879	12,880	12,992	13,134
N	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519	17,519

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Mean equation includes exogenous controls (not shown). `vce(robust)` corresponds to QML sandwich standard errors.

[†] ARMA(3,2) and ARMA(3,3) exhibit AR–MA root cancellation with near-explosive AR polynomial roots; fit improvement is likely spurious.

A.5 Structural Breaks on Rolling Window Coefficients

SE1: Structural Break Anaylsis

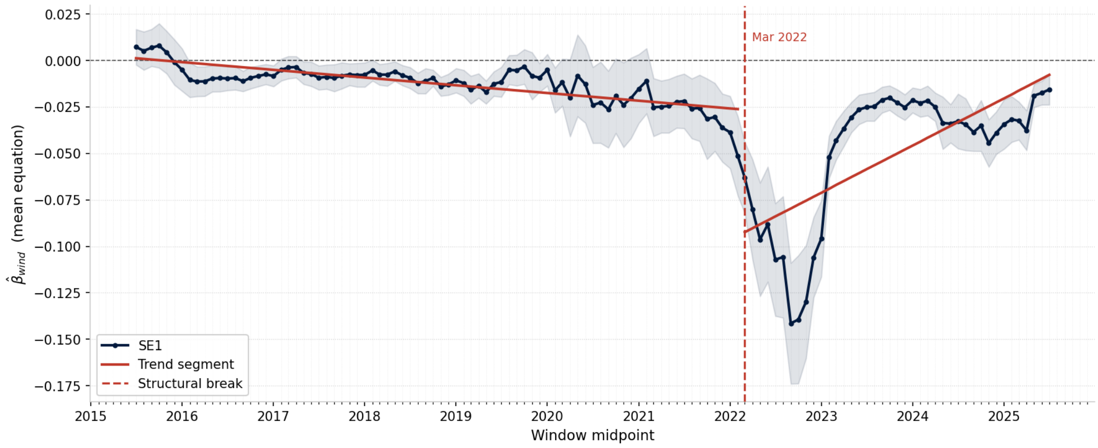


Figure A.5.1: SE1: 1 structural break, rolling-window level coefficient (1 year window, 1 month shift)

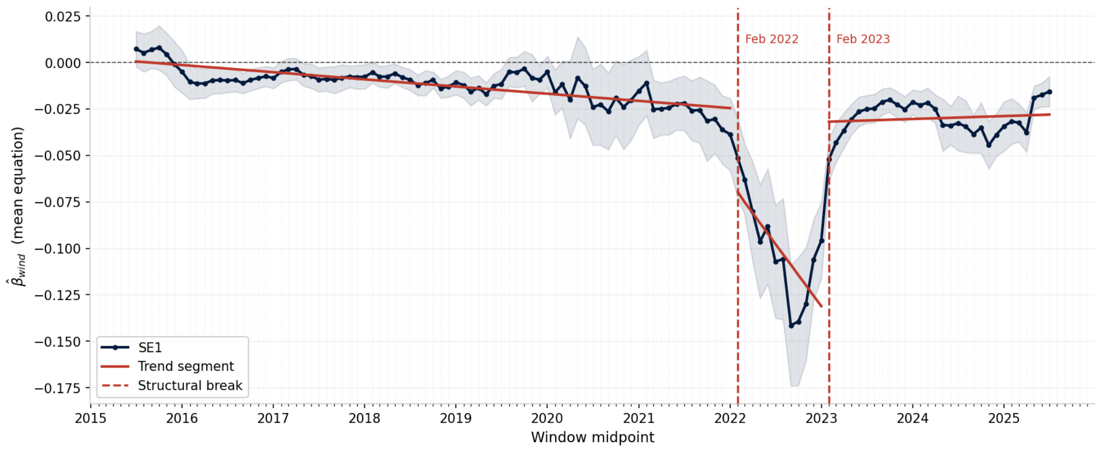


Figure A.5.2: SE1: 2 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

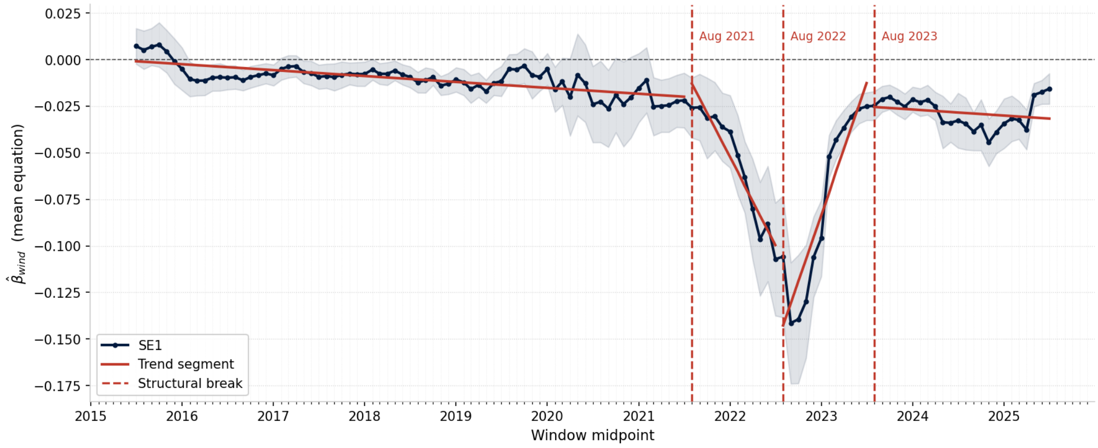


Figure A.5.3: SE1: 3 structural breaks, rolling-window level coefficient (1 year window, 1 month shift, chosen specification)

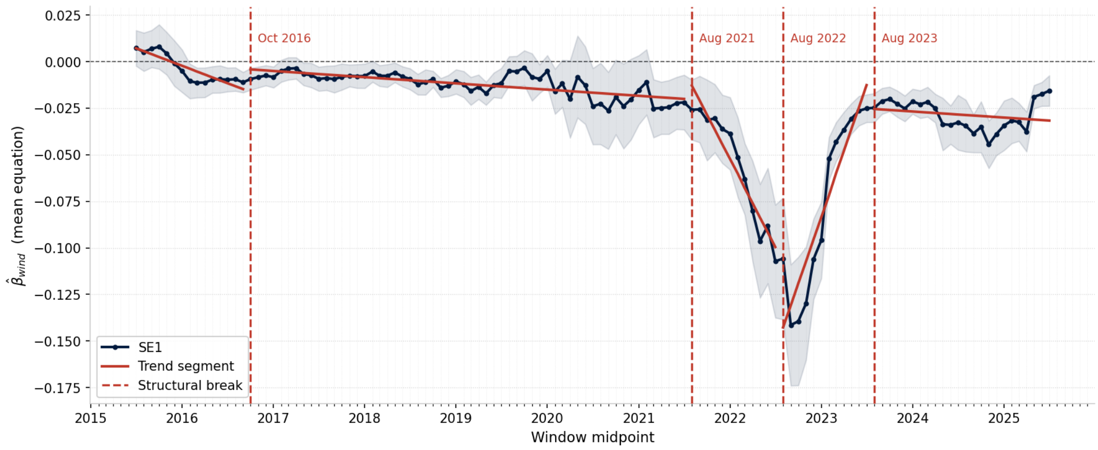


Figure A.5.4: SE1: 4 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

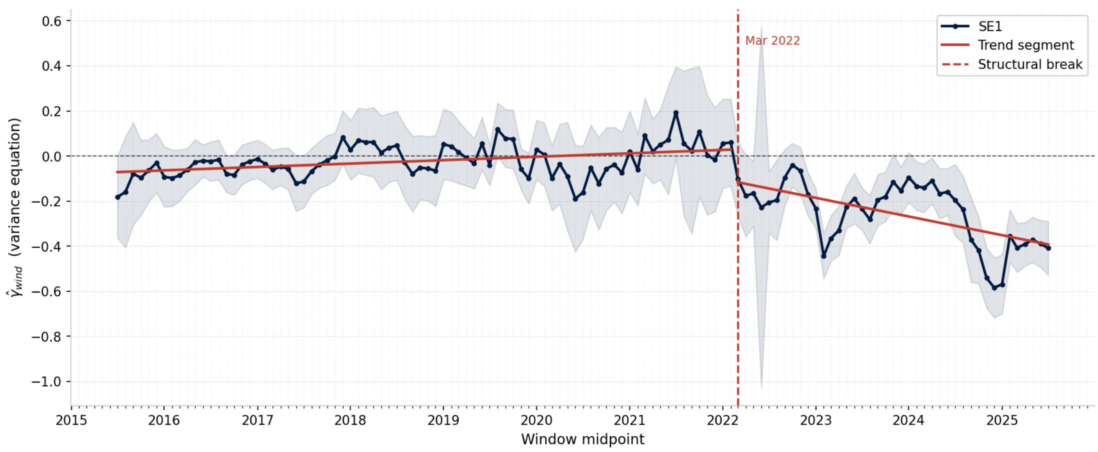


Figure A.5.5: SE1: 1 structural break, rolling-window variance coefficient (1 year window, 1 month shift)

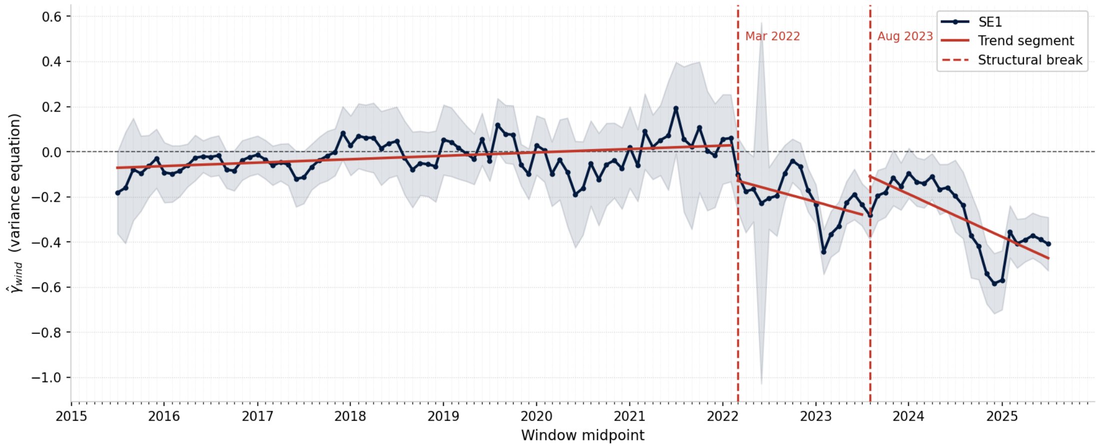


Figure A.5.6: SE1: 2 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift)

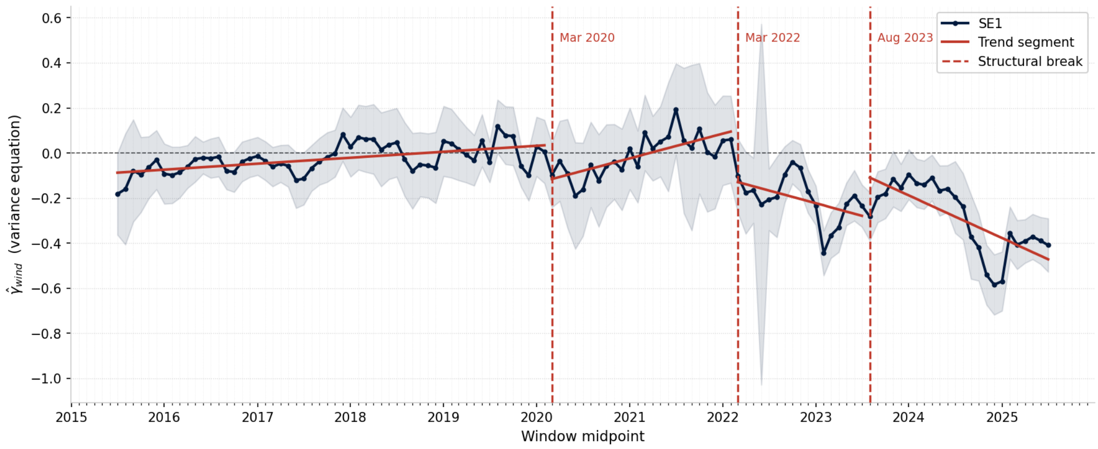


Figure A.5.7: SE1: 3 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift, chosen specification)

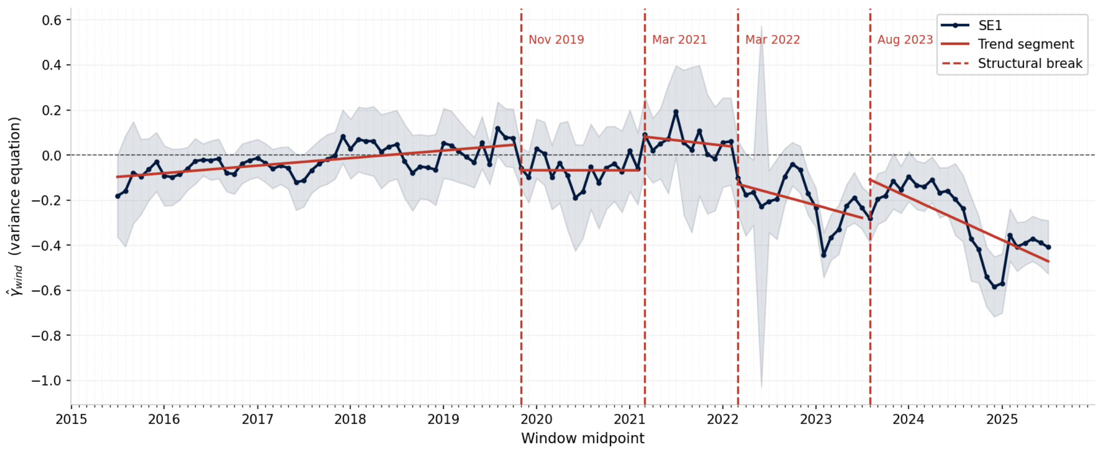


Figure A.5.8: SE1: 4 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift)

SE2: Structural Break Anaylsis

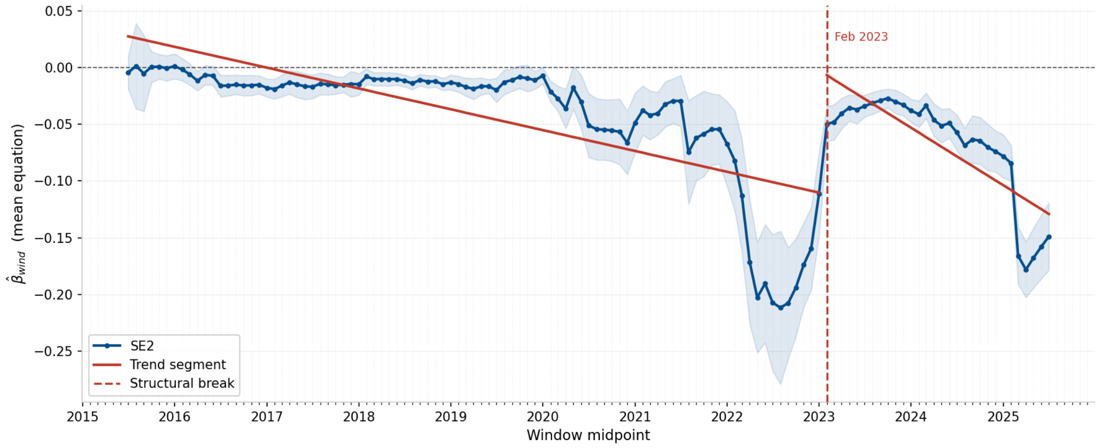


Figure A.5.9: SE2: 1 structural break, rolling-window level coefficient (1 year window, 1 month shift)

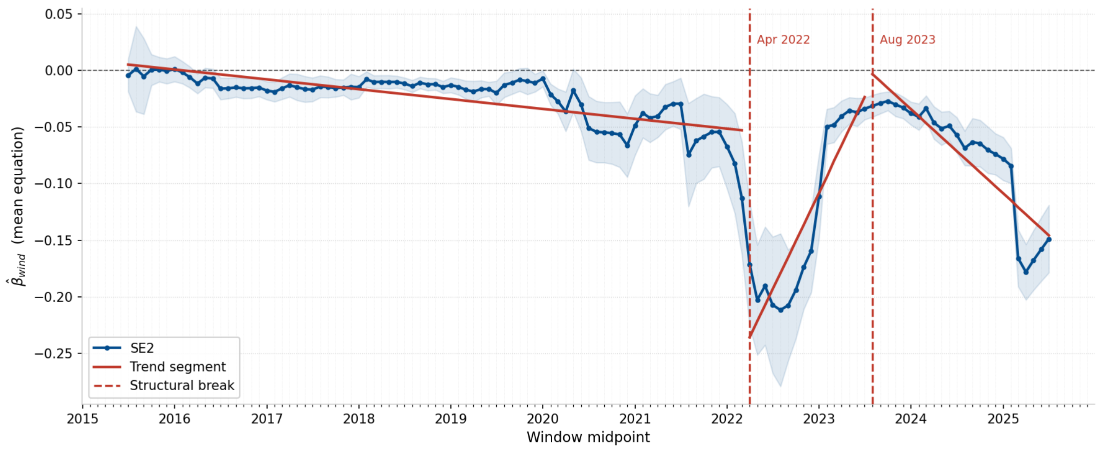


Figure A.5.10: SE2: 2 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

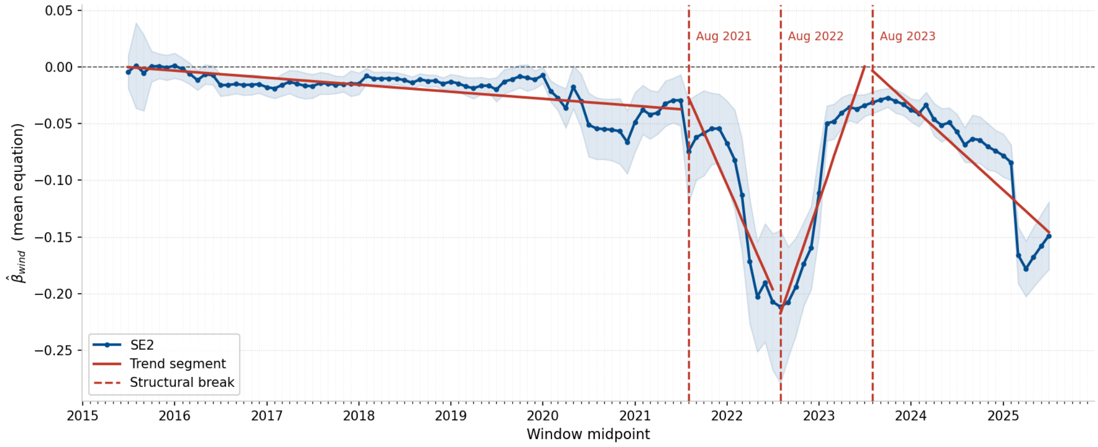


Figure A.5.11: SE2: 3 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

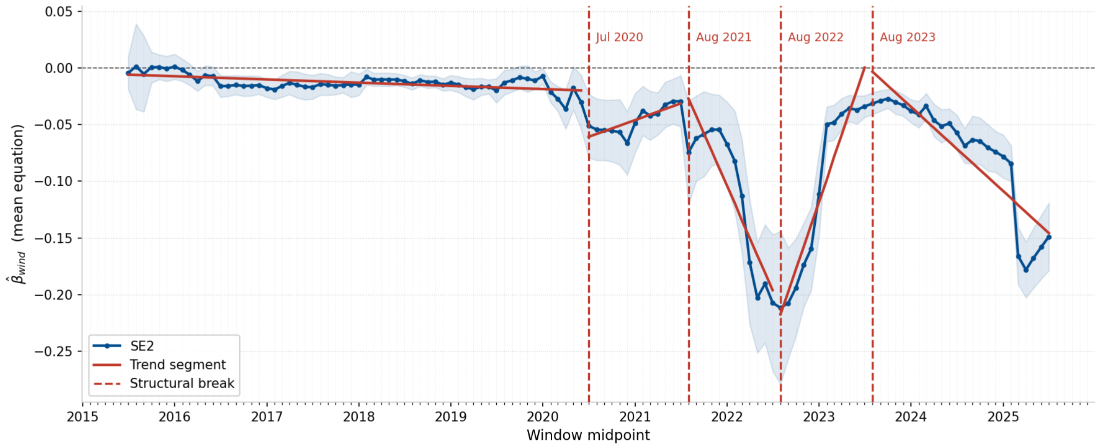


Figure A.5.12: SE2: 4 structural breaks, rolling-window level coefficient (1 year window, 1 month shift, chosen specification)

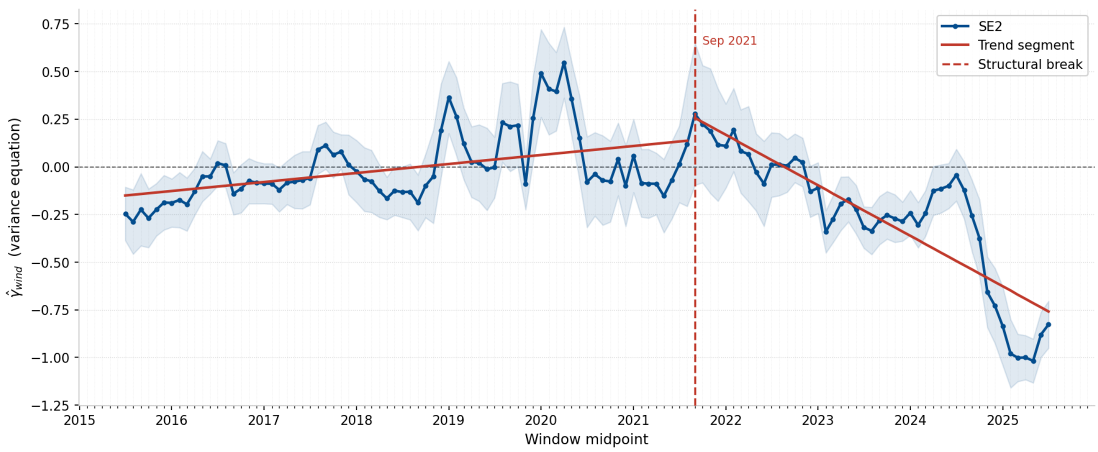


Figure A.5.13: SE2: 1 structural break, rolling-window variance coefficient (1 year window, 1 month shift)

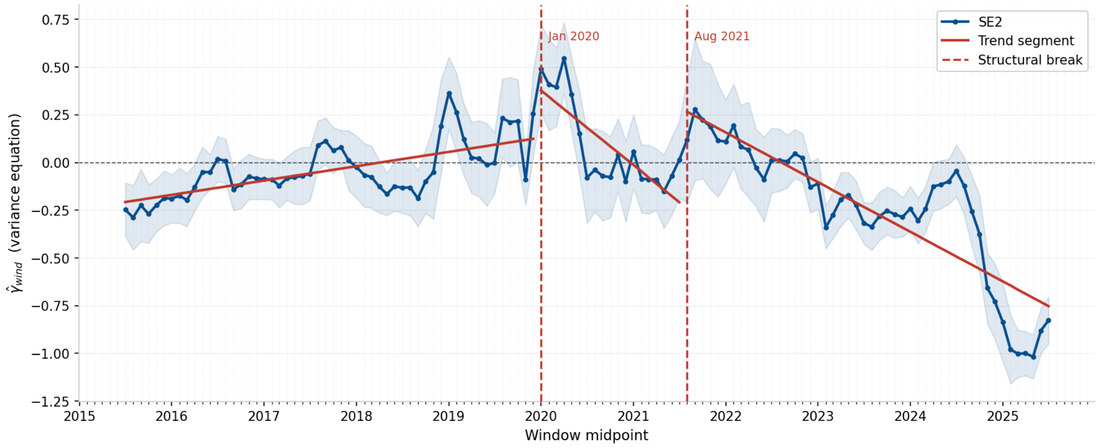


Figure A.5.14: SE2: 2 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift)

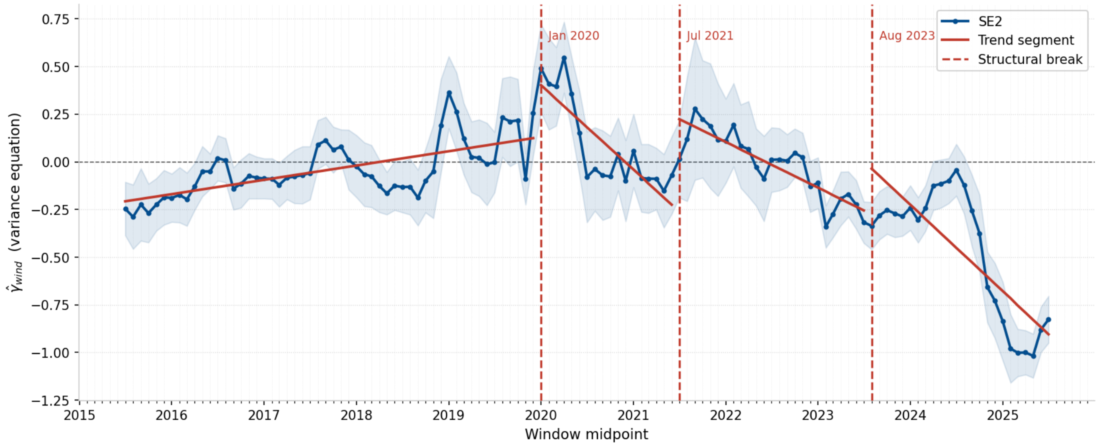


Figure A.5.15: SE2: 3 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift)

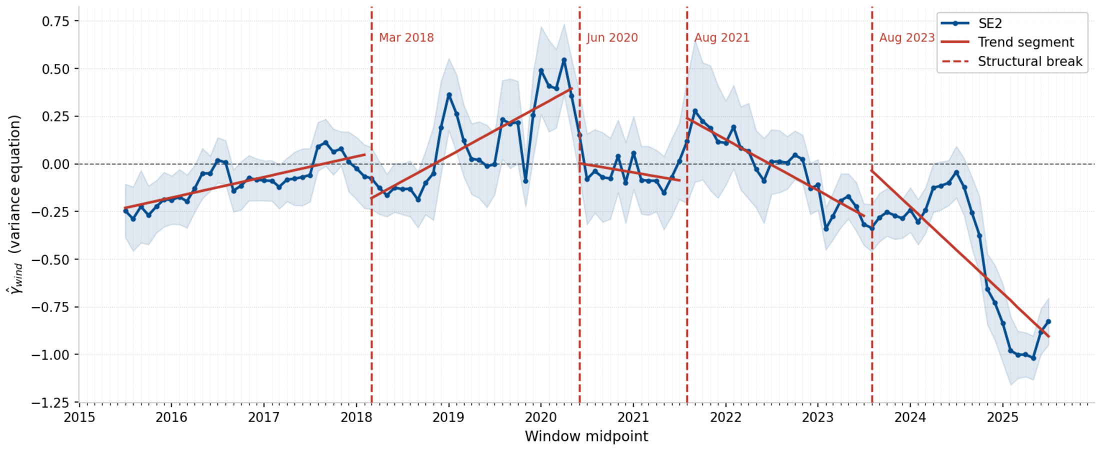


Figure A.5.16: SE2: 4 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift, chosen specification)

SE3: Structural Break Anaylsis

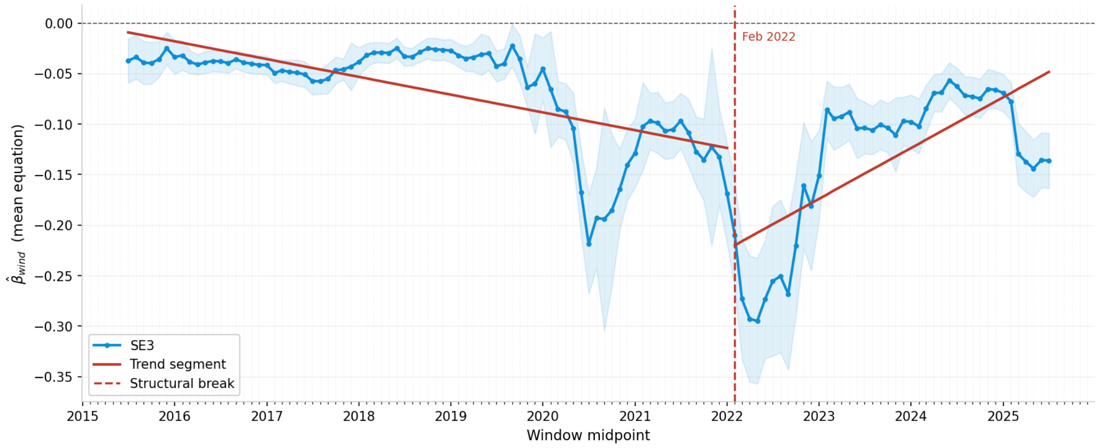


Figure A.5.17: SE3: 1 structural break, rolling-window level coefficient (1 year window, 1 month shift)

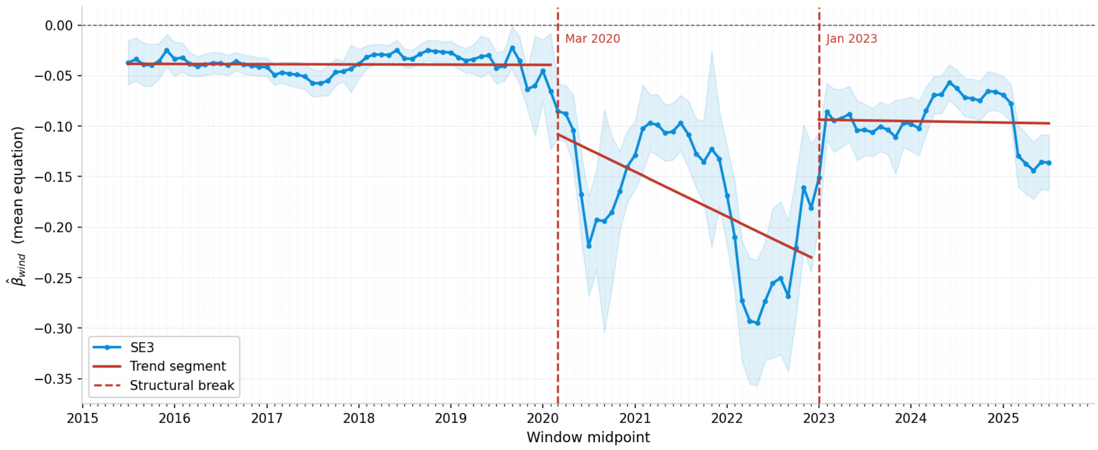


Figure A.5.18: SE3: 2 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

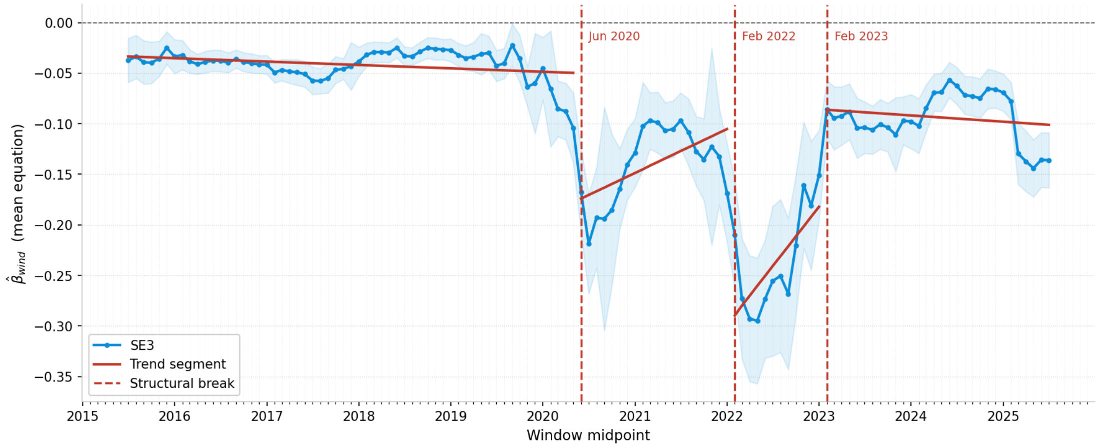


Figure A.5.19: SE3: 3 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

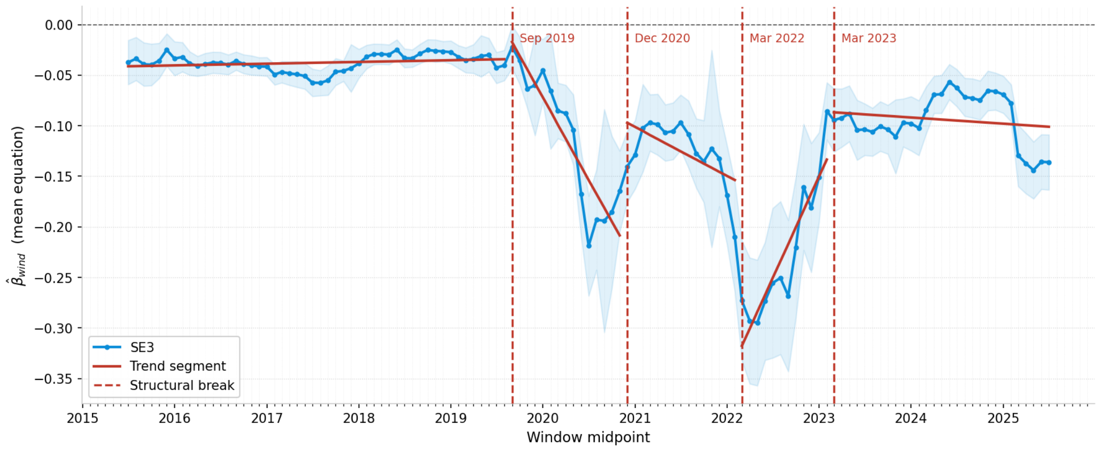


Figure A.5.20: SE3: 4 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

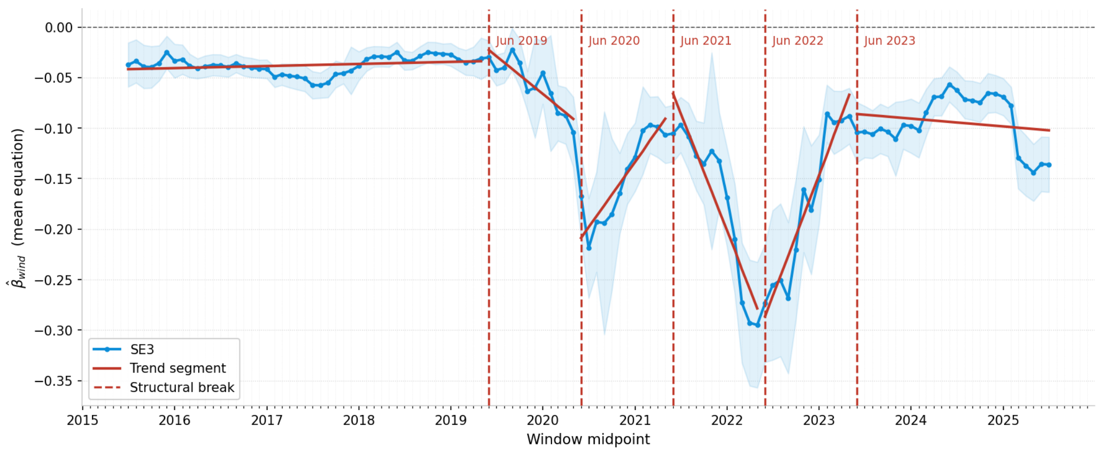


Figure A.5.21: SE3: 5 structural breaks, rolling-window level coefficient (1 year window, 1 month shift, chosen specification)

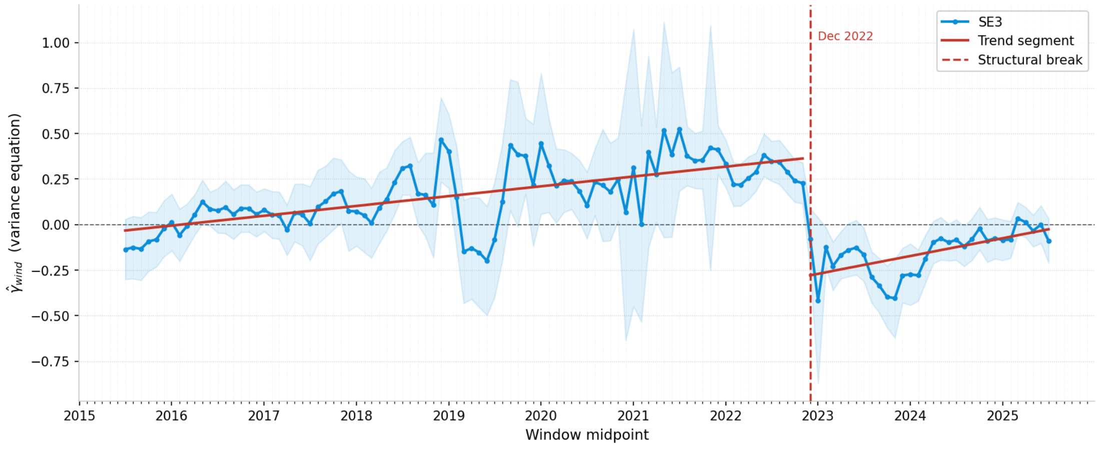


Figure A.5.22: SE3: 1 structural break, rolling-window variance coefficient (1 year window, 1 month shift)

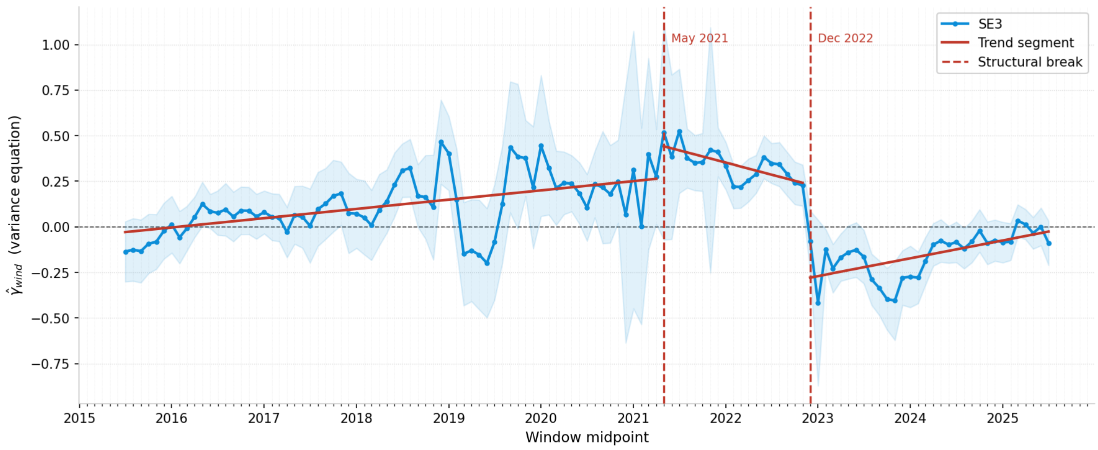


Figure A.5.23: SE3: 2 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift)

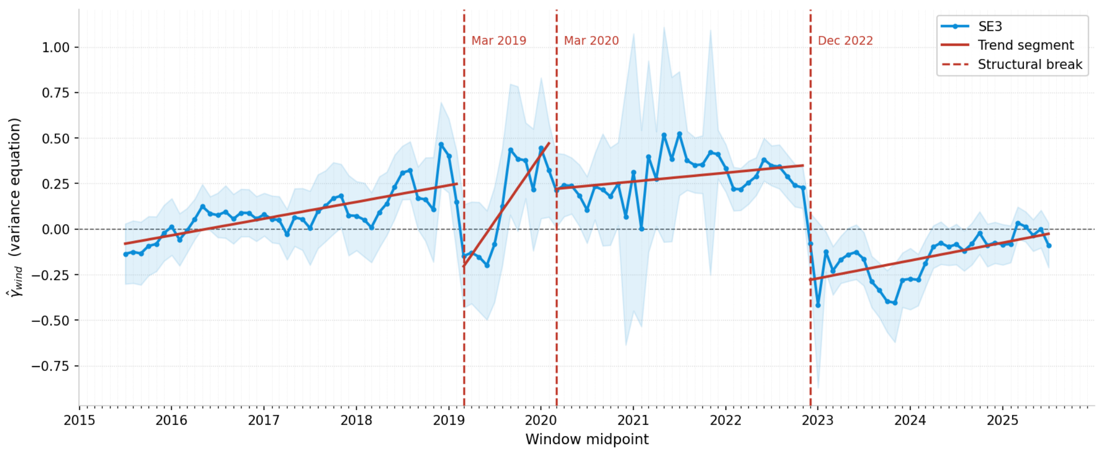


Figure A.5.24: SE3: 3 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift)

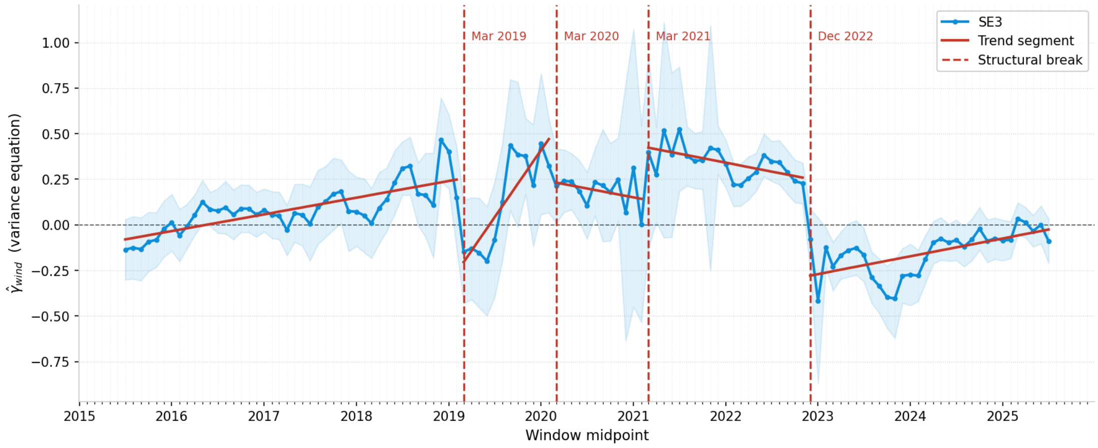


Figure A.5.25: SE3: 4 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift, chosen specification)

SE4: Structural Break Anaylsis

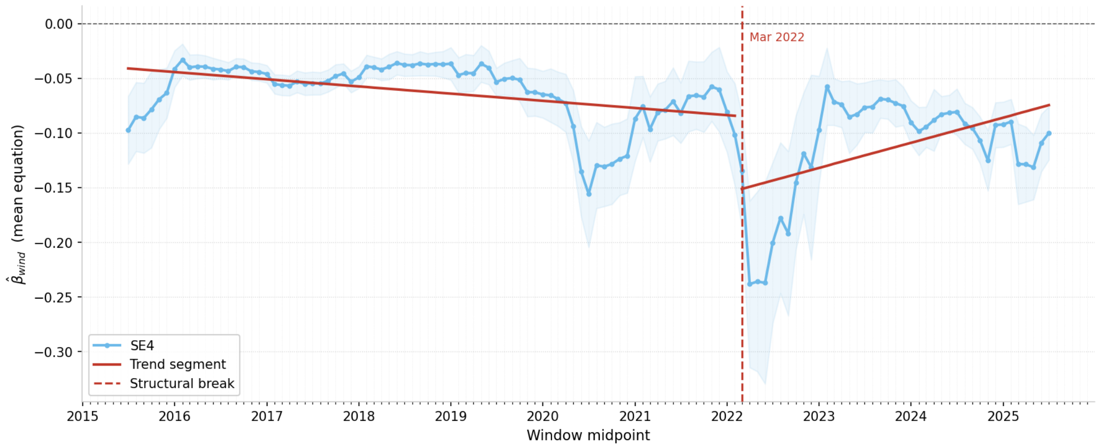


Figure A.5.26: SE4: 1 structural break, rolling-window level coefficient (1 year window, 1 month shift)

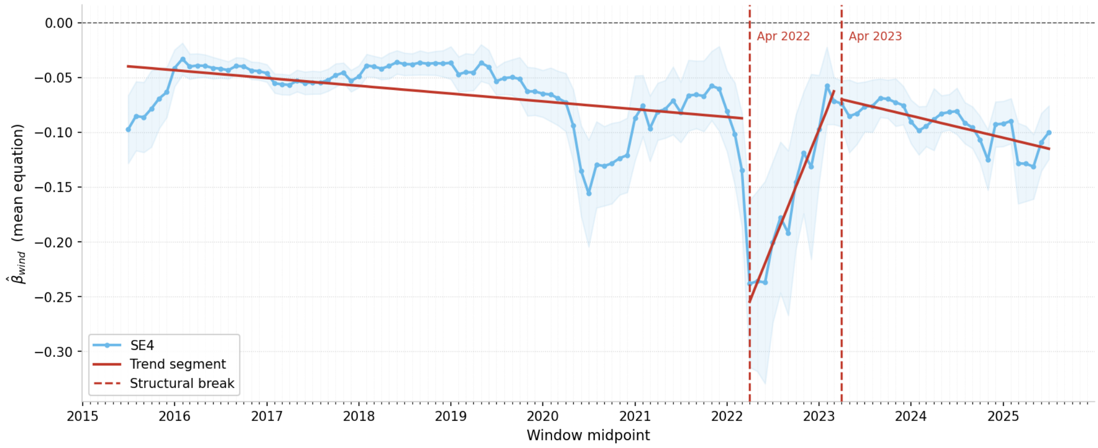


Figure A.5.27: SE4: 2 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

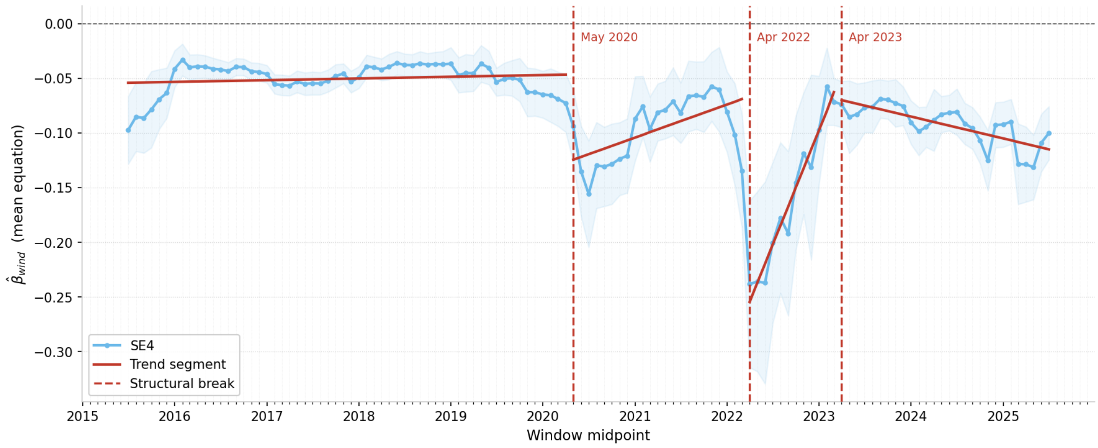


Figure A.5.28: SE4: 3 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

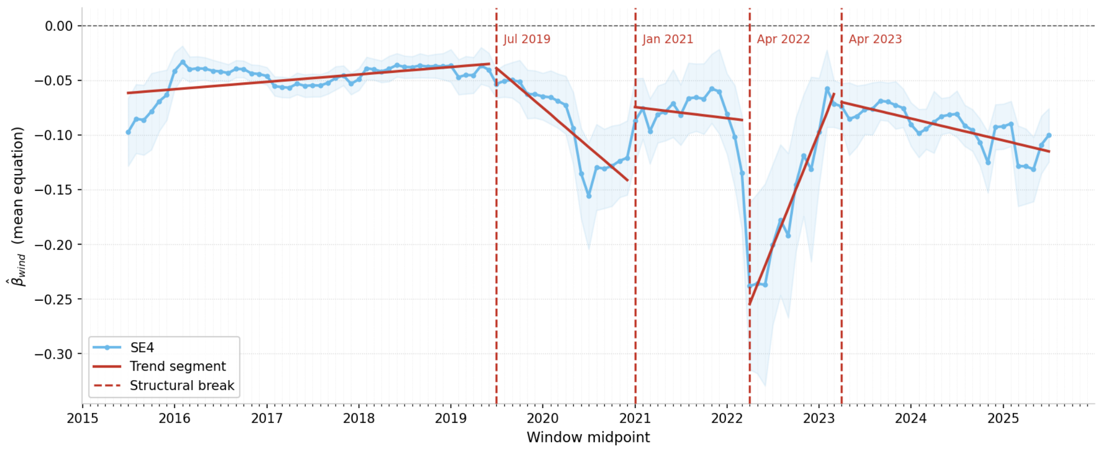


Figure A.5.29: SE4: 4 structural breaks, rolling-window level coefficient (1 year window, 1 month shift)

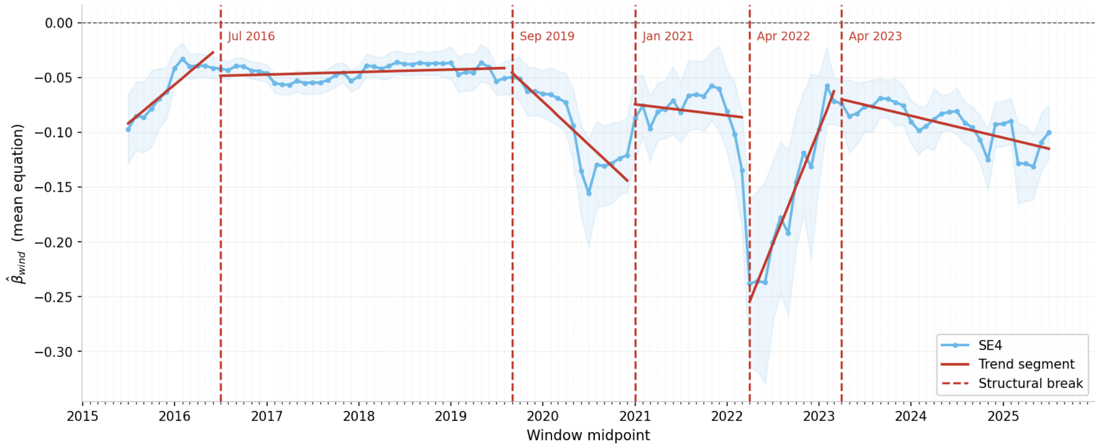


Figure A.5.30: SE4: 5 structural breaks, rolling-window level coefficient (1 year window, 1 month shift, chosen specification)

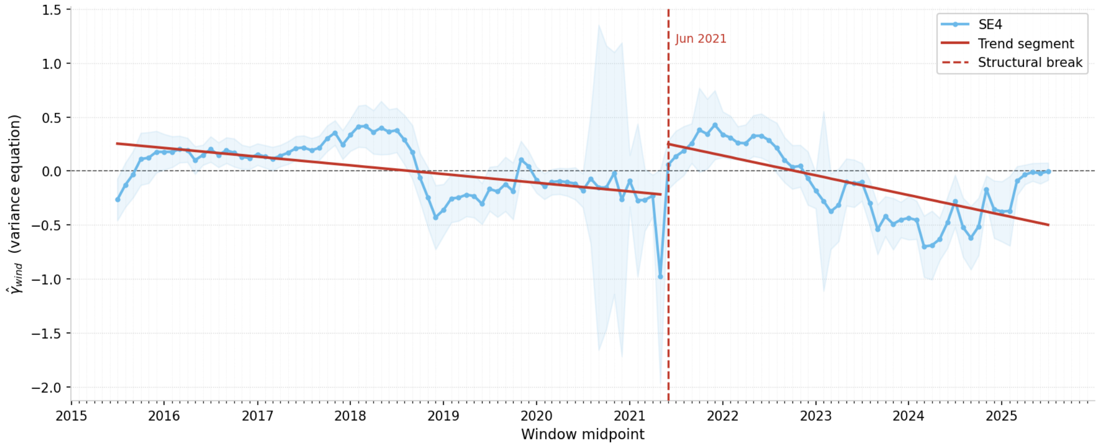


Figure A.5.31: SE4: 1 structural break, rolling-window variance coefficient (1 year window, 1 month shift)

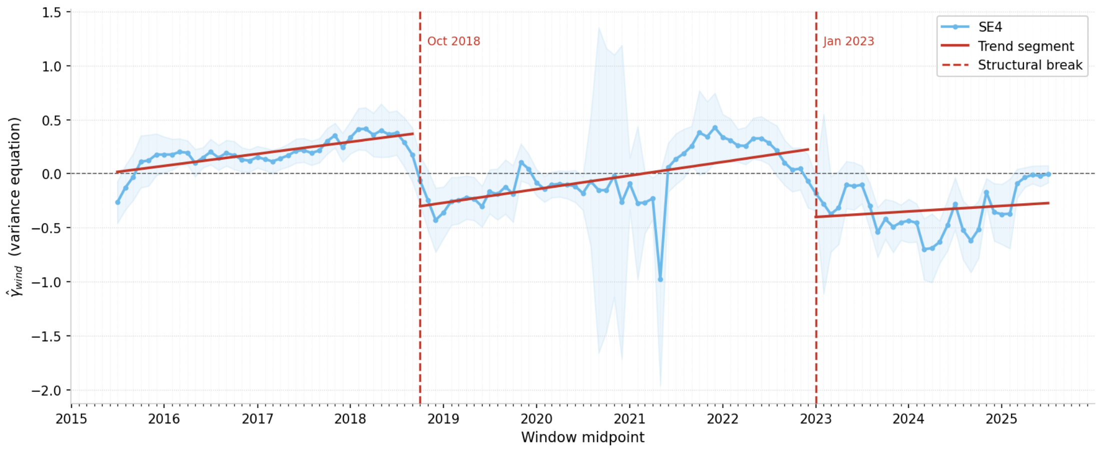


Figure A.5.32: SE4: 2 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift)

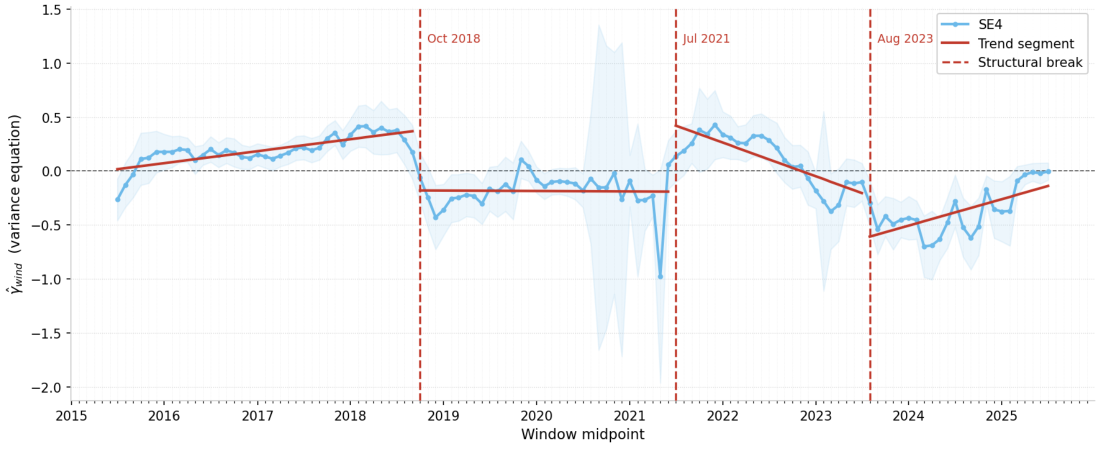


Figure A.5.33: SE4: 3 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift)

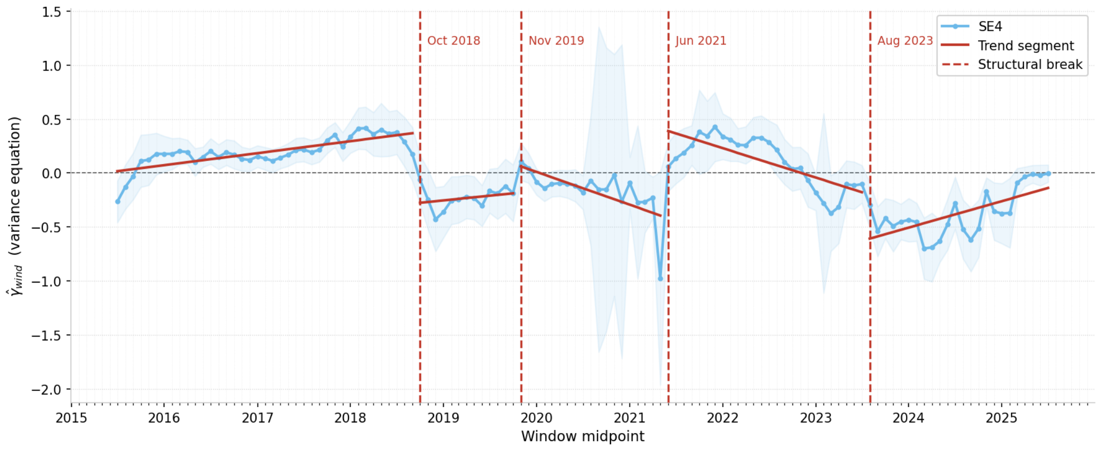


Figure A.5.34: SE4: 4 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift)

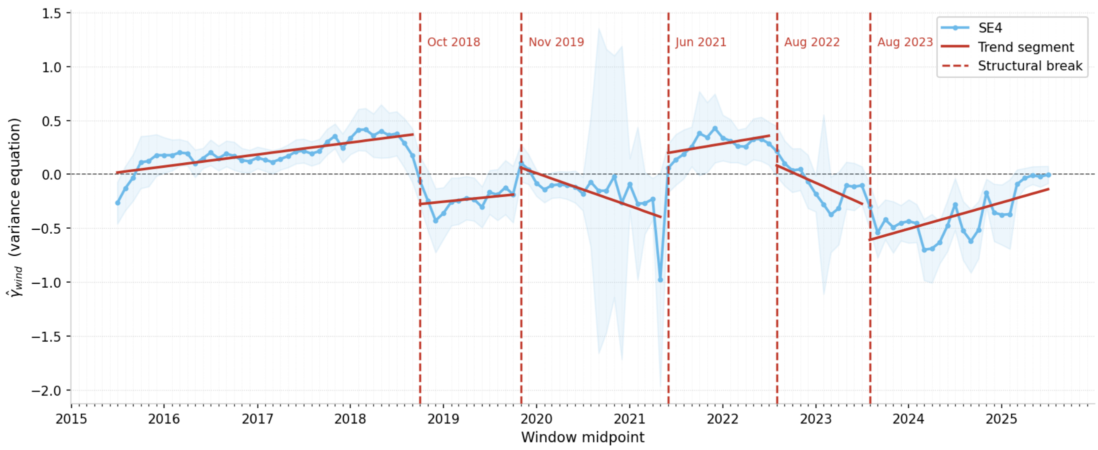


Figure A.5.35: SE4: 5 structural breaks, rolling-window variance coefficient (1 year window, 1 month shift, chosen specification)

A.6 Quantile Regression

Table A.6.1: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$, SE1

Year	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
2015	0.0079^{***}	0.0023^{***}	0.0009^{**}	-0.0005	-0.0043^{***}
2016	-0.0028^{***}	-0.0013^{***}	-0.0011^{***}	-0.0006	-0.0003
2017	-0.0019^{**}	-0.0012^{**}	-0.0011^{***}	-0.0008	-0.0017^{**}
2018	-0.0032^{***}	-0.0027^{***}	-0.0016^{***}	-0.0015^{***}	-0.0034^{***}
2019	-0.0026^{**}	-0.0020^{***}	-0.0022^{***}	-0.0019^{***}	-0.0053^{***}
2020	0.0028	0.0019^{*}	-0.0003	-0.0021^{**}	-0.0104^{***}
2021	-0.0039^{***}	-0.0010	-0.0005	-0.0021^{***}	-0.0045^{***}
2022	-0.0017	-0.0019	-0.0019	-0.0071^{***}	-0.0192^{***}
2023	-0.0029^{***}	-0.0017^{***}	-0.0017^{***}	-0.0031^{***}	-0.0093^{***}
2024	-0.0077^{***}	-0.0043^{***}	-0.0029^{***}	-0.0037^{***}	-0.0073^{***}
2025	-0.0027^{***}	-0.0016^{***}	-0.0025^{***}	-0.0053^{***}	-0.0121^{***}
Overall	-0.0000	-0.0001	-0.0005^{***}	-0.0008^{***}	-0.0029^{***}

Notes: Each entry reports $\hat{\beta}_{\text{wind}}(\tau)$ from a quantile regression of the log day-ahead spot price on log wind production and the full set of controls, estimated over the calendar year indicated. Overall reports the estimate over the full 2015–2025 sample. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.6.2: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$, SE2

Year	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
2015	0.0090^{***}	0.0029^{***}	0.0013^{**}	0.0002	-0.0035^{***}
2016	-0.0044^{***}	-0.0017^{***}	-0.0006	0.0000	-0.0003
2017	-0.0023^{**}	-0.0015^{**}	-0.0007	-0.0007	-0.0013
2018	-0.0006	-0.0015^{***}	-0.0015^{***}	-0.0019^{***}	-0.0046^{***}
2019	-0.0025^{**}	-0.0016^{*}	-0.0026^{***}	-0.0025^{***}	-0.0073^{***}
2020	-0.0074^{**}	-0.0027^{*}	-0.0019^{*}	-0.0019	-0.0101^{***}
2021	-0.0017	-0.0026^{**}	-0.0025^{***}	-0.0046^{***}	-0.0108^{***}
2022	-0.0315^{***}	-0.0158^{***}	-0.0112^{***}	-0.0150^{***}	-0.0235^{***}
2023	-0.0052^{***}	-0.0018^{***}	-0.0026^{***}	-0.0047^{***}	-0.0114^{***}
2024	-0.0156^{***}	-0.0079^{***}	-0.0050^{***}	-0.0075^{***}	-0.0129^{***}
2025	-0.0077^{**}	-0.0049^{***}	-0.0089^{***}	-0.0212^{***}	-0.0492^{***}
Overall	-0.0002	-0.0003^{*}	-0.0007^{***}	-0.0014^{***}	-0.0045^{***}

Notes: See Table A.6.1.

Table A.6.3: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$, SE3

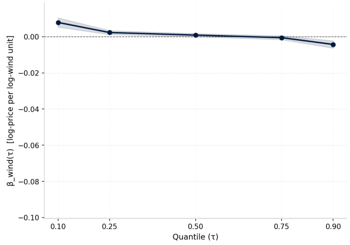
Year	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
2015	-0.0006	-0.0008	-0.0003	-0.0014	-0.0062^{***}
2016	-0.0074^{***}	-0.0038^{***}	-0.0025^{***}	-0.0017^{***}	-0.0021^{**}
2017	-0.0103^{***}	-0.0072^{***}	-0.0044^{***}	-0.0040^{***}	-0.0034^{**}
2018	-0.0090^{***}	-0.0043^{***}	-0.0025^{***}	-0.0018^{***}	0.0007
2019	-0.0023	-0.0035^{***}	-0.0046^{***}	-0.0069^{***}	-0.0100^{***}
2020	-0.0420^{***}	-0.0184^{***}	-0.0119^{***}	-0.0152^{***}	-0.0383^{***}
2021	-0.0289^{***}	-0.0176^{***}	-0.0089^{***}	-0.0070^{***}	-0.0068^{**}
2022	-0.0906^{***}	-0.0540^{***}	-0.0322^{***}	-0.0337^{***}	-0.0517^{***}
2023	-0.0177^{***}	-0.0095^{***}	-0.0061^{***}	-0.0059^{***}	-0.0127^{***}
2024	-0.0157^{***}	-0.0096^{***}	-0.0058^{***}	-0.0062^{***}	-0.0086^{***}
2025	-0.0261^{***}	-0.0180^{***}	-0.0138^{***}	-0.0165^{***}	-0.0275^{***}
Overall	-0.0098^{***}	-0.0052^{***}	-0.0035^{***}	-0.0026^{***}	-0.0046^{***}

Notes: See Table A.6.1.

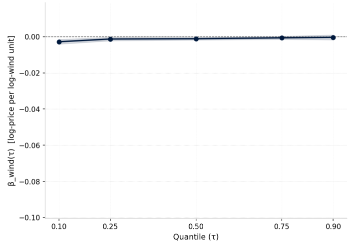
Table A.6.4: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$, SE4

Year	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
2015	-0.0048	-0.0058^{***}	-0.0034^{***}	-0.0039^{***}	-0.0079^{***}
2016	-0.0121^{***}	-0.0067^{***}	-0.0043^{***}	-0.0016^{**}	-0.0013
2017	-0.0177^{***}	-0.0104^{***}	-0.0053^{***}	-0.0030^{***}	-0.0042^{**}
2018	-0.0087^{***}	-0.0049^{***}	-0.0028^{***}	-0.0014^{*}	-0.0001
2019	-0.0028	-0.0050^{***}	-0.0068^{***}	-0.0083^{***}	-0.0113^{***}
2020	-0.0226^{***}	-0.0073^{***}	-0.0072^{***}	-0.0149^{***}	-0.0316^{***}
2021	-0.0296^{***}	-0.0116^{***}	-0.0057^{***}	-0.0070^{***}	-0.0074^{*}
2022	-0.0736^{***}	-0.0388^{***}	-0.0187^{***}	-0.0133^{***}	-0.0264^{***}
2023	-0.0096^{***}	-0.0071^{***}	-0.0052^{***}	-0.0056^{***}	-0.0118^{***}
2024	-0.0118^{***}	-0.0093^{***}	-0.0069^{***}	-0.0084^{***}	-0.0139^{***}
2025	-0.0162^{***}	-0.0159^{***}	-0.0127^{***}	-0.0166^{***}	-0.0295^{***}
Overall	-0.0080^{***}	-0.0044^{***}	-0.0035^{***}	-0.0038^{***}	-0.0095^{***}

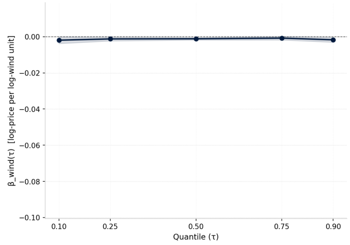
Notes: See Table A.6.1.



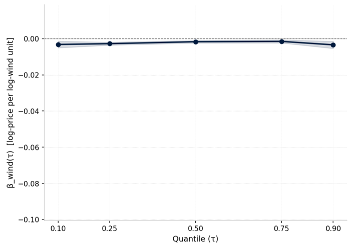
(a) SE1, 2015



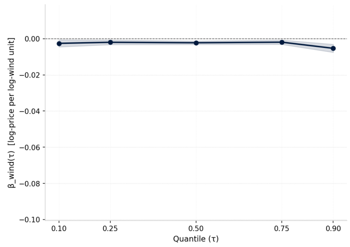
(b) SE1, 2016



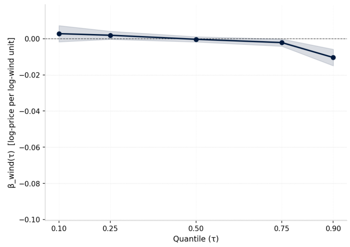
(c) SE1, 2017



(d) SE1, 2018



(e) SE1, 2019



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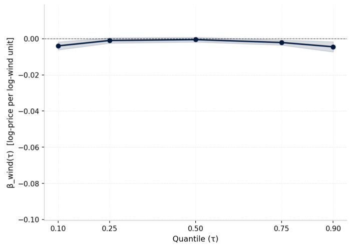
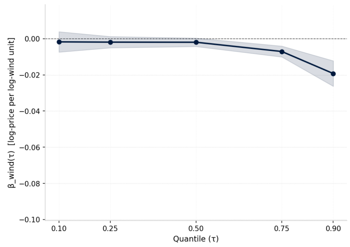
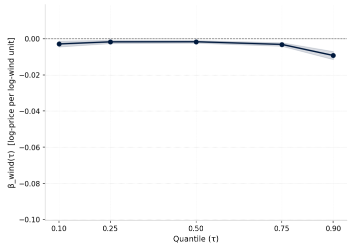


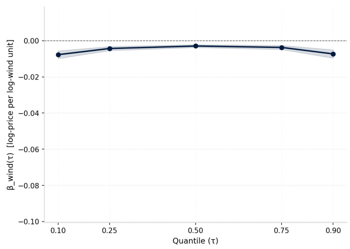
(g) SE1, 2021



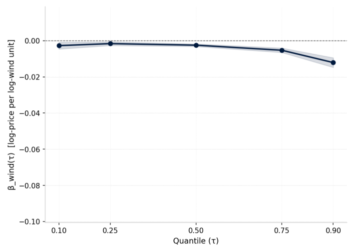
(h) SE1, 2022



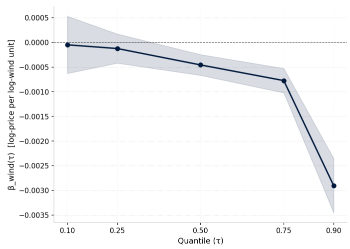
(i) SE1, 2023



(j) SE1, 2024

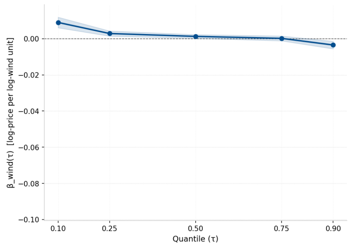


(k) SE1, 2025

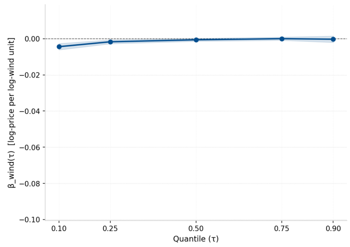


(l) SE1, 2015–2025

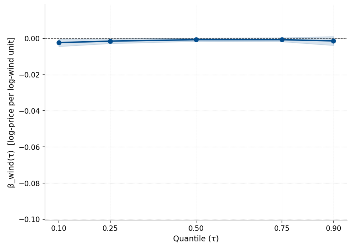
Figure A.6.1: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$ for SE1, all years 2015–2025 and full sample period.



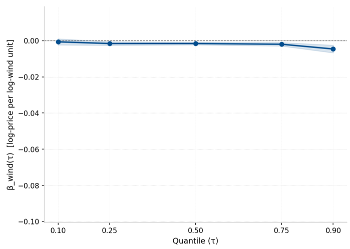
(a) SE2, 2015



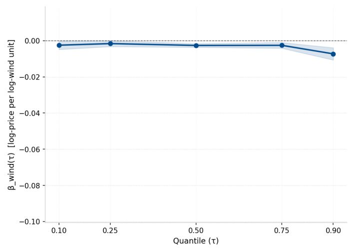
(b) SE2, 2016



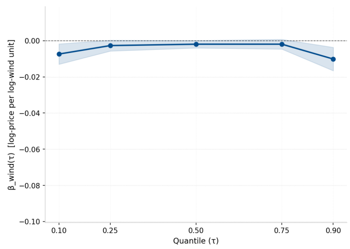
(c) SE2, 2017



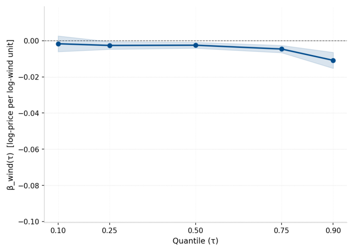
(d) SE2, 2018



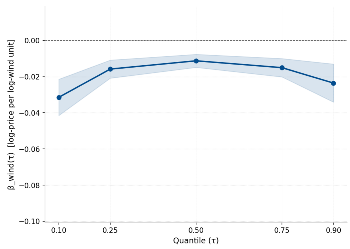
(e) SE2, 2019



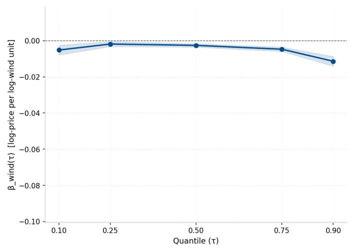
(f) SE2, 2020



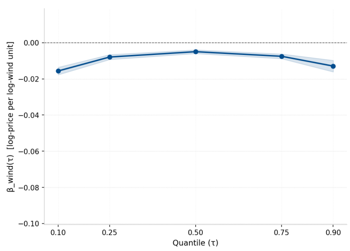
(g) SE2, 2021



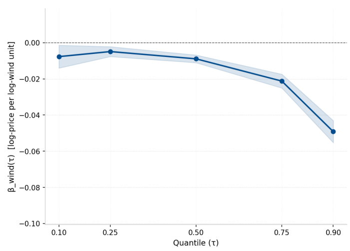
(h) SE2, 2022



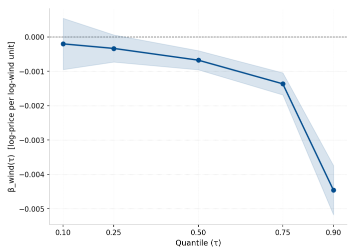
(i) SE2, 2023



(j) SE2, 2024

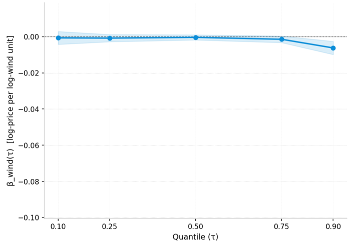


(k) SE2, 2025

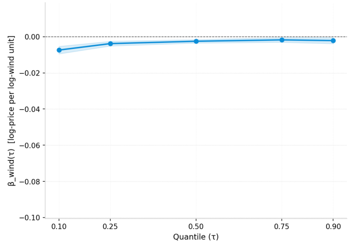


(l) SE2, 2015–2025

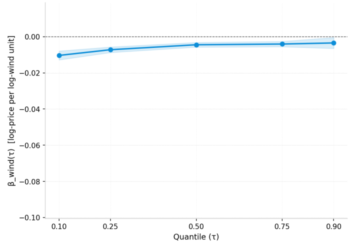
Figure A.6.2: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$ for SE2, all years 2015–2025 and full sample period.



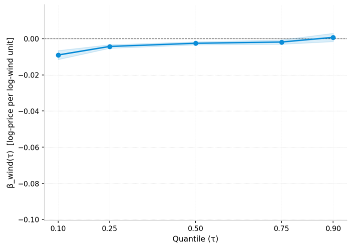
(a) SE3, 2015



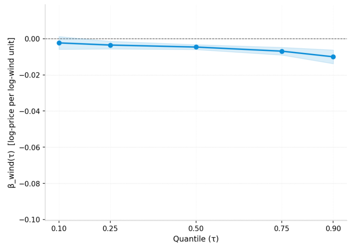
(b) SE3, 2016



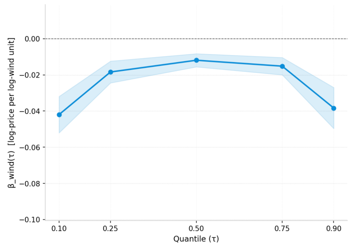
(c) SE3, 2017



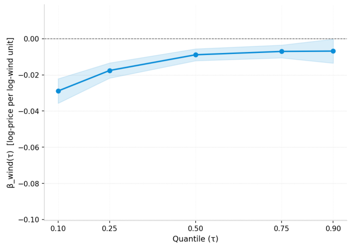
(d) SE3, 2018



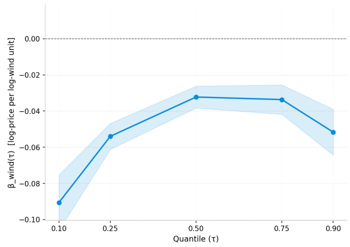
(e) SE3, 2019



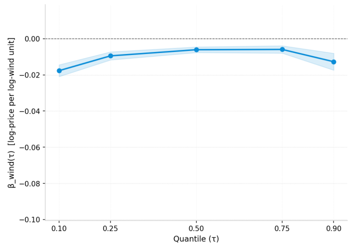
(f) SE3, 2020



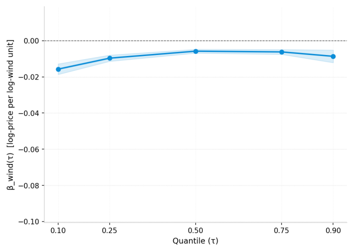
(g) SE3, 2021



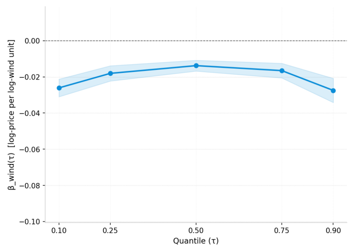
(h) SE3, 2022



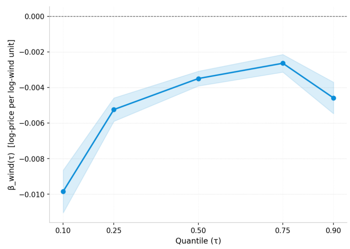
(i) SE3, 2023



(j) SE3, 2024

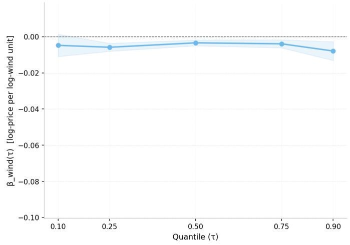


(k) SE3, 2025

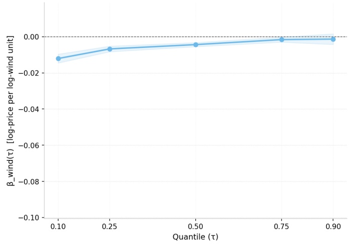


(l) SE3, 2015–2025

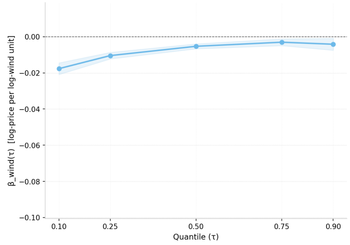
Figure A.6.3: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$ for SE3, all years 2015–2025 and full sample period.



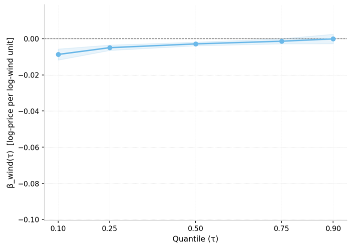
(a) SE4, 2015



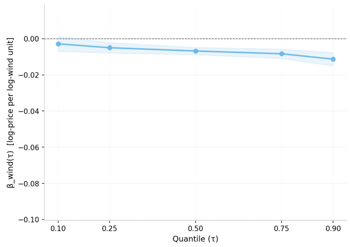
(b) SE4, 2016



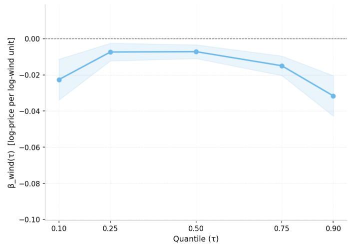
(c) SE4, 2017



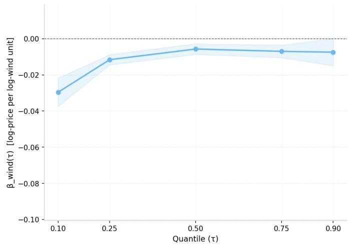
(d) SE4, 2018



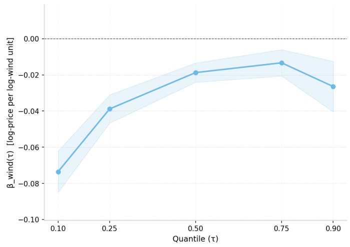
(e) SE4, 2019



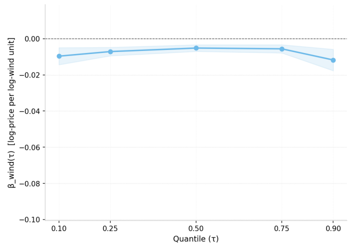
(f) SE4, 2020



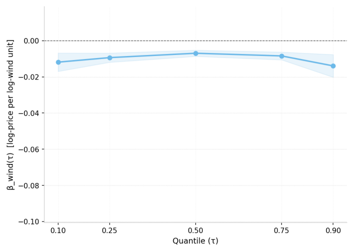
(g) SE4, 2021



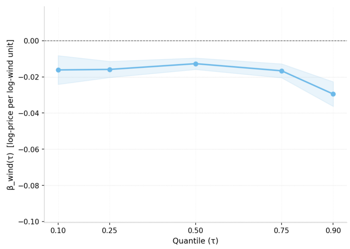
(h) SE4, 2022



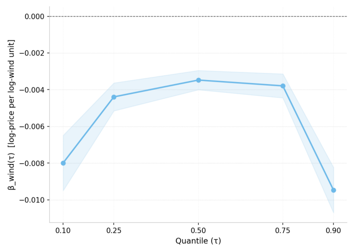
(i) SE4, 2023



(j) SE4, 2024



(k) SE4, 2025



(l) SE4, 2015–2025

Figure A.6.4: Quantile regression estimates of $\hat{\beta}_{\text{wind}}(\tau)$ for SE4, all years 2015–2025 and full sample period.